

Rewiring Immune Regulation Through Acute Exercise

**A Systematic Review and Meta-Analysis on TH1 and TH2 Cell Responses -
Documentation**

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Table of contents

Background	1
Downloads	1
1. Summary of Formulae	3
1.1. Introduction	3
1.2. Data Extraction and Computation Strategy	3
1.3. Estimating mean and SD from median/IQR/range	4
1.4. Mean and Variance of a Ratio	5
1.5. Effect Size Calculation	6
1.6. References	7
2. Workspace Setup	9
2.1. Packages	9
2.2. Outlier Detection Functions	9
3. Risk of Bias Assessment (ROBINS-I)	11
I. Th1	13
4. Th1 — Immediate Post-Exercise	15
4.1. Data Import	15
4.2. Primary Meta-Analysis	15
4.3. Outlier Detection	17
4.4. Funnel Plot & Influential Cases	18
4.5. Forest Plot	19
4.6. Asymmetry Test	20
4.7. Moderator Analyses	21
4.8. Multivariate Models	36
4.9. Interaction Surface (IFN- $\times$ IL-2)	38
5. Th1 — ~2 h Post-Exercise	39
5.1. Data Import	39
5.2. Primary Meta-Analysis	39
5.3. Outlier Detection	40
5.4. Funnel Plot & Influential Cases	41
5.5. Forest Plot	41
5.6. Asymmetry Test	42
6. Th1 — 17 h Post-Exercise	43
6.1. Data Import	43
6.2. Primary Meta-Analysis	43
6.3. Outlier Detection	44
6.4. Funnel Plot & Influential Cases	46
6.5. Forest Plot	48
6.6. Asymmetry Test	49
6.7. Moderator Analyses	49
6.8. Multivariate Models	59

II. Th2	61
7. Th2 — Immediate Post-Exercise	63
7.1. Data Import	63
7.2. Primary Meta-Analysis	63
7.3. Outlier Detection	65
7.4. Funnel Plot & Influential Cases	66
7.5. Forest Plot	67
7.6. Asymmetry Test	67
7.7. Moderator Analyses	68
7.8. Multivariate Models	85
8. Th2 — ~2 h Post-Exercise	87
8.1. Data Import	87
8.2. Primary Meta-Analysis	87
8.3. Outlier Detection	88
8.4. Funnel Plot & Influential Cases	89
8.5. Forest Plot	90
8.6. Asymmetry Test	90
9. Th2 — 17 h Post-Exercise	91
9.1. Data Import	91
9.2. Primary Meta-Analysis	91
9.3. Outlier Detection	92
9.4. Funnel Plot & Influential Cases	94
9.5. Forest Plot	95
9.6. Asymmetry Test	95
9.7. Moderator Analyses	96
9.8. Multivariate Models	106
III. Th1/Th2 Ratio	107
10. Th1/Th2 Ratio — Immediate Post-Exercise	109
10.1. Data Import	109
10.2. Primary Meta-Analysis	109
10.3. Outlier Detection	111
10.4. Funnel Plot & Influential Cases	112
10.5. Forest Plot	113
10.6. Asymmetry Test	113
10.7. Moderator Analyses	114
10.8. Multivariate Models	129
11. Th1/Th2 Ratio — ~2 h Post-Exercise	133
11.1. Data Import	133
11.2. Primary Meta-Analysis	133
11.3. Outlier Detection	134
11.4. Funnel Plot & Influential Cases	135
11.5. Forest Plot	136
11.6. Asymmetry Test	136
12. Th1/Th2 Ratio — 17 h Post-Exercise	137
12.1. Data Import	137
12.2. Primary Meta-Analysis	137
12.3. Outlier Detection	138
12.4. Funnel Plot & Influential Cases	140
12.5. Forest Plot	141
12.6. Asymmetry Test	141
12.7. Moderator Analyses	142
12.8. Multivariate Models	152

References

155

Background

This site documents the complete analysis pipeline behind our systematic review and meta-analysis of T-helper-cell responses to acute exercise. For each outcome — **Th1**, **Th2**, and the **Th1/Th2 ratio** — the chapters reproduce the random-effects meta-analysis across three post-exercise time windows (immediate, ~2 h, and ~17 h), together with outlier and influence diagnostics, small-study/publication-bias checks, and moderator and meta-regression analyses. Study quality is appraised with ROBINS-I, and the underlying effect-size computations are documented throughout. The raw data and the full analysis script are available below for reproducibility.

Downloads

Resource	File
Raw data — effect-size computation	<code>effect_computation_Th1Th2.xlsx</code>
Analysis script	<code>meta_Th1Th2.R</code>
Summary Formulae	<code>formulae.pdf</code>

1. Summary of Formulae

1.1. Introduction

Only formulas that were actually used in the calculations are highlighted by placing them in a box to improve readability and to make the key equations easy to identify. Other formulas are included for completeness and conceptual clarity.

Notation

Per study and time point

- n : sample size
- $\tilde{x}, \tilde{y}$: estimated means (approximated from published summary statistics)
- $\tilde{s}_x, \tilde{s}_y$: estimated standard deviations (approximated from published summary statistics)
- m : median
- q_1, q_3 : 1st and 3rd quartile
- a : minimum, b : maximum
- Φ^{-1} : inverse standard normal cumulative distribution function (e.g., R: `qnorm`)
- ρ : Correlation between Th1 and Th2 measurements

1.2. Data Extraction and Computation Strategy

1.2.1. Overview

This analysis requires extracting or computing the mean Th1/Th2 ratio and its associated standard deviation for each study and time point. However, the included studies reported their descriptive statistics in heterogeneous formats. To enable uniform computation of effect sizes, we developed a systematic approach:

1. **Classify how each study reported data** (see Section 1.2.2)
2. **Estimate means and SDs** when not directly reported (see Section 1.3)
3. **Compute the Th1/Th2 ratio** and its variance (see Section 1.4)
4. **Calculate standardized effect sizes** with small-sample correction (see Section 1.5)

The **computation table** (see Supplementary Materials) documents which approach (A1, A2, C1, C2, or C3) was applied to each study, providing full methodological transparency.

1.2.2. Data Reporting Types

Studies fell into two categories based on how Th1 and Th2 values were reported:

A1: Summary Statistics Directly Reported - Article reported means and standard deviations (or sufficient summary statistics to estimate them) - No access to individual participant data

1. Summary of Formulae

A2: Raw Data Available - Individual participant-level data were available (either directly from the article or upon request from authors) - We computed means, SDs, and participant-level correlations directly from the raw data

1.2.3. Summary Statistics Estimation Scenarios

When articles reported summary statistics other than means and SDs, we estimated means and SDs using sample-size-adjusted formulas (Wan et al. 2014) depending on which statistics were available:

Table 1.1.: Summary Statistics Estimation Scenarios and Corresponding Formulas

Scenario	Available Data	$\bar{x}_{estimated}$	$s_{estimated}$
C1	Min, Max, Med., n	(1)	(2)
C2	Q1, Med., Q3, n	(3)	(4)
C3	Mean, SE, n	Direct	(5)

1.3. Estimating mean and SD from median/IQR/range

Table 1.1 above summarizes the estimation scenarios.

The detailed formulas for each scenario are presented below, following the methodology of Wan et al. (2014).

1.3.1. Scenario C1: Min + Max + Median + n (a, m, b, n)

$$\bar{x} \approx \frac{a + 2m + b}{4} \quad (1)$$

$$s \approx \frac{b - a}{2\Phi^{-1}\left(\frac{n-0.375}{n+0.25}\right)} \quad (2)$$

1.3.2. Scenario C2: Median + IQR + n (q_1, m, q_3, n)

$$\bar{x} \approx \frac{q_1 + m + q_3}{3} \quad (3)$$

$$s \approx \frac{q_3 - q_1}{2\Phi^{-1}\left(\frac{0.75n-0.125}{n+0.25}\right)} \quad (4)$$

1.3.3. Scenario C3: Mean, SE

$$s = SE\sqrt{n} \quad (5)$$

1.4. Mean and Variance of a Ratio

Because several studies did not report raw data, the Th1/Th2 ratio could not be calculated directly. For these studies, we first approximated the mean and standard deviation from the published summary statistics (see Section 1.3). Using the estimated mean and SD, we then computed the Th1/Th2 ratio for studies that did not publish the ratio but provided sufficient information to derive it.

1.4.1. Calculation of the mean ratio

Let $\tilde{x}_R$ denote the mean Th1/Th2 ratio, let $\tilde{x}_{\text{Th1}}$ and $\tilde{y}_{\text{Th2}}$ denote the means of Th1 and Th2, respectively, obtained either by estimation from published summary statistics or by calculation using raw data.

$$\boxed{\tilde{x}_R = \frac{\tilde{x}_{\text{Th1}}}{\tilde{y}_{\text{Th2}}} \quad (\tilde{y}_{\text{Th2}} \neq 0)} \quad (6)$$

1.4.2. Calculation of the variance (and standard deviation) of a ratio

It is important to understand that Th1 and Th2 are two dependent variables. Taking this into account, we have used the following formula, published by Van Kempen and Van Vliet (2000):

Their Formula 7 approximates the variance of a ratio using a first-order Taylor expansion (Van Kempen and Van Vliet 2000, 301).

$$\text{var}\left(\frac{\tilde{x}}{\tilde{y}}\right) \approx \frac{\text{Var}(\tilde{x})}{\tilde{y}^2} + \frac{\tilde{x}^2 \text{Var}(\tilde{y})}{\tilde{y}^4} - 2 \frac{\tilde{x}}{\tilde{y}^3} \text{cov}(\tilde{x}, \tilde{y}) \quad (7)$$

Since $\text{var}() = s^2$ and $\text{cov}(\tilde{x}, \tilde{y}) = \rho s_{\tilde{x}} s_{\tilde{y}}$ (by rearranging) this can be written as:

$$\text{var}\left(\frac{\tilde{x}}{\tilde{y}}\right) \approx \frac{s_{\tilde{x}}^2}{\tilde{y}^2} + \frac{\tilde{x}^2 s_{\tilde{y}}^2}{\tilde{y}^4} - 2 \frac{\tilde{x} \rho s_{\tilde{x}} s_{\tilde{y}}}{\tilde{y}^3} \quad (8)$$

which can be rewritten as:

$$\text{var}\left(\frac{\tilde{x}}{\tilde{y}}\right) \approx \left(\frac{s_{\tilde{x}}}{\tilde{y}}\right)^2 + \left(\frac{\tilde{x} s_{\tilde{y}}}{\tilde{y}^2}\right)^2 - 2 \rho \frac{\tilde{x} s_{\tilde{x}} s_{\tilde{y}}}{\tilde{y}^3} \quad (9)$$

The corresponding standard deviation is obtained by

$$\boxed{\begin{aligned} SD\left(\frac{\tilde{x}}{\tilde{y}}\right) &\approx \sqrt{\left(\frac{s_{\tilde{x}}}{\tilde{y}}\right)^2 + \left(\frac{\tilde{x} s_{\tilde{y}}}{\tilde{y}^2}\right)^2 - 2 \rho \frac{\tilde{x} s_{\tilde{x}} s_{\tilde{y}}}{\tilde{y}^3}} \\ \rho &= 0.5 \end{aligned}} \quad (1.1)$$

1.5. Effect Size Calculation

1.5.1. Unstandardized mean difference

- $\tilde{x}_{\text{pre}}, s_{\text{pre}}$
- $\tilde{x}_{\text{post}}, s_{\text{post}}$
- r : Pre–Post correlation (outcome-specific)
- n : is the number of pairs

Note: A fixed pre–post correlation of $r = 0.5$ was applied uniformly across all effect size calculations.

Mean Change

$$D = \tilde{x}_{\text{post}} - \tilde{x}_{\text{pre}} \quad (11)$$

(Borenstein 2013, 22)

Standard deviation of D

$$SD_{\text{diff}} = \sqrt{s_{\text{pre}}^2 + s_{\text{post}}^2 - 2r s_{\text{pre}} s_{\text{post}}} \quad (12)$$

(Borenstein 2013, 24)

Variance of D

$$V_D = \frac{SD_{\text{diff}}^2}{n} \quad (13)$$

(Borenstein 2013, 24)

Standard error of D

$$SE_D = \sqrt{V_D} \quad (14)$$

(Borenstein 2013, 24)

Within-subject SD

$$S_{\text{within}} = \frac{SD_{\text{diff}}}{\sqrt{2(1-r)}} \quad (15)$$

(Borenstein 2013, 29)

1.5.2. Standardized mean difference

$$d = \frac{D}{S_{\text{within}}} \quad (16)$$

(Borenstein 2013, 29)

Variance and standard error of d

$$V_d = \left(\frac{1}{n} + \frac{d^2}{2n} \right) 2(1-r) \quad (17)$$

(Borenstein 2013, 29)

$$SE_d = \sqrt{V_d} \quad (18)$$

(Borenstein 2013, 29)

1.5.3. Small-sample correction (Hedges' g)

Degrees of freedom for pre-post measures (df):

$$df = n - 1 \quad (19)$$

(Borenstein 2013, 29)

Correction factor (J)

$$J = 1 - \frac{3}{4df - 1} \quad (20)$$

(Borenstein 2013, 27)

Hedges' g :

$$g = J d \quad (21)$$

(Borenstein 2013, 27)

Variance and SE of g

$$V_g = J^2 V_d \quad (22)$$

(Borenstein 2013, 27)

$$SE_g = \sqrt{V_g} \quad (23)$$

(Borenstein 2013, 27)

1.6. References

2. Workspace Setup

All chapters load a shared setup file (`_common.R`) that installs/loads the required packages, sets a consistent global `ggplot2` theme, defines the leave-one-out outlier-detection functions, and exposes small helpers for the standard funnel, forest, and bubble plots used throughout. This keeps every analysis chapter self-contained and reproducible regardless of render order.

2.1. Packages

2.2. Outlier Detection Functions

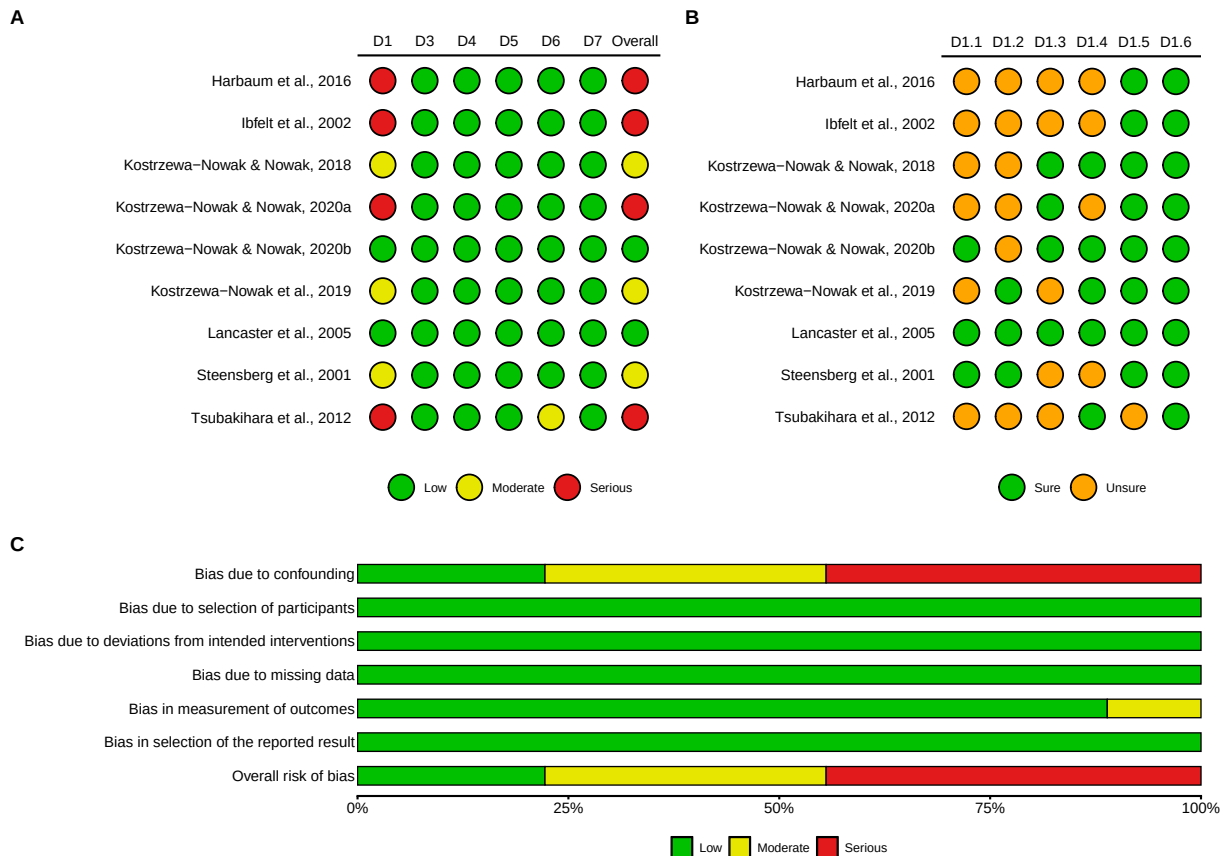
Custom functions — required for all outlier analyses below.

The following four functions implement the leave-one-out standardised residual approach described in Hedges and Olkin (1985) and Viechtbauer and Cheung (2010). A study is classified as an outlier if its standardised residual exceeds $|3|$; values above $|1.96|$ warrant closer inspection. The model (fixed- vs. random-effects) is selected automatically from the robust heterogeneity index I_r^2 ($< 30\% \rightarrow$ fixed-effects, otherwise random-effects). The outlier routine is always passed the **variance** (`seTE2`) as its `s2` argument.

The full definitions live in `_common.R` and are sourced automatically at the top of every chapter.

3. Risk of Bias Assessment (ROBINS-I)

Downloads: ROBINS-I Data (xlsx) · ROBINS-I R-Script



Confounding-factor items (Panel B).

- **D1.1** — No exercise 24–48 h pre-intervention?
- **D1.2** — Standardized food intake among participants (e.g. all fasted or standardized breakfast)?
- **D1.3** — No caffeine, acute medication, or alcohol 24 h before exercise?
- **D1.4** — Identical warm-up between participants?
- **D1.5** — Good way to assess the controlled intensity (HR control, VO₂, VT)?
- **D1.6** — Homogeneity of the group (sex, age, BMI, training status)?

Part I.

Th1

4. Th1 — Immediate Post-Exercise

4.1. Data Import

The base analysis uses the pre-computed Hedges' g file for Th1 at $r = 0.5$ (`effect_size/Th1/pre_post/Th1_pre_post_r_050`). Sensitivity analyses can substitute the $r = 0.3 / 0.4 / 0.6 / 0.7$ files in the same folder. The pipeline column names are mapped to the analysis conventions (VE_g, Sex, Age, BMI) by `load_es()`.

4.2. Primary Meta-Analysis

`metagen()` fits a random-effects model on the pre-computed effect sizes (g , SE_g) using the **Hedges–Olkin (HE)** ² estimator and the **Knapp–Hartung** small-sample correction (`hakn = TRUE`). After inspecting the full set, **Kostrzewa-Nowak et al. (2019)** was detected as very different across meta-analysis and while not influential on the overall effect, its values were unrealistic and brought substantial misleading variance.

4.2.1. Full Modell {meta}

Number of studies: $k = 18$

	SMD	95% CI	t
Random effects model (HK)	0.8982	[0.3316; 1.4649]	3.34
Prediction interval		[-1.5159; 3.3124]	
	p-value		
Random effects model (HK)	0.0038		
Prediction interval			

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 1.2328$ [0.5533; 2.8692]; $\tau = 1.1103$ [0.7438; 1.6939]
 $I^2 = 86.8\%$ [80.7%; 91.0%]; $H = 2.76$ [2.27; 3.34]

Test of heterogeneity:

Q	d.f.	p-value
129.23	17	< 0.0001

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model ($df = 17$)
- Prediction interval based on t-distribution ($df = 17$)

4.2.2. Adapted Model {meta}

Caution

Kostrzewa-Nowak et al. (2019) removed (unrealistic values / misleading variance); re-fit

Number of studies: $k = 17$

	SMD	95% CI	t
Random effects model (HK)	0.9980	[0.4524; 1.5436]	3.88
Prediction interval		[-1.2488; 3.2448]	
	p-value		
Random effects model (HK)	0.0013		
Prediction interval			

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 1.0535$ [0.4647; 2.5537]; $\tau = 1.0264$ [0.6817; 1.5980]

$I^2 = 86.4\%$ [79.8%; 90.9%]; $H = 2.72$ [2.22; 3.32]

Test of heterogeneity:

Q	d.f.	p-value
117.96	16	< 0.0001

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 16)
- Prediction interval based on t-distribution (df = 16)

4.2.3. Adapted Model {metafor}

Random-Effects Model ($k = 17$; τ^2 estimator: HE)

τ^2 (estimated amount of total heterogeneity): 1.0535 (SE = 0.4224)

τ (square root of estimated τ^2 value): 1.0264

I^2 (total heterogeneity / total variability): 90.76%

H^2 (total variability / sampling variability): 10.82

Test for Heterogeneity:

$Q(df = 16) = 117.9586$, $p\text{-val} < .0001$

Model Results:

estimate	se	tval	df	pval	ci.lb	ci.ub	
0.9980	0.2574	3.8774	16	0.0013	0.4524	1.5436	**

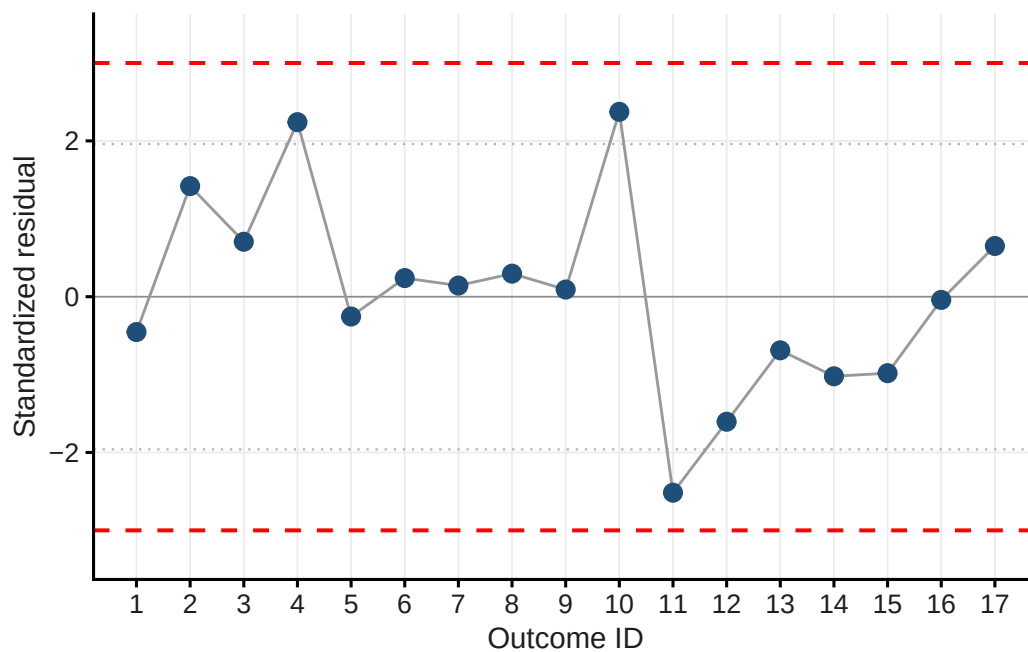
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

4.3. Outlier Detection

Leave-one-out standardised residuals are computed via `metaoutliers()`, which is passed the **variance** (`m1$seTE^2`). Studies with $|z| > 3$ are flagged as outliers; $|z| > 1.96$ warrants inspection.

Table 4.1.: Standardised residuals

Short_reference	g	std_res
Ibfelt et al. (2002)	0.577	-0.454
Kostrzewa-Nowak & Nowak (2018)a	2.289	1.420
Kostrzewa-Nowak & Nowak (2018)b	1.613	0.705
Kostrzewa-Nowak & Nowak (2020a)a	2.781	2.241
Kostrzewa-Nowak & Nowak (2020a)b	0.761	-0.255
Kostrzewa-Nowak & Nowak (2020b)a	1.195	0.239
Kostrzewa-Nowak & Nowak (2020b)b	1.111	0.143
Kostrzewa-Nowak & Nowak (2020b)c	1.245	0.296
Kostrzewa-Nowak & Nowak (2020b)d	1.066	0.092
Kostrzewa-Nowak & Nowak (2020b)e	3.282	2.374
Steensberg et al. (2001)	-1.092	-2.516
Tsubakihara et al. (2012)	-0.276	-1.605
Lancaster et al. (2005)a	0.373	-0.688
Lancaster et al. (2005)b	0.091	-1.022
Lancaster et al. (2005)c	0.124	-0.982
Harbaum et al. (2016)a	0.949	-0.041
Harbaum et al. (2016)b	1.584	0.651



! Important

No outcome has standardized residuals above 3 or below -3.

4.4. Funnel Plot & Influential Cases

The contour-enhanced funnel plot shows study estimates against their standard errors; shaded regions mark significance zones ($\alpha = 0.10, 0.05, 0.01$). The `dfbetas` diagnostic from `metafor::rma.mv()` identifies **influential cases** — distinct from statistical outliers and not necessarily excluded.

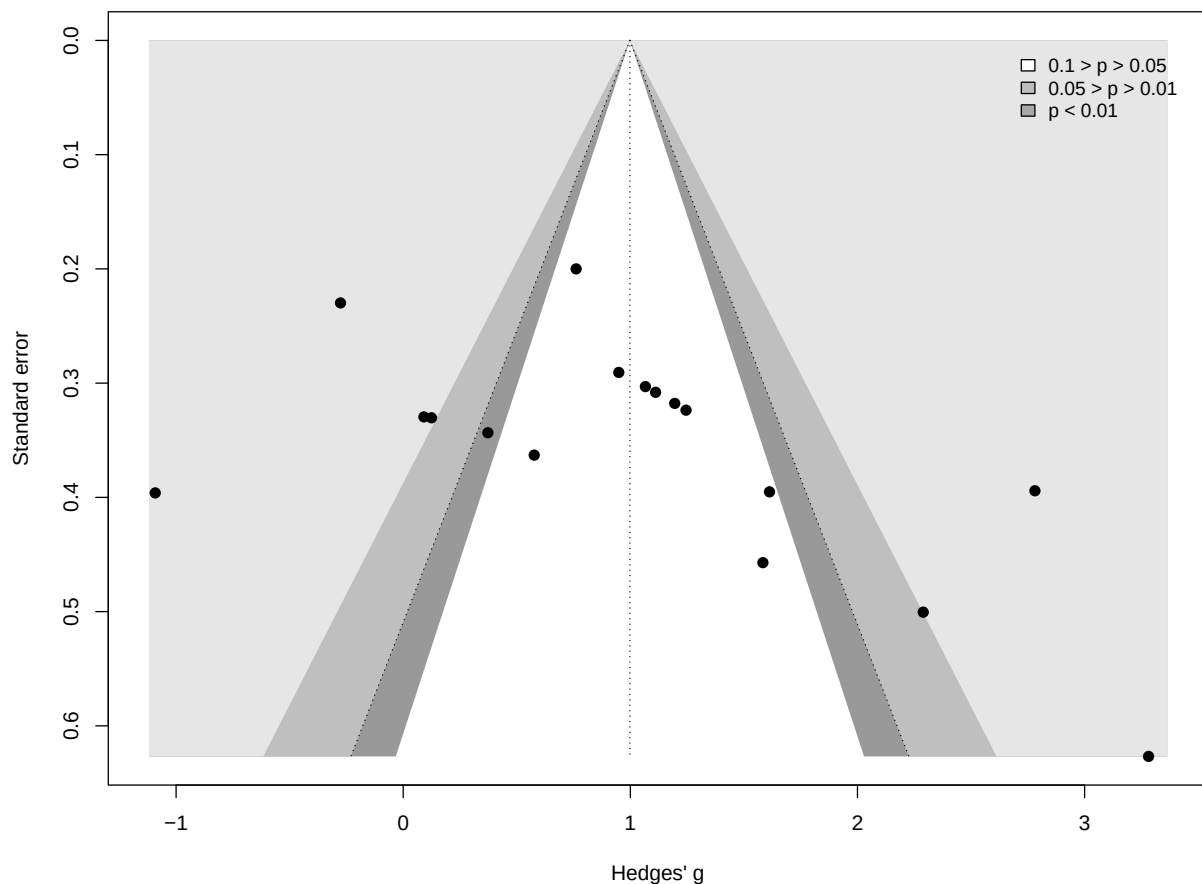


Table 4.2.: DFBETAS influence diagnostics

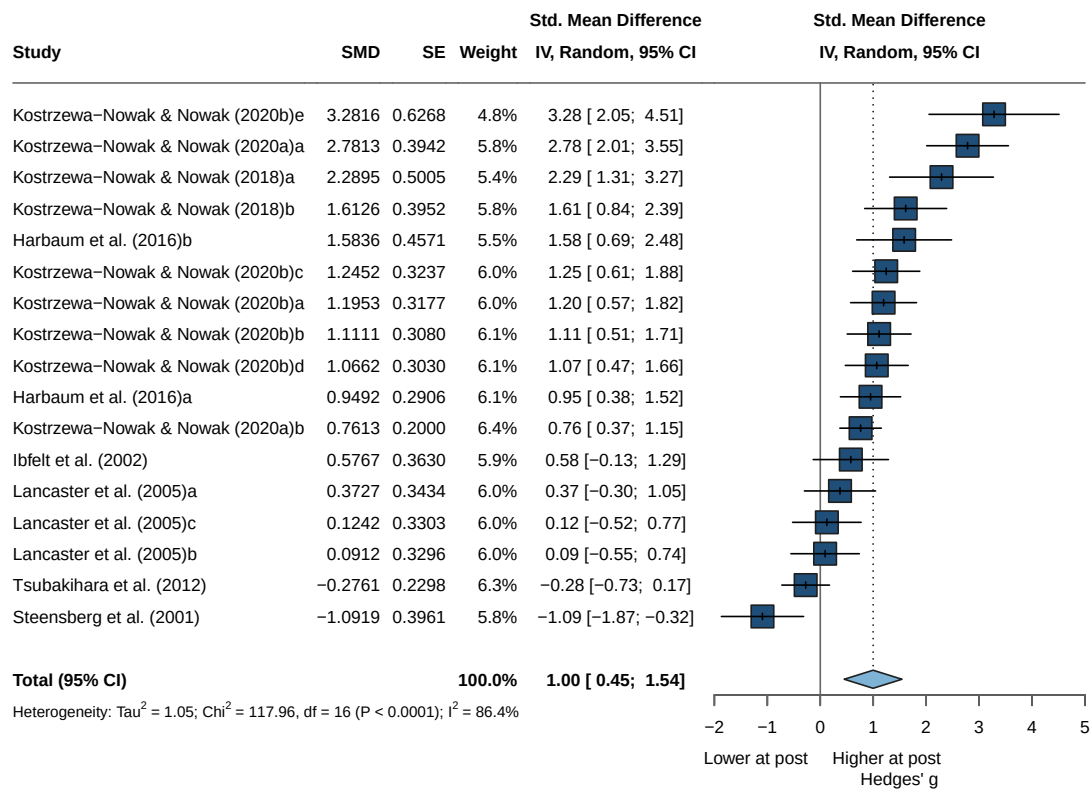
Short_reference	g	dfbetas
Ibfelt et al. (2002)	0.577	-0.13459692
Kostrzewska-Nowak & Nowak (2018)a	2.289	0.48230850
Kostrzewska-Nowak & Nowak (2018)b	1.613	0.43065936
Kostrzewska-Nowak & Nowak (2020a)a	2.781	1.04875284
Kostrzewska-Nowak & Nowak (2020a)b	0.761	-0.07030521
Kostrzewska-Nowak & Nowak (2020b)a	1.195	0.33513405
Kostrzewska-Nowak & Nowak (2020b)b	1.111	0.28334594
Kostrzewska-Nowak & Nowak (2020b)c	1.245	0.36199846
Kostrzewska-Nowak & Nowak (2020b)d	1.066	0.25226569

Table 4.2.: DFBETAS influence diagnostics

Short_reference	g	dfbetas
Kostrzewa-Nowak & Nowak (2020b)e	3.282	0.50656391
Steensberg et al. (2001)	-1.092	-0.98309619
Tsubakihara et al. (2012)	-0.276	-1.80121819
Lancaster et al. (2005)a	0.373	-0.29482591
Lancaster et al. (2005)b	0.091	-0.53785734
Lancaster et al. (2005)c	0.124	-0.50993030
Harbaum et al. (2016)a	0.949	0.15837850
Harbaum et al. (2016)b	1.584	0.30720822

One value was below -1 and one above 1 , so both were identified as influential cases. However, neither is an outlier (Kostrzewa-Nowak & Nowak (2020a)a and Tsubakihara et al. (2012)). Accordingly, all outcomes are retained, and the sources of heterogeneity are examined in the moderator analysis.

4.5. Forest Plot



4.6. Asymmetry Test

Egger's regression test (`method = "linreg"`) evaluates funnel asymmetry as a proxy for small-study/publication effects (one-tailed, $p < .10$). Where asymmetry is suggested, a trim-and-fill analysis estimates potentially missing studies.

4.6.1. Egger's regression test

Linear regression test of funnel plot asymmetry

Test result: $t = 2.11$, $df = 15$, $p\text{-value} = 0.0523$

Bias estimate: 4.9260 (SE = 2.3369)

Details:

- multiplicative residual heterogeneity variance ($\tau^2 = 6.0667$)
- predictor: standard error
- weight: inverse variance
- reference: Egger et al. (1997), BMJ

4.6.2. Trim and Fill

Estimated number of missing studies on the left side: 0 (SE = 2.5907)

Random-Effects Model ($k = 17$; τ^2 estimator: HE)

τ^2 (estimated amount of total heterogeneity): 1.0535 (SE = 0.4224)

τ (square root of estimated τ^2 value): 1.0264

I^2 (total heterogeneity / total variability): 90.76%

H^2 (total variability / sampling variability): 10.82

Test for Heterogeneity:

$Q(df = 16) = 117.9586$, $p\text{-val} < .0001$

Model Results:

estimate	se	zval	pval	ci.lb	ci.ub	
0.9980	0.2643	3.7765	0.0002	0.4800	1.5159	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i No missing studies

4.7. Moderator Analyses

4.7.1. Risk of Bias

Subgroup analysis by overall RoB rating tests whether study quality moderates the pooled effect; the between-subgroup statistic (Q_b) indicates whether subgroup means differ.

4.7.1.1. Model

Number of studies: $k = 17$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 1.0535$ [0.4647; 2.5537]; $\tau = 1.0264$ [0.6817; 1.5980]

$I^2 = 86.4\%$ [79.8%; 90.9%]; $H = 2.72$ [2.22; 3.32]

Test of heterogeneity:

Q	d.f.	p-value
117.96	16	< 0.0001

Results for subgroups (random effects model (HK)):

	k	SMD	95% CI	τ^2	τ
RoB = Serious	6	1.0287	[-0.0523; 2.1097]	0.9592	0.9794
RoB = Moderate	3	0.9237	[-3.5234; 5.3709]	3.0131	1.7358
RoB = Low	8	0.9996	[0.2061; 1.7932]	0.8995	0.9484

	Q	I^2
RoB = Serious	50.50	90.1%
RoB = Moderate	35.85	94.4%
RoB = Low	31.00	77.4%

Test for subgroup differences (random effects model (HK)):

	Q	d.f.	p-value
Between groups	0.01	2	0.9952

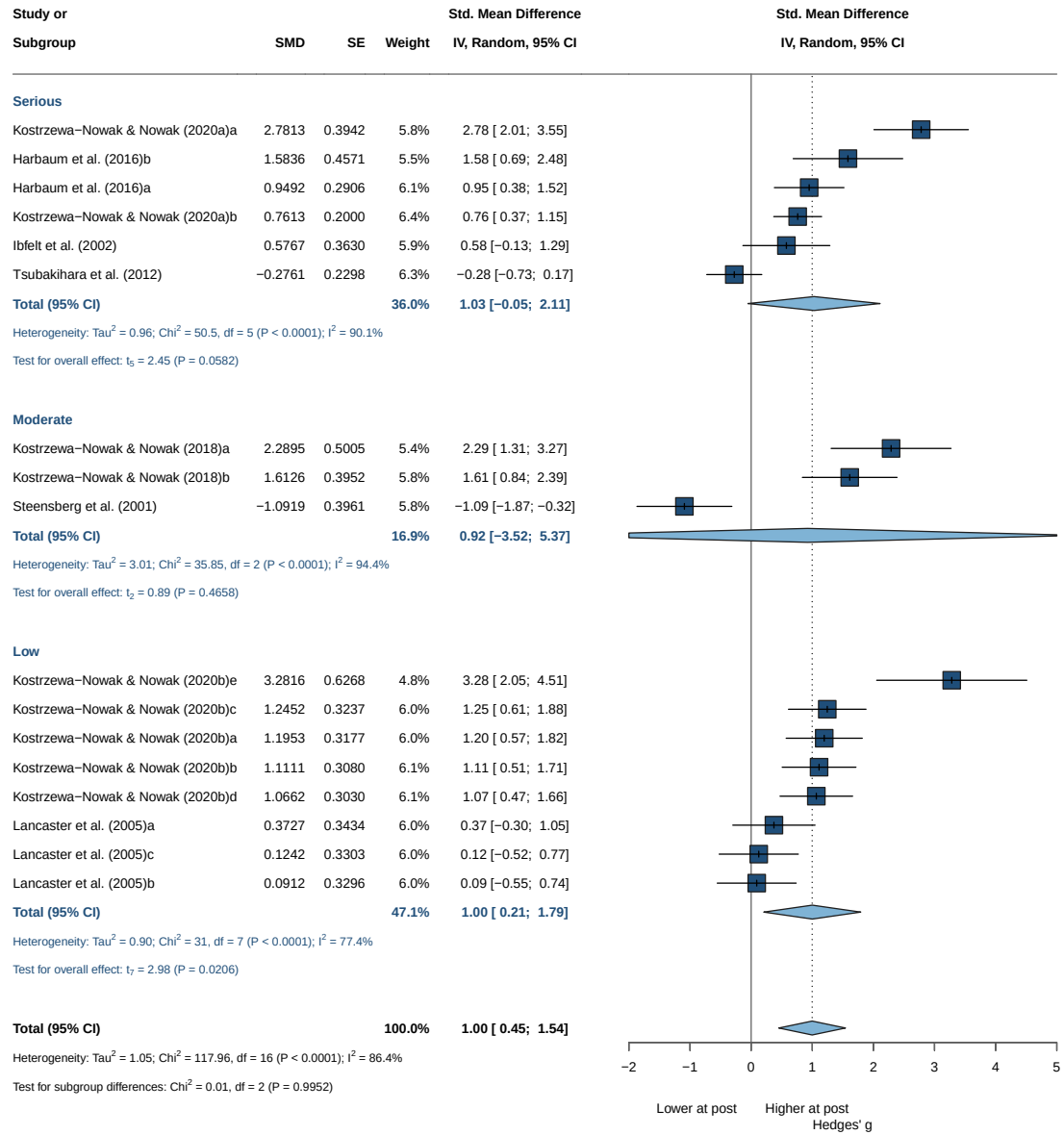
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

Note

Not significant results

4.7.1.2. Forrestplot



4.7.2. Staining

Number of studies: $k = 17$

Quantifying heterogeneity (with 95% CIs):

```
tau^2 = 1.0535 [0.4647; 2.5537]; tau = 1.0264 [0.6817; 1.5980]
I^2 = 86.4% [79.8%; 90.9%]; H = 2.72 [2.22; 3.32]
```

Test of heterogeneity:

```
Q d.f. p-value
117.96 16 < 0.0001
```

Results for subgroups (random effects model (HK)):

	k	SMD	95% CI	tau^2	tau
staining = intra	14	1.0632	[0.4218; 1.7046]	1.1630	1.0784
staining = extra	3	0.7028	[-1.6445; 3.0501]	0.7783	0.8822

	Q	I^2
staining = intra	91.40	85.8%
staining = extra	18.92	89.4%

Test for subgroup differences (random effects model (HK)):

```
Q d.f. p-value
Between groups 0.34 1 0.5617
```

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

i Note

Not significant results

4.7.3. Sex

4.7.3.1. Model

Number of studies: k = 17

Quantifying heterogeneity (with 95% CIs):

```
tau^2 = 1.0535 [0.4647; 2.5537]; tau = 1.0264 [0.6817; 1.5980]
I^2 = 86.4% [79.8%; 90.9%]; H = 2.72 [2.22; 3.32]
```

Test of heterogeneity:

```
Q d.f. p-value
117.96 16 < 0.0001
```

Results for subgroups (random effects model (HK)):

	k	SMD	95% CI	tau^2	tau
sex = male	14	1.0632	[0.4218; 1.7046]	1.1630	1.0784

4. Th1 — Immediate Post-Exercise

```
sex = female    1 -0.2761 [-0.7265; 0.1743]    --    --
sex = mixed     2  1.1682 [-2.6647; 5.0012]  0.0545 0.2335
               Q    I2
sex = male     91.40 85.8%
sex = female    0.00    --
sex = mixed     1.37 27.1%
```

Test for subgroup differences (random effects model (HK)):

```
           Q d.f.  p-value
Between groups 19.89    2 < 0.0001
```

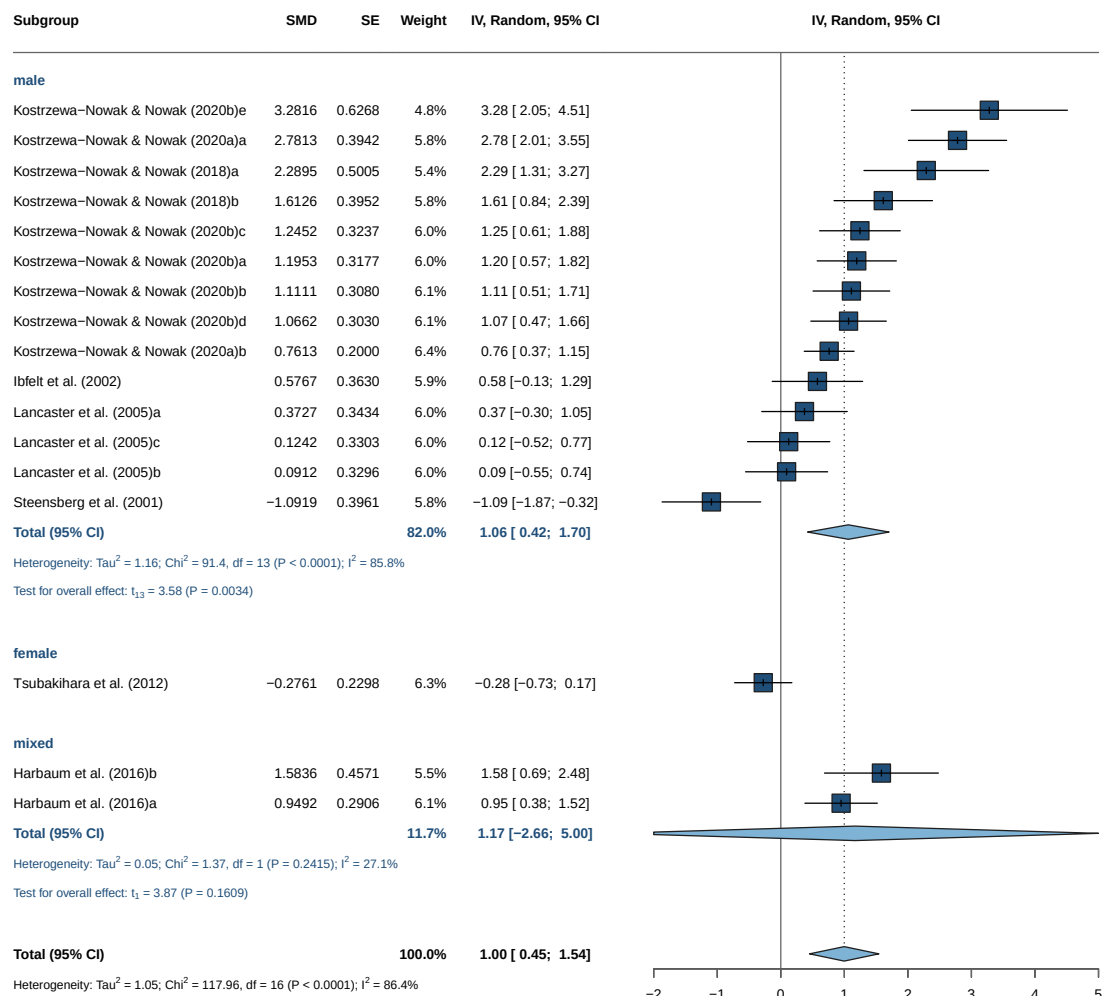
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

! Important

Significant results but 14 vs 1 vs 2

4.7.3.2. Forrest Plot



4.7.4. Intensity

Number of studies: $k = 16$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 1.0050$ [0.4207; 2.5358]; $\tau = 1.0025$ [0.6486; 1.5924]

$I^2 = 84.0\%$ [75.3%; 89.6%]; $H = 2.50$ [2.01; 3.10]

Test of heterogeneity:

Q	d.f.	p-value
93.51	15	< 0.0001

Results for subgroups (random effects model (HK)):

	k	SMD	95% CI	τ^2
intensity = Vigorous	5	0.0310	[-0.7527; 0.8147]	0.2967
intensity = Maximal	11	1.5400	[1.0222; 2.0579]	0.5104

	τ	Q	I^2
intensity = Vigorous	0.5447	11.36	64.8%
intensity = Maximal	0.7145	38.82	74.2%

4. Th1 — Immediate Post-Exercise

Test for subgroup differences (random effects model (HK)):

	Q	d.f.	p-value
Between groups	17.03	1	< 0.0001

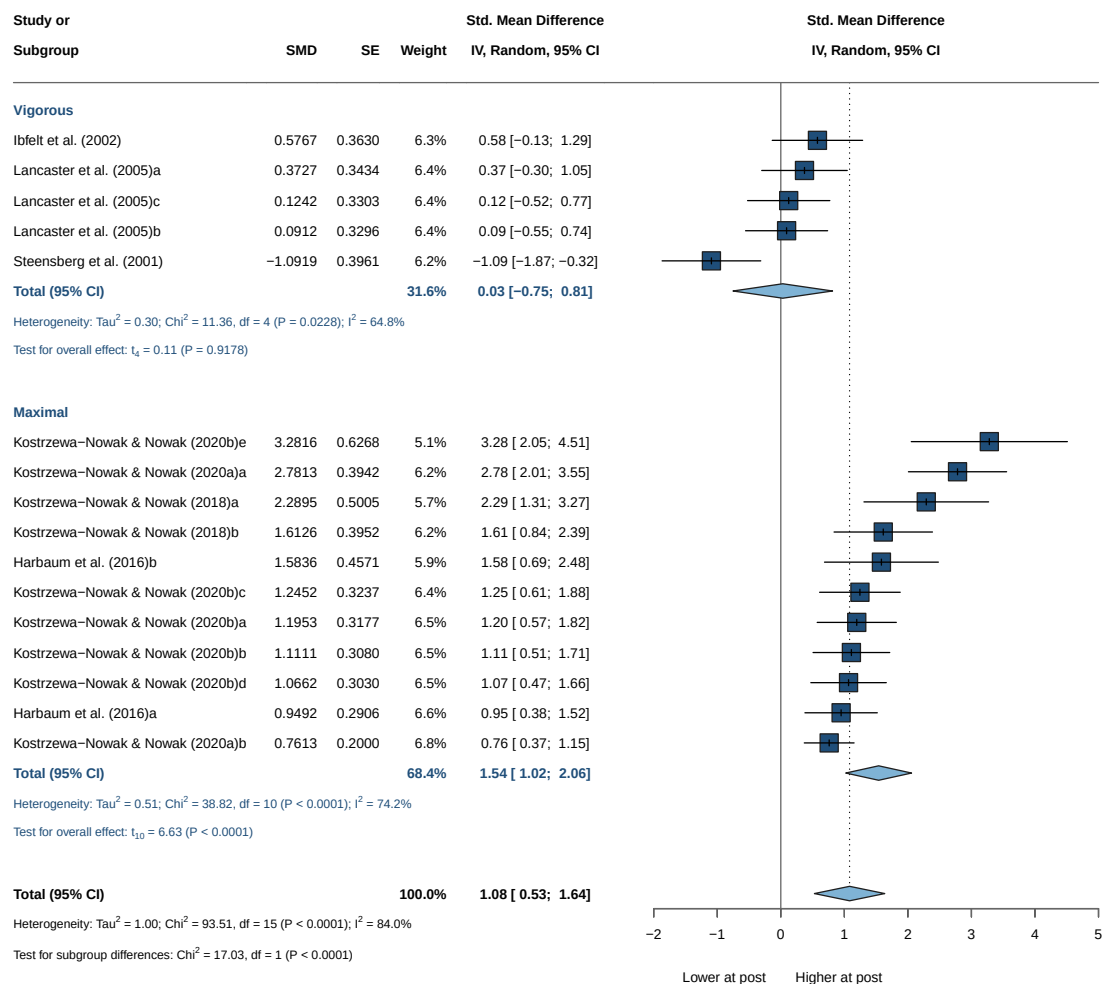
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

! Important

Significant results

4.7.4.1. Forrest Plot



4.7.5. Continuous Moderators (Meta-Regression)

Continuous moderators are tested via weighted meta-regression (`metareg()`). A significant slope indicates the moderator explains part of the between-study heterogeneity (τ^2); bubble plots show the fitted line with

95% CI.

4.7.5.1. Age

Mixed-Effects Model (k = 16; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-22.6865	49.0041	51.3730	53.6908	53.3730

tau ² (estimated amount of residual heterogeneity):	1.0410 (SE = 0.4487)
tau (square root of estimated tau ² value):	1.0203
I ² (residual heterogeneity / unaccounted variability):	89.97%
H ² (unaccounted variability / sampling variability):	9.97
R ² (amount of heterogeneity accounted for):	0.00%

Test for Residual Heterogeneity:

QE(df = 14) = 91.8671, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 14) = 0.4880, p-val = 0.4963

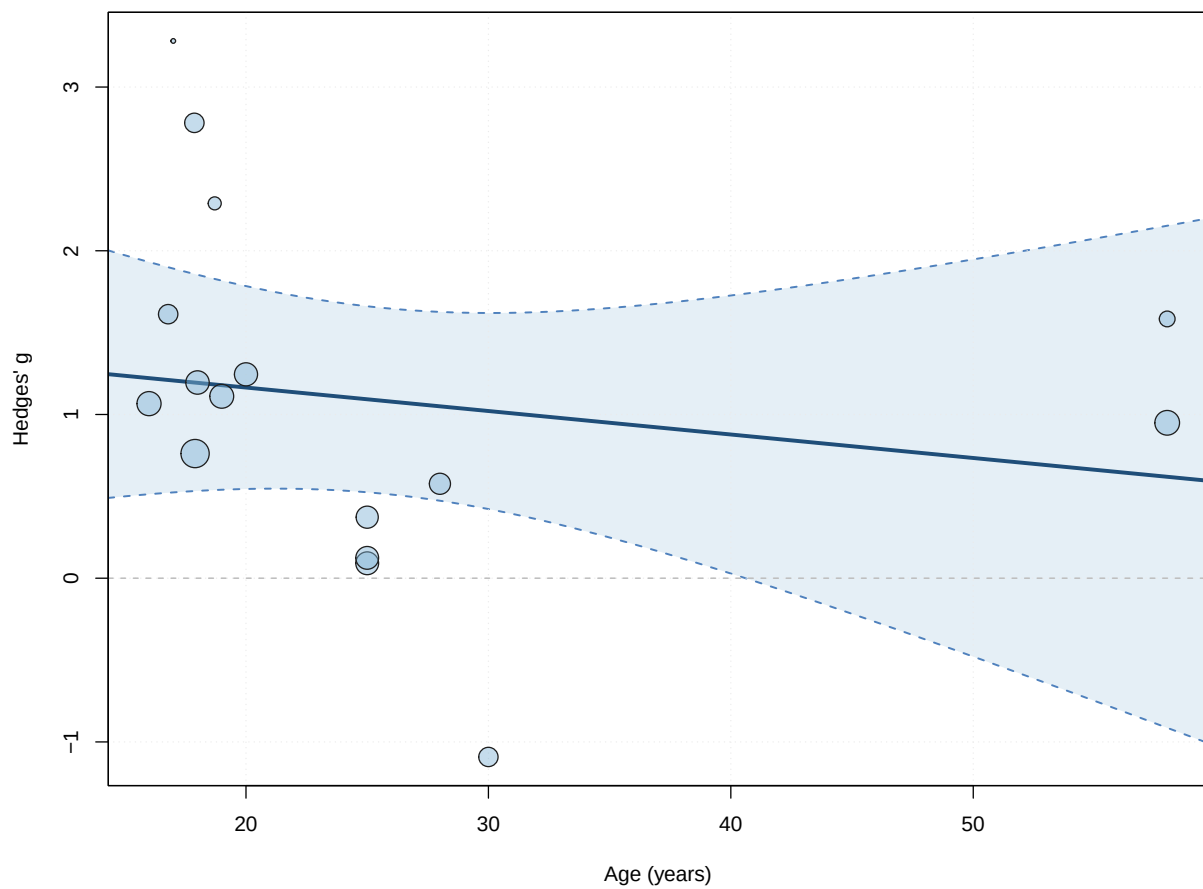
Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	1.4515	0.5896	2.4621	14	0.0274	0.1871	
age	-0.0143	0.0205	-0.6986	14	0.4963	-0.0584	
intrcpt	2.7160						*
age	0.0297						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant



4.7.5.2. BMI

Mixed-Effects Model (k = 12; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-15.7587	34.9893	37.5174	38.9721	40.5174

tau² (estimated amount of residual heterogeneity): 0.8458 (SE = 0.4461)
 tau (square root of estimated tau² value): 0.9197
 I² (residual heterogeneity / unaccounted variability): 89.09%
 H² (unaccounted variability / sampling variability): 9.17
 R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 10) = 77.5843, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 10) = 0.0134, p-val = 0.9102

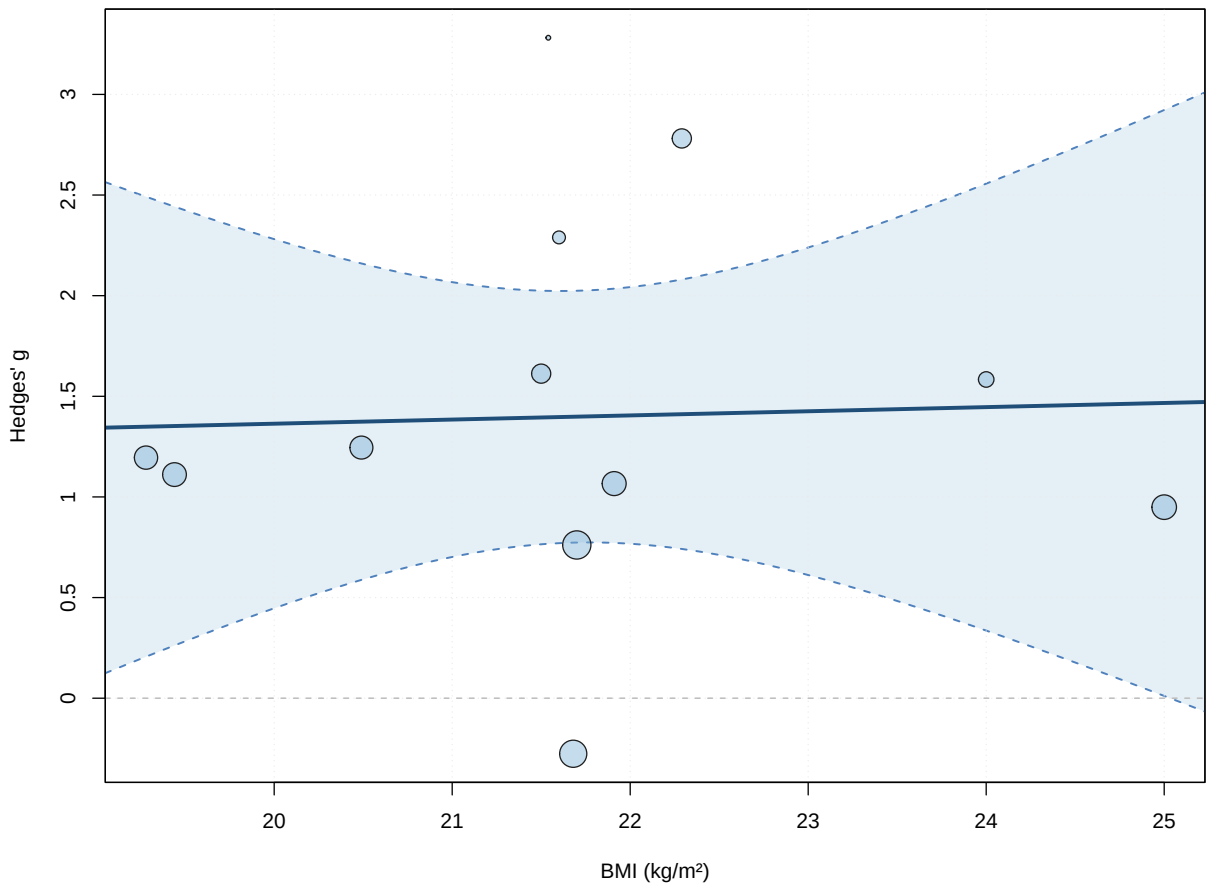
Model Results:

	estimate	se	tval	df	pval	ci.lb
intrcpt	0.9518	3.8746	0.2457	10	0.8109	-7.6813
bmi	0.0206	0.1782	0.1156	10	0.9102	-0.3764
	ci.ub					
intrcpt	9.5849					
bmi	0.4176					

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant



4.7.5.3. Duration

Mixed-Effects Model (k = 15; tau^2 estimator: HE)

logLik	deviance	AIC	BIC	AICc
-13.7530	30.8355	33.5060	35.6301	35.6878

4. Th1 — Immediate Post-Exercise

```
tau^2 (estimated amount of residual heterogeneity):    0.3425 (SE = 0.1937)
tau (square root of estimated tau^2 value):           0.5853
I^2 (residual heterogeneity / unaccounted variability): 74.35%
H^2 (unaccounted variability / sampling variability):   3.90
R^2 (amount of heterogeneity accounted for):           65.26%
```

Test for Residual Heterogeneity:

QE(df = 13) = 31.7293, p-val = 0.0026

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 13) = 20.8420, p-val = 0.0005

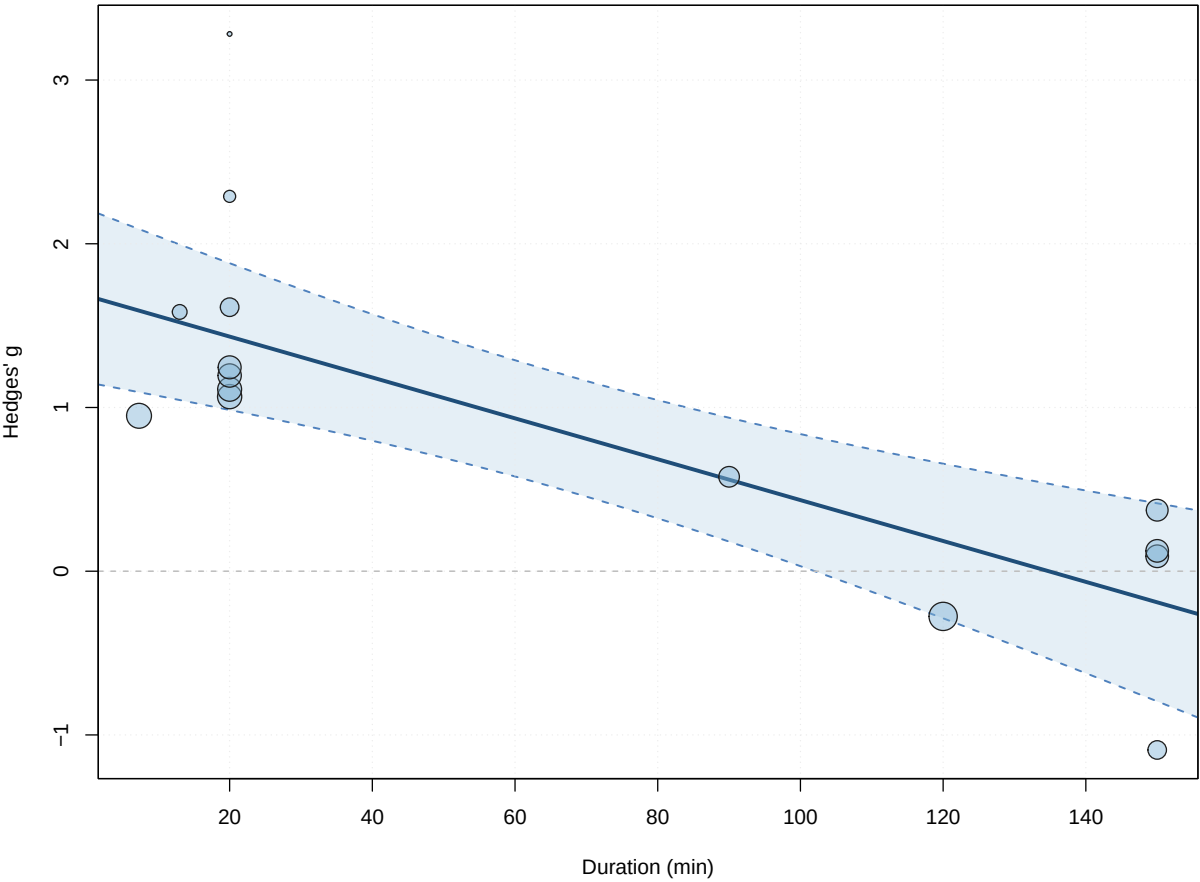
Model Results:

	estimate	se	tval	df	pval	ci.lb
intrcpt	1.6827	0.2451	6.8662	13	<.0001	1.1532
duration	-0.0125	0.0027	-4.5653	13	0.0005	-0.0184
	ci.ub					
intrcpt	2.2121	***				
duration	-0.0066	***				

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

! Important

Significant



4.7.5.4. Dose

Mixed-Effects Model (k = 14; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-12.5188	27.2639	31.0376	32.9547	33.4376

tau² (estimated amount of residual heterogeneity): 0.3246 (SE = 0.1970)
tau (square root of estimated tau² value): 0.5697
I² (residual heterogeneity / unaccounted variability): 71.72%
H² (unaccounted variability / sampling variability): 3.54
R² (amount of heterogeneity accounted for): 65.62%

Test for Residual Heterogeneity:
QE(df = 12) = 25.4779, p-val = 0.0127

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 12) = 19.2815, p-val = 0.0009

Model Results:

4. Th1 — Immediate Post-Exercise

```

      estimate      se      tval  df      pval      ci.lb
intrcpt    1.7943  0.2547   7.0457  12  <.0001    1.2394
dose      -0.0020  0.0005  -4.3911  12  0.0009   -0.0030

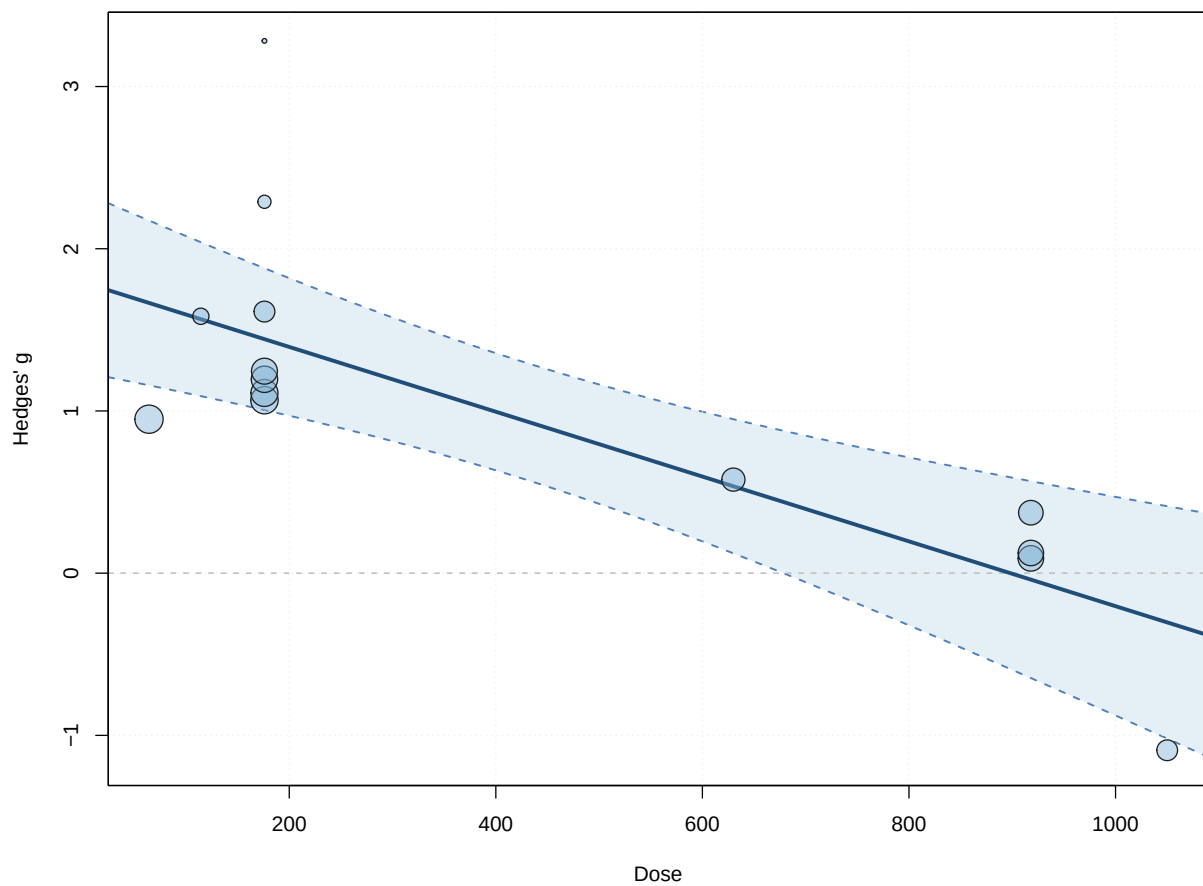
      ci.ub
intrcpt    2.3491  ***
dose      -0.0010  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

! Important

Significant — collinear with duration



4.7.5.5. IFN

Mixed-Effects Model (k = 7; tau² estimator: HE)

```

logLik  deviance      AIC      BIC     AICc
-9.1465  20.7348  24.2929  24.1307  32.2929

```

```

tau^2 (estimated amount of residual heterogeneity):    0.9848 (SE = 0.7226)
tau (square root of estimated tau^2 value):           0.9924
I^2 (residual heterogeneity / unaccounted variability): 88.95%
H^2 (unaccounted variability / sampling variability):  9.05
R^2 (amount of heterogeneity accounted for):           0.00%

```

Test for Residual Heterogeneity:

QE(df = 5) = 27.1266, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 0.0809, p-val = 0.7875

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	2.0560	1.7626	1.1665	5	0.2960	-2.4748	
IFN	-0.2718	0.9556	-0.2845	5	0.7875	-2.7282	
intrcpt	6.5869						
IFN	2.1846						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

4.7.5.6. IL-4

Mixed-Effects Model (k = 9; tau^2 estimator: HE)

logLik	deviance	AIC	BIC	AICc
-10.3229	23.4488	26.6457	27.2374	31.4457

```

tau^2 (estimated amount of residual heterogeneity):    0.6578 (SE = 0.4320)
tau (square root of estimated tau^2 value):           0.8111
I^2 (residual heterogeneity / unaccounted variability): 86.45%
H^2 (unaccounted variability / sampling variability):  7.38
R^2 (amount of heterogeneity accounted for):           0.00%

```

Test for Residual Heterogeneity:

QE(df = 7) = 30.1320, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 7) = 0.8032, p-val = 0.3999

Model Results:

4. Th1 — Immediate Post-Exercise

```

      estimate      se    tval  df    pval    ci.lb
intrcpt    1.3165  0.3308  3.9793   7  0.0053   0.5342
IL_4       0.2506  0.2796  0.8962   7  0.3999  -0.4105
      ci.ub
intrcpt    2.0988  **
IL_4       0.9117

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Note

Not significant

4.7.5.7. IL-6

Mixed-Effects Model (k = 14; tau² estimator: HE)

```

      logLik  deviance      AIC      BIC      AICc
-18.0578   39.5434   42.1155   44.0327   44.5155

```

```

tau^2 (estimated amount of residual heterogeneity):    0.8055 (SE = 0.3889)
tau (square root of estimated tau^2 value):           0.8975
I^2 (residual heterogeneity / unaccounted variability): 88.00%
H^2 (unaccounted variability / sampling variability):  8.34
R^2 (amount of heterogeneity accounted for):           0.00%

```

Test for Residual Heterogeneity:

QE(df = 12) = 63.3467, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 12) = 0.1327, p-val = 0.7220

Model Results:

```

      estimate      se    tval  df    pval    ci.lb
intrcpt    1.2063  0.2945  4.0964  12  0.0015   0.5647
IL_6       0.0604  0.1659  0.3642  12  0.7220  -0.3010
      ci.ub
intrcpt    1.8479  **
IL_6       0.4219

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```


i Note

Not significant

4.7.5.8. IL_10Mixed-Effects Model (k = 11; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-12.3204	27.0095	30.6409	31.8345	34.0694

tau ² (estimated amount of residual heterogeneity):	0.5518 (SE = 0.3375)
tau (square root of estimated tau ² value):	0.7428
I ² (residual heterogeneity / unaccounted variability):	81.00%
H ² (unaccounted variability / sampling variability):	5.26
R ² (amount of heterogeneity accounted for):	0.00%

Test for Residual Heterogeneity:

QE(df = 9) = 35.0975, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 9) = 0.2716, p-val = 0.6148

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	1.5574	0.2433	6.3998	9	0.0001	1.0069	
IL_10	-0.3495	0.6706	-0.5211	9	0.6148	-1.8664	
intrcpt	2.1078	***					
IL_10	1.1675						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

4.7.5.9. IL_2Mixed-Effects Model (k = 7; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-8.6034	19.6487	23.2069	23.0446	31.2069

tau ² (estimated amount of residual heterogeneity):	0.8092 (SE = 0.5809)
--	----------------------

4. Th1 — Immediate Post-Exercise

```
tau (square root of estimated tau^2 value):      0.8996
I^2 (residual heterogeneity / unaccounted variability): 89.58%
H^2 (unaccounted variability / sampling variability): 9.59
R^2 (amount of heterogeneity accounted for):      0.73%
```

Test for Residual Heterogeneity:

```
QE(df = 5) = 32.1011, p-val < .0001
```

Test of Moderators (coefficient 2):

```
F(df1 = 1, df2 = 5) = 0.8054, p-val = 0.4106
```

Model Results:

	estimate	se	tval	df	pval	ci.lb
intrcpt	0.2553	1.4923	0.1711	5	0.8709	-3.5808
IL_2	0.7369	0.8211	0.8974	5	0.4106	-1.3739
	ci.ub					
intrcpt	4.0915					
IL_2	2.8477					

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

4.8. Multivariate Models

Multivariate meta-regression tests interactions between candidate moderators. Neither model below yields independent predictors.

4.8.1. Intensity * Duration

Mixed-Effects Model (k = 14; tau^2 estimator: HE)

```
tau^2 (estimated amount of residual heterogeneity): 0.3855 (SE = 0.2452)
tau (square root of estimated tau^2 value):         0.6209
I^2 (residual heterogeneity / unaccounted variability): 74.36%
H^2 (unaccounted variability / sampling variability): 3.90
R^2 (amount of heterogeneity accounted for):         59.17%
```

Test for Residual Heterogeneity:

```
QE(df = 10) = 24.8856, p-val = 0.0056
```

Test of Moderators (coefficients 2:4):

```
F(df1 = 3, df2 = 10) = 5.5862, p-val = 0.0164
```

Model Results:

	estimate	se	tval	df
intrcpt	0.7778	0.9262	0.8398	10
intensityVigorous	0.8233	1.9758	0.4167	10
duration	0.0403	0.0508	0.7938	10
intensityVigorous:duration	-0.0517	0.0523	-0.9887	10
	pval	ci.lb	ci.ub	
intrcpt	0.4206	-1.2859	2.8415	
intensityVigorous	0.6857	-3.5792	5.2257	
duration	0.4457	-0.0728	0.1534	
intensityVigorous:duration	0.3461	-0.1681	0.0648	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

4.8.2. IFN * IL_2

Mixed-Effects Model (k = 7; tau² estimator: HE)

tau ² (estimated amount of residual heterogeneity):	0.8792 (SE = 0.8102)
tau (square root of estimated tau ² value):	0.9376
I ² (residual heterogeneity / unaccounted variability):	88.83%
H ² (unaccounted variability / sampling variability):	8.95
R ² (amount of heterogeneity accounted for):	0.00%

Test for Residual Heterogeneity:

QE(df = 3) = 21.9341, p-val < .0001

Test of Moderators (coefficients 2:4):

F(df1 = 3, df2 = 3) = 0.7599, p-val = 0.5866

Model Results:

	estimate	se	tval	df	pval	ci.lb
intrcpt	17.1616	16.2628	1.0553	3	0.3688	-34.5937
IFN	-9.6844	9.0286	-1.0726	3	0.3620	-38.4174
IL_2	-8.3898	9.5134	-0.8819	3	0.4428	-38.6658
IFN:IL_2	5.1176	5.1389	0.9959	3	0.3927	-11.2366
	ci.ub					
intrcpt	68.9169					
IFN	19.0485					
IL_2	21.8862					

4. Th1 — Immediate Post-Exercise

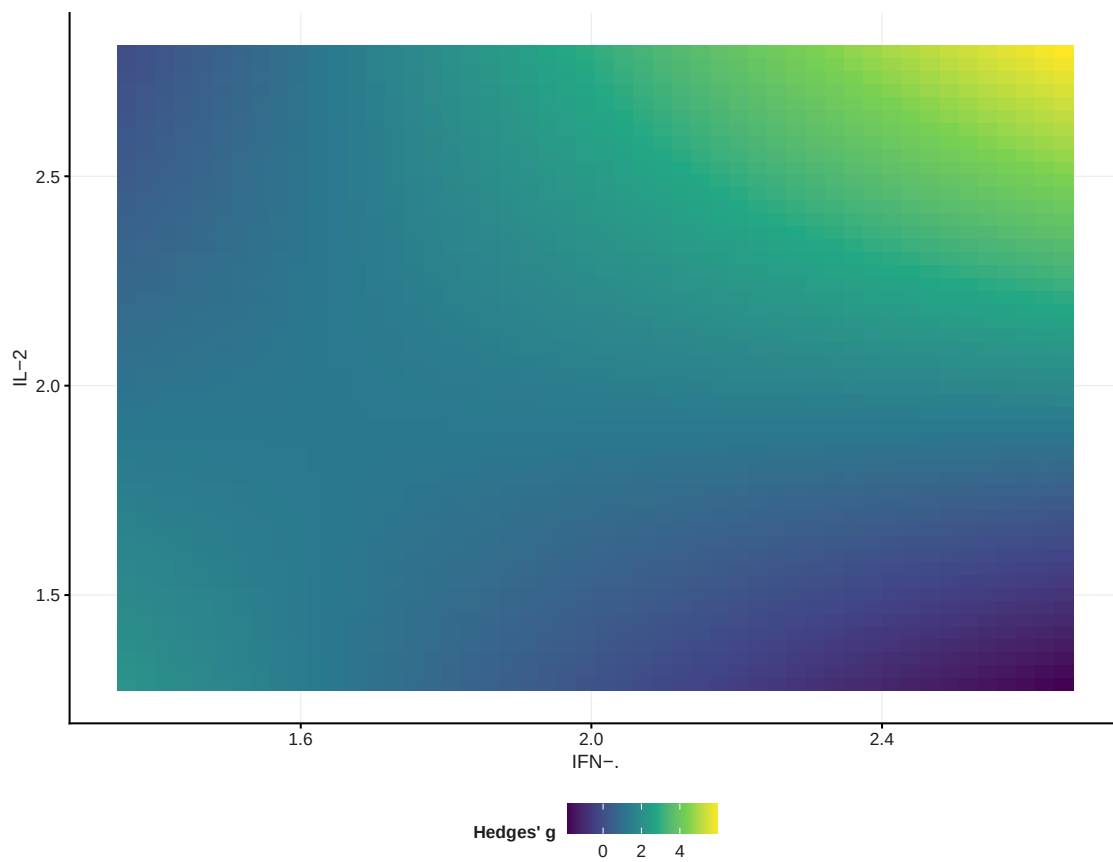
IFN:IL_2 21.4719

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

4.9. Interaction Surface (IFN- $\times$ IL-2)



5. Th1 — ~2 h Post-Exercise

5.1. Data Import

The ~2 h window (`effect_size/Th1/pre_post+2h/Th1_pre_post+2h_r_050.csv`) contains few arms and no cytokine/BMI coverage, so it is reported for the main analysis, outlier check, funnel, forest, and asymmetry only — no moderator modelling is possible. No study is excluded a priori in this window.

5.2. Primary Meta-Analysis

5.2.1. {meta}

Number of studies: $k = 5$

	SMD	95% CI	t
Random effects model (HK)	-0.4137	[-1.0063; 0.1789]	-1.94
Prediction interval		[-1.5190; 0.6917]	
	p-value		
Random effects model (HK)	0.1247		
Prediction interval			

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.1110$ [0.0000; 1.8375]; $\tau = 0.3332$ [0.0000; 1.3555]
 $I^2 = 42.5\%$ [0.0%; 78.9%]; $H = 1.32$ [1.00; 2.17]

Test of heterogeneity:

Q	d.f.	p-value
6.96	4	0.1382

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model ($df = 4$)
- Prediction interval based on t-distribution ($df = 4$)

5.2.2. {metafor}

Random-Effects Model ($k = 5$; τ^2 estimator: HE)

```

tau^2 (estimated amount of total heterogeneity): 0.1110 (SE = 0.1699)
tau (square root of estimated tau^2 value):      0.3332
I^2 (total heterogeneity / total variability):    46.86%
H^2 (total variability / sampling variability):    1.88

```

Test for Heterogeneity:

Q(df = 4) = 6.9560, p-val = 0.1382

Model Results:

```

estimate      se      tval  df    pval    ci.lb    ci.ub
-0.4137  0.2134  -1.9381   4   0.1247  -1.0063   0.1789

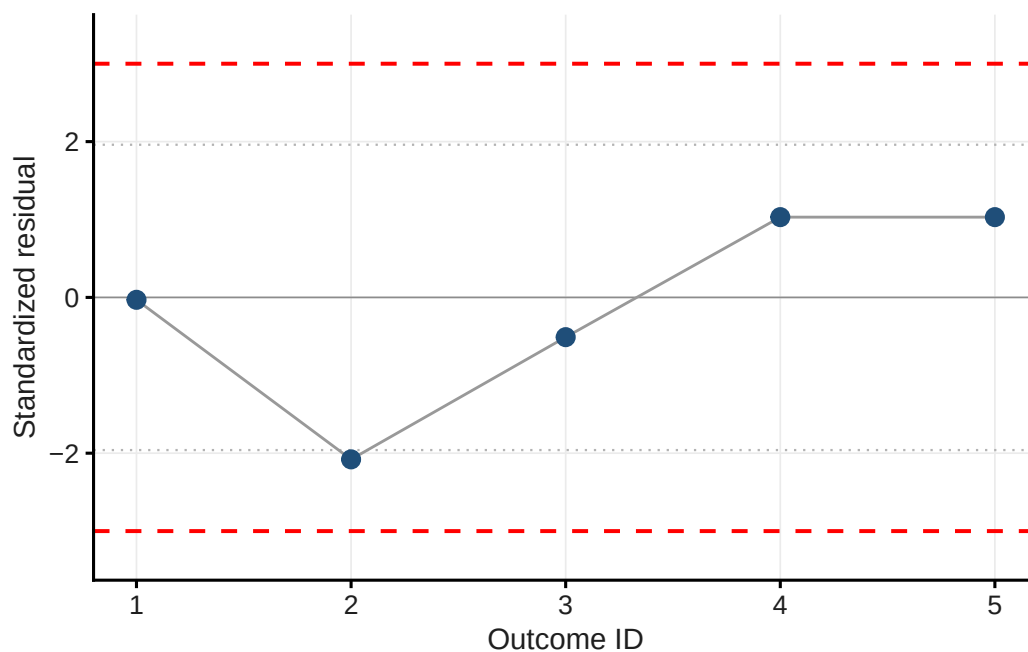
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

5.3. Outlier Detection

Table 5.1.: Standardised residuals

Short_reference	g	std_res
Ibfelt et al. (2002)	-0.435	-0.031
Steensberg et al. (2001)	-1.164	-2.079
Lancaster et al. (2005)a	-0.657	-0.510
Lancaster et al. (2005)b	0.000	1.031
Lancaster et al. (2005)c	0.000	1.031



i Note

No outcome has standardized residuals above 3 or below -3.

5.4. Funnel Plot & Influential Cases

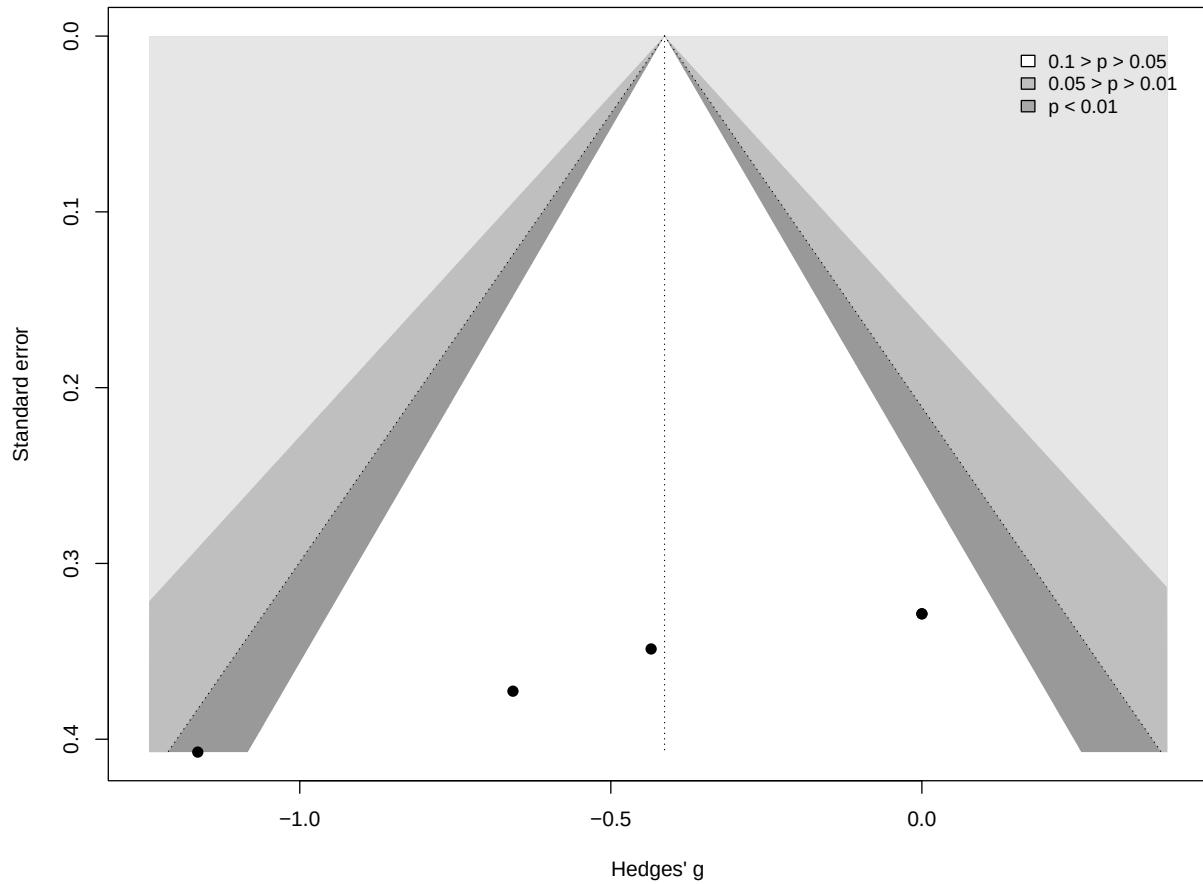


Table 5.2.: DFBETAS influence diagnostics

Short_reference	g	dfbetas
Ibfelt et al. (2002)	-0.435	-0.08474579
Steensberg et al. (2001)	-1.164	-0.87614442
Lancaster et al. (2005)a	-0.657	-0.38036241
Lancaster et al. (2005)b	0.000	0.73086010
Lancaster et al. (2005)c	0.000	0.73086010

One value below -1 and one above 1 were identified as influential cases. Neither is an outlier (Kostrzewa-Nowak & Nowak (2020a)a and Tsubakihara et al. (2012)); all outcomes are retained.

5.5. Forest Plot

```
#| fig-width: 8 #| fig-height: 4 #| out-width: 100% forest_main(m1)
```

5.6. Asymmetry Test

Note: the number of studies ($k = 5$) is too small to reliably test for small-study effects (minimum $k = 10$; Borenstein et al., 2009). No Egger's test or trim-and-fill is performed.

6. Th1 — 17 h Post-Exercise

6.1. Data Import

6.2. Primary Meta-Analysis

As in the immediate window, **Kostrzewa-Nowak et al. (2019)** is removed before pooling. Outlier detection then flags **Steensberg et al. (2001)**, which is both an outlier and an influential case and is therefore withdrawn from the downstream analyses (m2_1).

6.2.1. {meta}

Number of studies: k = 11

	SMD	95% CI	t
Random effects model (HK)	1.8184	[0.8759; 2.7610]	4.30
Prediction interval		[-2.1567; 5.7935]	
	p-value		
Random effects model (HK)	0.0016		
Prediction interval			

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 2.8848$ [0.3797; 7.3113]; $\tau = 1.6985$ [0.6162; 2.7039]
 $I^2 = 83.2\%$ [71.3%; 90.1%]; $H = 2.44$ [1.87; 3.18]

Test of heterogeneity:

Q	d.f.	p-value
59.37	10	< 0.0001

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 10)
- Prediction interval based on t-distribution (df = 10)

6.2.2. {metafor}

Random-Effects Model (k = 11; τ^2 estimator: HE)

```
tau^2 (estimated amount of total heterogeneity): 2.8848 (SE = 2.0141)
tau (square root of estimated tau^2 value):      1.6985
I^2 (total heterogeneity / total variability):    95.32%
H^2 (total variability / sampling variability):   21.39
```

Test for Heterogeneity:

```
Q(df = 10) = 59.3676, p-val < .0001
```

Model Results:

```
estimate      se      tval  df      pval   ci.lb   ci.ub
    1.8184   0.4230   4.2986  10   0.0016   0.8759   2.7610  **
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

6.2.3. Adapted Modell {meta}

Caution

Re-Fit: Removed Konstrezw-Nowak et al. (2019)

Number of studies: k = 10

```

                        SMD                95% CI      t
Random effects model (HK) 1.6316 [ 0.8788; 2.3845] 4.90
Prediction interval                [-0.8087; 4.0720]
                        p-value
Random effects model (HK) 0.0008
Prediction interval
```

Quantifying heterogeneity (with 95% CIs):

```
tau^2 = 1.0428 [0.3167; 3.7647]; tau = 1.0212 [0.5628; 1.9403]
I^2 = 83.8% [71.7%; 90.7%]; H = 2.48 [1.88; 3.28]
```

Test of heterogeneity:

```
Q d.f.  p-value
55.52   9 < 0.0001
```

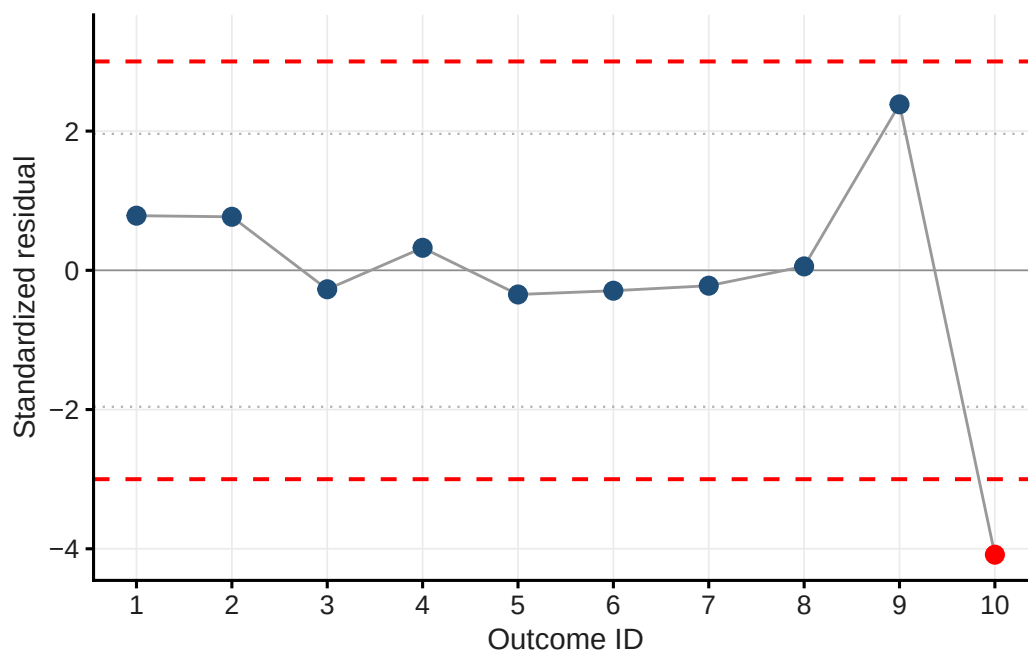
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 9)
- Prediction interval based on t-distribution (df = 9)

6.3. Outlier Detection

Table 6.1.: Standardised residuals

Short_reference	g	std_res
Kostrzewska-Nowak & Nowak (2018)a	2.320	0.785
Kostrzewska-Nowak & Nowak (2018)b	2.305	0.769
Kostrzewska-Nowak & Nowak (2020a)a	1.374	-0.272
Kostrzewska-Nowak & Nowak (2020a)b	1.889	0.323
Kostrzewska-Nowak & Nowak (2020b)a	1.305	-0.347
Kostrzewska-Nowak & Nowak (2020b)b	1.352	-0.293
Kostrzewska-Nowak & Nowak (2020b)c	1.416	-0.221
Kostrzewska-Nowak & Nowak (2020b)d	1.660	0.056
Kostrzewska-Nowak & Nowak (2020b)e	3.959	2.384
Steensberg et al. (2001)	-0.454	-4.085



Note

One outcome has standardized residuals below -3 (Steensberg et al. (2001)).

6.4. Funnel Plot & Influential Cases

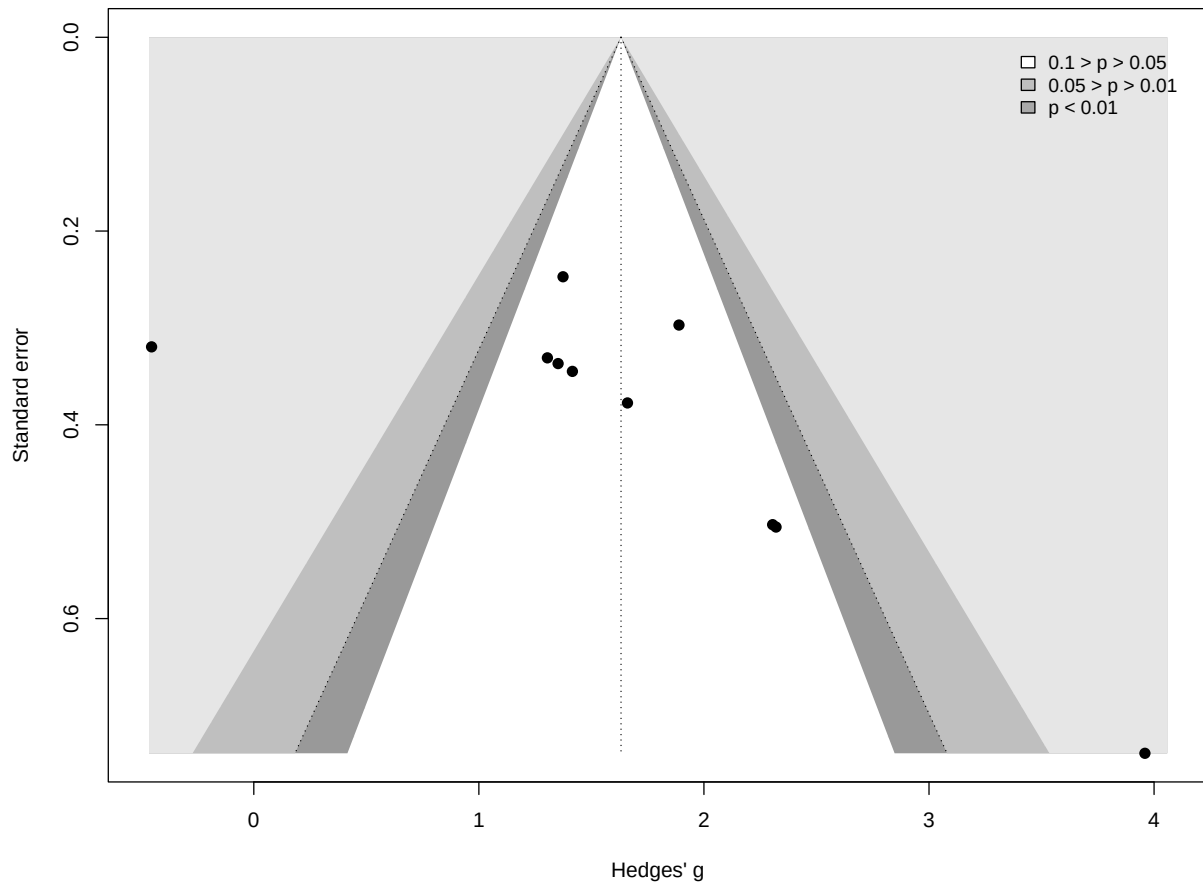


Table 6.2.: DFBETAS influence diagnostics

Short_reference	g	dfbetas
Kostrzewa-Nowak & Nowak (2018)a	2.320	0.42461320
Kostrzewa-Nowak & Nowak (2018)b	2.305	0.42202631
Kostrzewa-Nowak & Nowak (2020a)a	1.374	-0.04511648
Kostrzewa-Nowak & Nowak (2020a)b	1.889	0.72827130
Kostrzewa-Nowak & Nowak (2020b)a	1.305	-0.10227485
Kostrzewa-Nowak & Nowak (2020b)b	1.352	-0.04589921
Kostrzewa-Nowak & Nowak (2020b)c	1.416	0.02298307
Kostrzewa-Nowak & Nowak (2020b)d	1.660	0.22848951
Kostrzewa-Nowak & Nowak (2020b)e	3.959	0.53549707
Steensberg et al. (2001)	-0.454	-2.29509141

Caution

Steensberg et al. (2001) is < -1 (influential) and also an outlier $\rightarrow$ withdrawn.

6.4.1. Adapted Model

Number of studies: $k = 9$

	SMD	95% CI	t
Random effects model (HK)	1.8367	[1.2690; 2.4043]	7.46
Prediction interval		[0.0315; 3.6418]	
	p-value		
Random effects model (HK)	< 0.0001		
Prediction interval			

Quantifying heterogeneity (with 95% CIs):

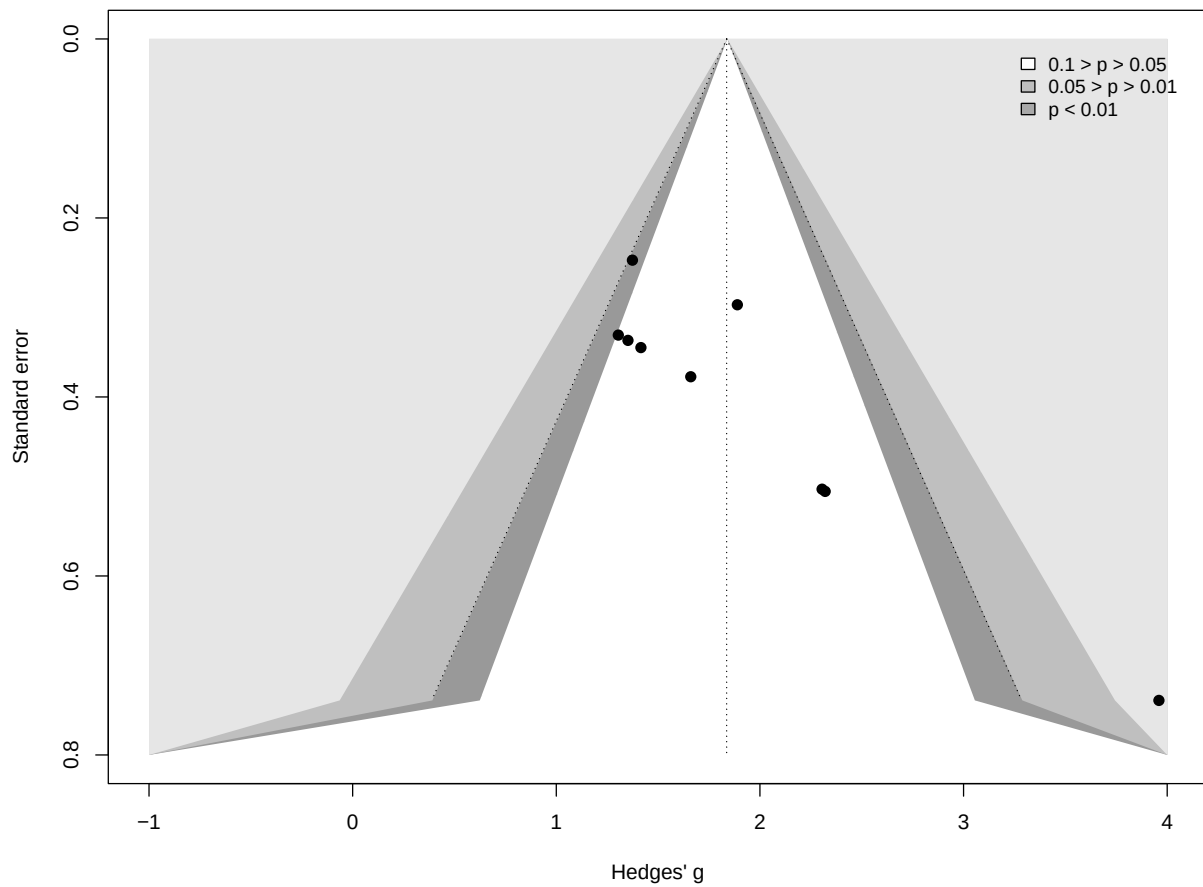
$\tau^2 = 0.5349$ [0.0000; 2.2412]; $\tau = 0.7314$ [0.0000; 1.4971]
 $I^2 = 54.2\%$ [2.8%; 78.4%]; $H = 1.48$ [1.01; 2.15]

Test of heterogeneity:

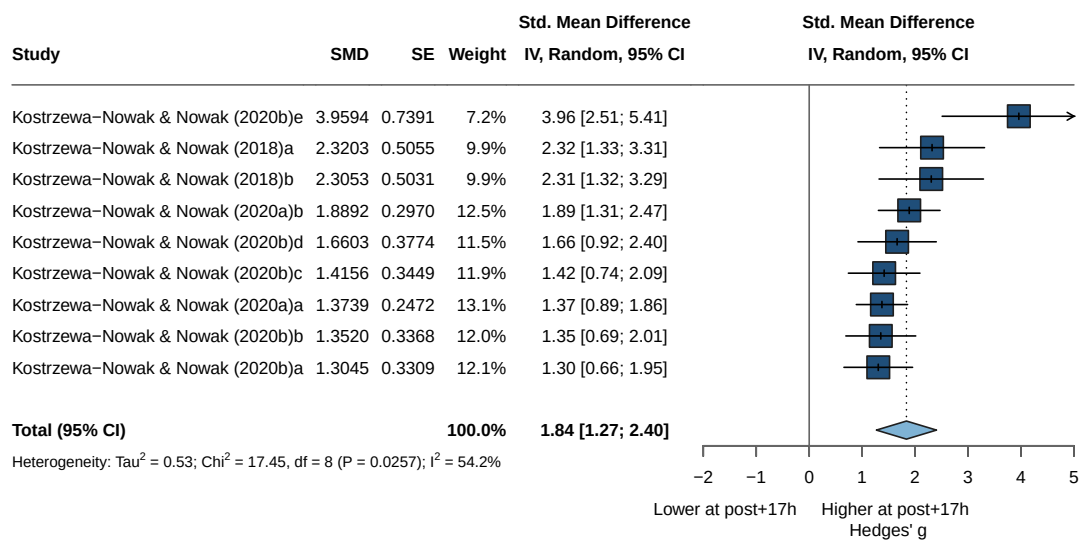
Q	d.f.	p-value
17.45	8	0.0257

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 8)
- Prediction interval based on t-distribution (df = 8)



6.5. Forest Plot



6.6. Asymmetry Test

6.6.1. Egger's regression test

6.6.2. Trim and Fill

Estimated number of missing studies on the left side: 0 (SE = 1.9045)

Random-Effects Model (k = 9; tau² estimator: HE)

tau² (estimated amount of total heterogeneity): 0.5349 (SE = 0.3673)

tau (square root of estimated tau² value): 0.7314

I² (total heterogeneity / total variability): 80.24%

H² (total variability / sampling variability): 5.06

Test for Heterogeneity:

Q(df = 8) = 17.4510, p-val = 0.0257

Model Results:

estimate	se	zval	pval	ci.lb	ci.ub	
1.8367	0.2791	6.5818	<.0001	1.2897	2.3836	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

 No missing studies

6.7. Moderator Analyses

6.7.1. Risk of Bias

6.7.1.1. Model

Number of studies: k = 9

Quantifying heterogeneity (with 95% CIs):

tau² = 0.5349 [0.0000; 2.2412]; tau = 0.7314 [0.0000; 1.4971]

I² = 54.2% [2.8%; 78.4%]; H = 1.48 [1.01; 2.15]

Test of heterogeneity:

Q	d.f.	p-value
17.45	8	0.0257

Results for subgroups (random effects model (HK)):

k	SMD	95% CI	tau ²	tau
---	-----	--------	------------------	-----

```

RoB = Moderate    2 2.3128 [ 2.2169; 2.4086]      0      0
RoB = Serious     2 1.6052 [-1.6511; 4.8615] 0.0581 0.2410
RoB = Low         5 1.8263 [ 0.5408; 3.1117] 1.0891 1.0436
               Q   I^2
RoB = Moderate    0.00  0.0%
RoB = Serious     1.78 43.8%
RoB = Low         11.76 66.0%

```

Test for subgroup differences (random effects model (HK)):

```

               Q d.f. p-value
Between groups 8.72    2  0.0128

```

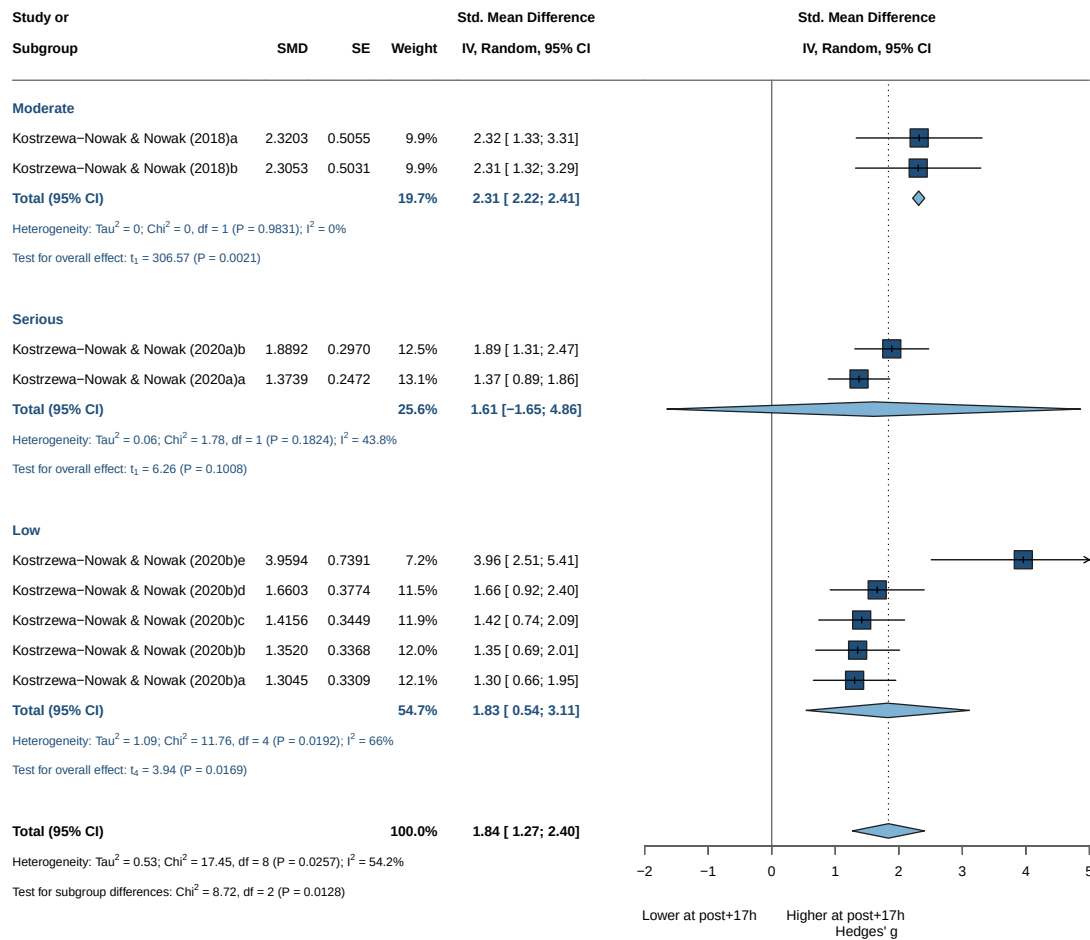
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

Note

Not significant results

6.7.1.2. Forrest Plot



Intensity: all retained arms share the same intensity, so no intensity subgroup analysis is performed.

6.7.2. Continuous Moderators (Meta-Regression)

6.7.2.1. Age

Mixed-Effects Model ($k = 9$; τ^2 estimator: HE)

logLik	deviance	AIC	BIC	AICc
-9.3856	19.2634	24.7713	25.3629	29.5713

τ^2 (estimated amount of residual heterogeneity): 0.5156 (SE = 0.3831)
 τ (square root of estimated τ^2 value): 0.7181
 I^2 (residual heterogeneity / unaccounted variability): 79.71%
 H^2 (unaccounted variability / sampling variability): 4.93
 R^2 (amount of heterogeneity accounted for): 3.61%

Test for Residual Heterogeneity:

QE($df = 7$) = 15.6877, p -val = 0.0281

6. Th1 — 17 h Post-Exercise

Test of Moderators (coefficient 2):

$F(df1 = 1, df2 = 7) = 1.0164, p\text{-val} = 0.3470$

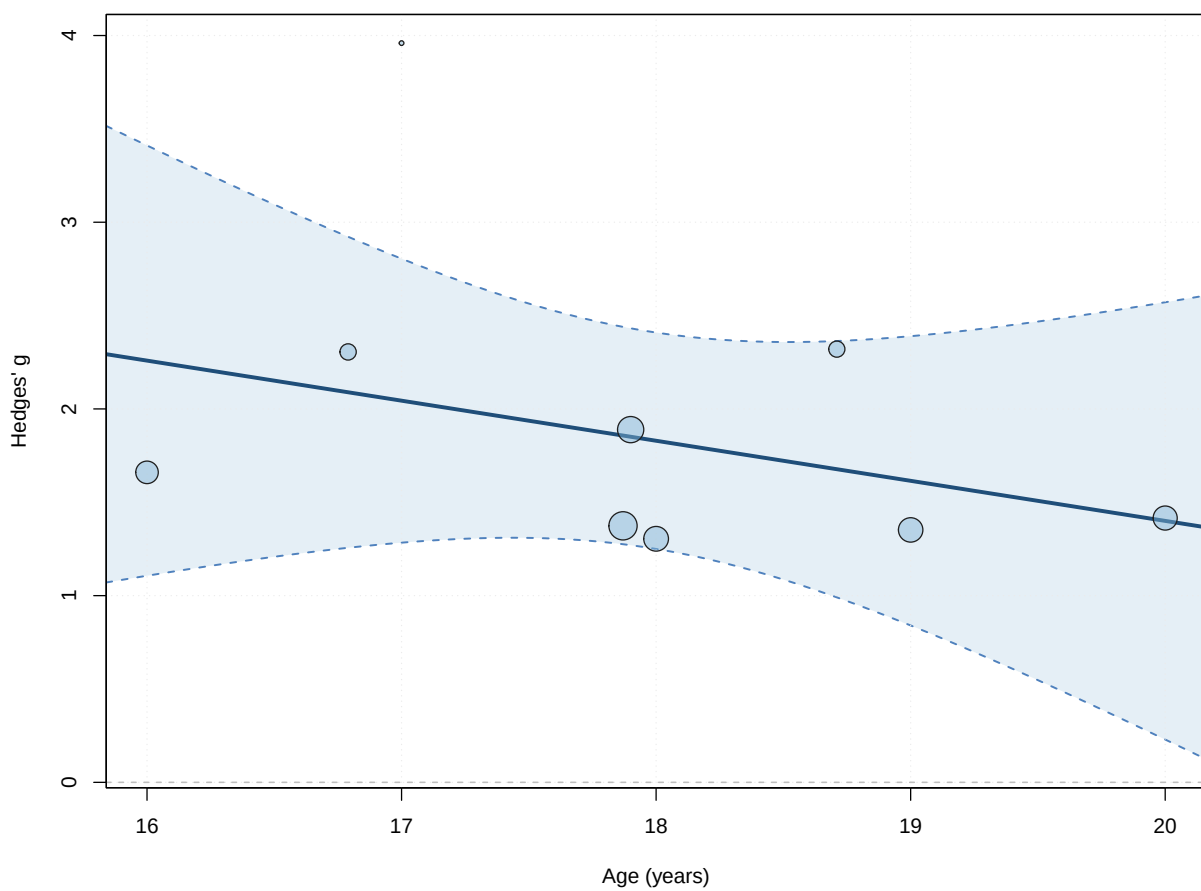
Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	5.6926	3.8350	1.4844	7	0.1813	-3.3758	
age	-0.2146	0.2129	-1.0082	7	0.3470	-0.7180	
intrcpt	14.7609						
age	0.2888						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant



6.7.2.2. BMI

Mixed-Effects Model (k = 9; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-9.3661	19.2242	24.7321	25.3238	29.5321

tau² (estimated amount of residual heterogeneity): 0.5147 (SE = 0.3872)
 tau (square root of estimated tau² value): 0.7175
 I² (residual heterogeneity / unaccounted variability): 78.61%
 H² (unaccounted variability / sampling variability): 4.68
 R² (amount of heterogeneity accounted for): 3.77%

Test for Residual Heterogeneity:

QE(df = 7) = 15.9237, p-val = 0.0258

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 7) = 1.0645, p-val = 0.3365

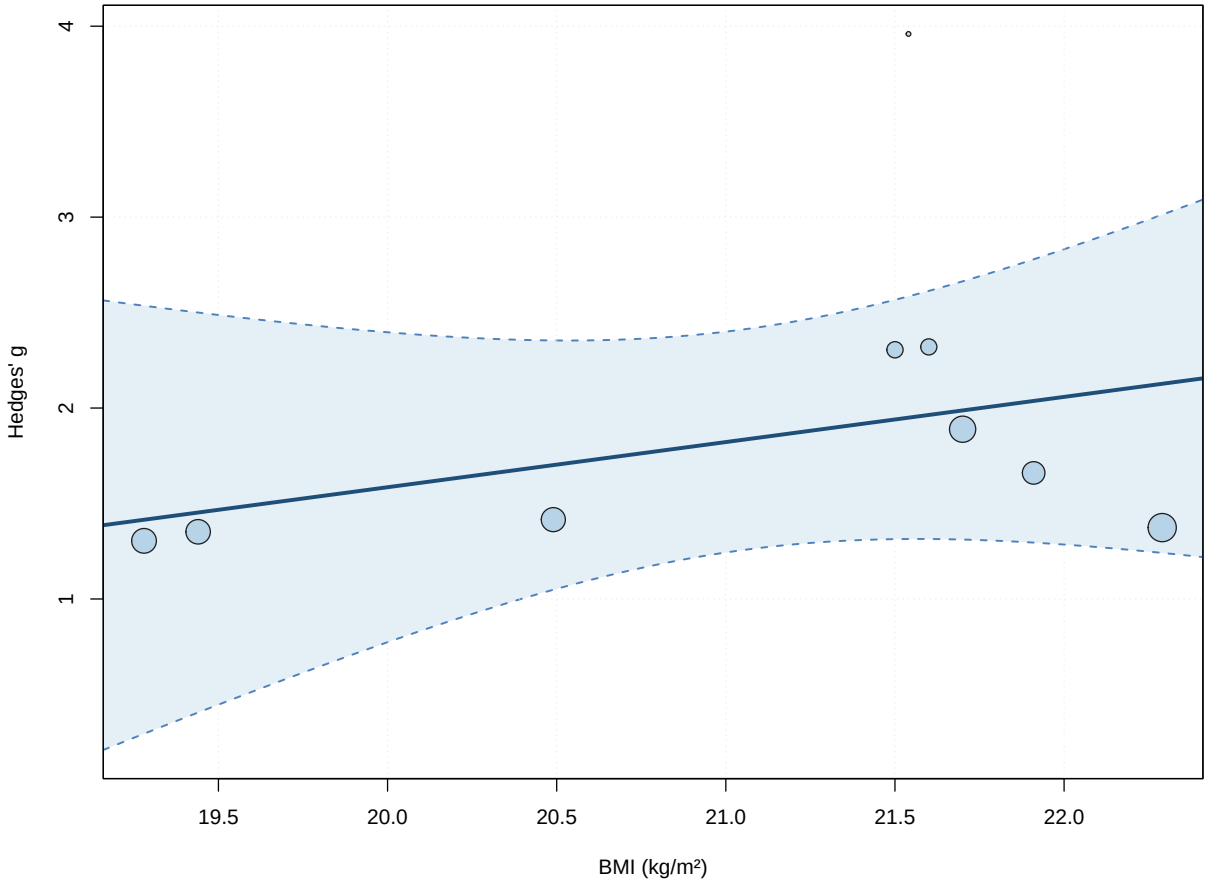
Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	-3.1460	4.8331	-0.6509	7	0.5359	-14.5743	
bmi	0.2366	0.2293	1.0317	7	0.3365	-0.3056	
intrcpt	8.2824						
bmi	0.7787						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant results



6.7.2.3. Duration

Random-Effects Model (k = 7; tau^2 estimator: HE)

logLik	deviance	AIC	BIC	AICc
-8.5222	15.9889	21.0443	20.9362	24.0443

tau^2 (estimated amount of total heterogeneity): 0.6770 (SE = 0.5235)
tau (square root of estimated tau^2 value): 0.8228
I^2 (total heterogeneity / total variability): 80.13%
H^2 (total variability / sampling variability): 5.03

Test for Heterogeneity:
Q(df = 6) = 15.4832, p-val = 0.0168

Model Results:

estimate	se	tval	df	pval	ci.lb	ci.ub	
1.9274	0.3239	5.9501	6	0.0010	1.1348	2.7200	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant results

6.7.2.4. Dose

Random-Effects Model (k = 7; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-8.5222	15.9889	21.0443	20.9362	24.0443

tau² (estimated amount of total heterogeneity): 0.6770 (SE = 0.5235)

tau (square root of estimated tau² value): 0.8228

I² (total heterogeneity / total variability): 80.13%

H² (total variability / sampling variability): 5.03

Test for Heterogeneity:

Q(df = 6) = 15.4832, p-val = 0.0168

Model Results:

estimate	se	tval	df	pval	ci.lb	ci.ub
1.9274	0.3239	5.9501	6	0.0010	1.1348	2.7200 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant results

6.7.2.5. IFN

Mixed-Effects Model (k = 7; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-8.6058	18.6411	23.2116	23.0493	31.2116

tau² (estimated amount of residual heterogeneity): 0.8890 (SE = 0.6816)

tau (square root of estimated tau² value): 0.9429

I² (residual heterogeneity / unaccounted variability): 87.53%

H² (unaccounted variability / sampling variability): 8.02

R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

6. Th1 — 17 h Post-Exercise

QE(df = 5) = 13.1061, p-val = 0.0224

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 0.0320, p-val = 0.8651

Model Results:

	estimate	se	tval	df	pval	ci.lb
intrcpt	1.9455	1.1277	1.7252	5	0.1451	-0.9533
IFN	-0.0645	0.3608	-0.1788	5	0.8651	-0.9920
	ci.ub					
intrcpt	4.8442					
IFN	0.8630					

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note

Not significant results

6.7.2.6. IL_4

Mixed-Effects Model (k = 7; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-8.6353	18.7000	23.2706	23.1083	31.2706

tau² (estimated amount of residual heterogeneity): 0.9003 (SE = 0.6946)
tau (square root of estimated tau² value): 0.9488
I² (residual heterogeneity / unaccounted variability): 87.21%
H² (unaccounted variability / sampling variability): 7.82
R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 5) = 13.3188, p-val = 0.0206

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 0.0105, p-val = 0.9225

Model Results:

	estimate	se	tval	df	pval	ci.lb
intrcpt	1.7842	0.4532	3.9366	5	0.0110	0.6191
IL_4	-0.0606	0.5927	-0.1023	5	0.9225	-1.5842
	ci.ub					
intrcpt	2.9494	*				
IL_4	1.4629					

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant results

6.7.2.7. IL_6Mixed-Effects Model (k = 9; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-8.2134	16.9190	22.4269	23.0185	27.2269

tau ² (estimated amount of residual heterogeneity):	0.3862 (SE = 0.3095)
tau (square root of estimated tau ² value):	0.6214
I ² (residual heterogeneity / unaccounted variability):	75.64%
H ² (unaccounted variability / sampling variability):	4.10
R ² (amount of heterogeneity accounted for):	27.80%

Test for Residual Heterogeneity:

QE(df = 7) = 12.0828, p-val = 0.0979

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 7) = 3.2234, p-val = 0.1157

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	1.4941	0.2761	5.4119	7	0.0010	0.8413	
IL_6	0.1709	0.0952	1.7954	7	0.1157	-0.0542	
intrcpt	2.1469	***					
IL_6	0.3960						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant results

6.7.2.8. IL_10Mixed-Effects Model (k = 9; tau² estimator: HE)

6. Th1 — 17 h Post-Exercise

logLik	deviance	AIC	BIC	AICc
-10.1243	20.7407	26.2486	26.8403	31.0486

```
tau^2 (estimated amount of residual heterogeneity):    0.6430 (SE = 0.4456)
tau (square root of estimated tau^2 value):           0.8019
I^2 (residual heterogeneity / unaccounted variability): 83.76%
H^2 (unaccounted variability / sampling variability):  6.16
R^2 (amount of heterogeneity accounted for):           0.00%
```

Test for Residual Heterogeneity:

QE(df = 7) = 17.0898, p-val = 0.0168

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 7) = 0.0868, p-val = 0.7768

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	1.7784	0.3578	4.9701	7	0.0016	0.9323	
IL_10	0.0714	0.2424	0.2946	7	0.7768	-0.5017	
intrcpt	2.6245						**
IL_10	0.6445						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note

Not significant results

6.7.2.9. IL_2

Mixed-Effects Model (k = 7; tau^2 estimator: HE)

logLik	deviance	AIC	BIC	AICc
-8.6267	18.6829	23.2535	23.0912	31.2535

```
tau^2 (estimated amount of residual heterogeneity):    0.9041 (SE = 0.6941)
tau (square root of estimated tau^2 value):           0.9509
I^2 (residual heterogeneity / unaccounted variability): 87.75%
H^2 (unaccounted variability / sampling variability):  8.16
R^2 (amount of heterogeneity accounted for):           0.00%
```

Test for Residual Heterogeneity:

QE(df = 5) = 11.8338, p-val = 0.0371

Test of Moderators (coefficient 2):

$F(df1 = 1, df2 = 5) = 0.0501, p\text{-val} = 0.8317$

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	1.6362	0.6316	2.5905	5	0.0488	0.0126	
IL_2	0.0662	0.2956	0.2239	5	0.8317	-0.6937	
intrcpt	3.2599						*
IL_2	0.8261						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant results

6.8. Multivariate Models

Mixed-Effects Model (k = 7; tau² estimator: HE)

tau ² (estimated amount of residual heterogeneity):	1.4592 (SE = 1.3845)
tau (square root of estimated tau ² value):	1.2080
I ² (residual heterogeneity / unaccounted variability):	90.49%
H ² (unaccounted variability / sampling variability):	10.52
R ² (amount of heterogeneity accounted for):	0.00%

Test for Residual Heterogeneity:

$QE(df = 3) = 11.4927, p\text{-val} = 0.0093$

Test of Moderators (coefficients 2:4):

$F(df1 = 3, df2 = 3) = 0.0705, p\text{-val} = 0.9719$

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	1.7103	2.6663	0.6415	3	0.5669	-6.7751	
IFN	-0.1372	0.8430	-0.1627	3	0.8811	-2.8200	
IL_2	0.5964	1.9677	0.3031	3	0.7816	-5.6658	
IFN:IL_2	-0.0966	0.4584	-0.2108	3	0.8466	-1.5554	
intrcpt	10.1958						
IFN	2.5457						
IL_2	6.8586						
IFN:IL_2	1.3621						

6. *Th1 — 17 h Post-Exercise*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

Part II.

Th2

7. Th2 — Immediate Post-Exercise

7.1. Data Import

The base analysis uses the pre-computed Hedges' g file for Th2 at $r = 0.5$ (`effect_size/Th2/pre_post/Th2_pre_post_r_050`). Sensitivity analyses can substitute the $r = 0.3 / 0.4 / 0.6 / 0.7$ files in the same folder. The pipeline column names are mapped to the analysis conventions (VE_g, Sex, Age, BMI) by `load_es()`.

7.2. Primary Meta-Analysis

`metagen()` fits a random-effects model on the pre-computed effect sizes (g , SE_g) using the **Hedges–Olkin (HE)** ² estimator and the **Knapp–Hartung** small-sample correction (`hakn = TRUE`). **Kostrzewa-Nowak et al. (2019)** is removed before pooling and the model re-fitted; `metafor::rma()` cross-validates.

7.2.1. Full Model {meta}

Number of studies: $k = 18$

	SMD	95% CI	t
Random effects model (HK)	0.3214	[-0.0871; 0.7298]	1.66
Prediction interval		[-1.9144; 2.5571]	
	p-value		
Random effects model (HK)	0.1152		
Prediction interval			

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 1.0569$ [0.1238; 1.5639]; $\tau = 1.0281$ [0.3518; 1.2506]
 $I^2 = 71.9\%$ [54.9%; 82.5%]; $H = 1.89$ [1.49; 2.39]

Test of heterogeneity:

Q	d.f.	p-value
60.51	17	< 0.0001

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model ($df = 17$)
- Prediction interval based on t-distribution ($df = 17$)

7.2.2. Full Model {metafor}

Random-Effects Model (k = 18; tau² estimator: HE)

tau² (estimated amount of total heterogeneity): 1.0569 (SE = 0.4699)
 tau (square root of estimated tau² value): 1.0281
 I² (total heterogeneity / total variability): 93.23%
 H² (total variability / sampling variability): 14.78

Test for Heterogeneity:

Q(df = 17) = 60.5052, p-val < .0001

Model Results:

estimate	se	tval	df	pval	ci.lb	ci.ub
0.3214	0.1936	1.6601	17	0.1152	-0.0871	0.7298

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

7.2.3. Adapted Model {meta}

Number of studies: k = 17

	SMD	95% CI	t
Random effects model (HK)	0.3791	[0.0865; 0.6718]	2.75
Prediction interval		[-0.7743; 1.5326]	
	p-value		
Random effects model (HK)	0.0143		
Prediction interval			

Quantifying heterogeneity (with 95% CIs):

tau² = 0.2749 [0.0825; 0.7087]; tau = 0.5243 [0.2873; 0.8418]
 I² = 70.2% [51.2%; 81.8%]; H = 1.83 [1.43; 2.35]

Test of heterogeneity:

Q	d.f.	p-value
53.73	16	< 0.0001

Details of meta-analysis methods:

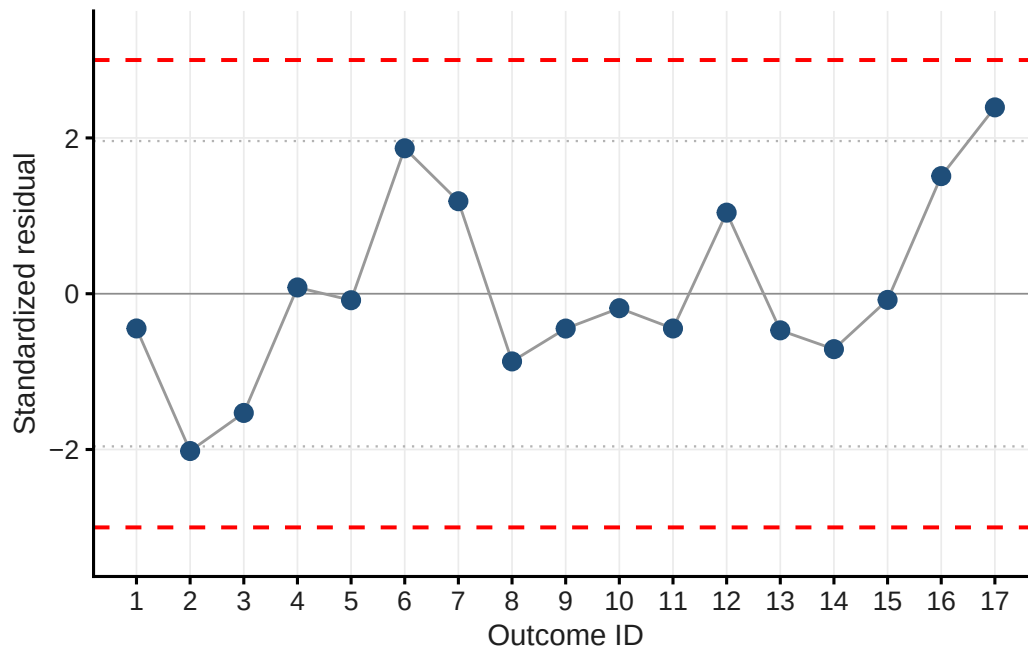
- Inverse variance method
- Hedges estimator for tau²
- Q-Profile method for confidence interval of tau² and tau
- Calculation of I² based on Q
- Hartung-Knapp adjustment for random effects model (df = 16)
- Prediction interval based on t-distribution (df = 16)

7.3. Outlier Detection

Leave-one-out standardised residuals are computed via `metaoutliers()`, which is passed the **variance** (`m3$seTE^2`). Studies with $|z| > 3$ are flagged as outliers; $|z| > 1.96$ warrants inspection.

Table 7.1.: Standardised residuals

Short_reference	g	std_res
Ibfelt et al. (2002)	0.137	-0.446
Kostrzewa-Nowak & Nowak (2018)a	-0.529	-2.019
Kostrzewa-Nowak & Nowak (2018)b	-0.330	-1.530
Kostrzewa-Nowak & Nowak (2020a)a	0.410	0.080
Kostrzewa-Nowak & Nowak (2020a)b	0.334	-0.084
Kostrzewa-Nowak & Nowak (2020b)a	1.272	1.867
Kostrzewa-Nowak & Nowak (2020b)b	0.941	1.187
Kostrzewa-Nowak & Nowak (2020b)c	-0.036	-0.869
Kostrzewa-Nowak & Nowak (2020b)d	0.158	-0.447
Kostrzewa-Nowak & Nowak (2020b)e	0.282	-0.186
Steensberg et al. (2001)	0.144	-0.446
Tsubakihara et al. (2012)	0.861	1.041
Lancaster et al. (2005)a	0.124	-0.470
Lancaster et al. (2005)b	0.000	-0.712
Lancaster et al. (2005)c	0.329	-0.079
Harbaum et al. (2016)a	1.096	1.510
Harbaum et al. (2016)b	1.829	2.392



Caution

One outcome has standardized residuals around -2 (Kostrzewa-Nowak et al. (2019)).

7.4. Funnel Plot & Influential Cases

The contour-enhanced funnel plot shows study estimates against their standard errors; shaded regions mark significance zones ($\alpha = 0.10, 0.05, 0.01$). The `dfbetas` diagnostic from `metafor::rma.mv()` identifies **influential cases** — distinct from statistical outliers and not necessarily excluded.

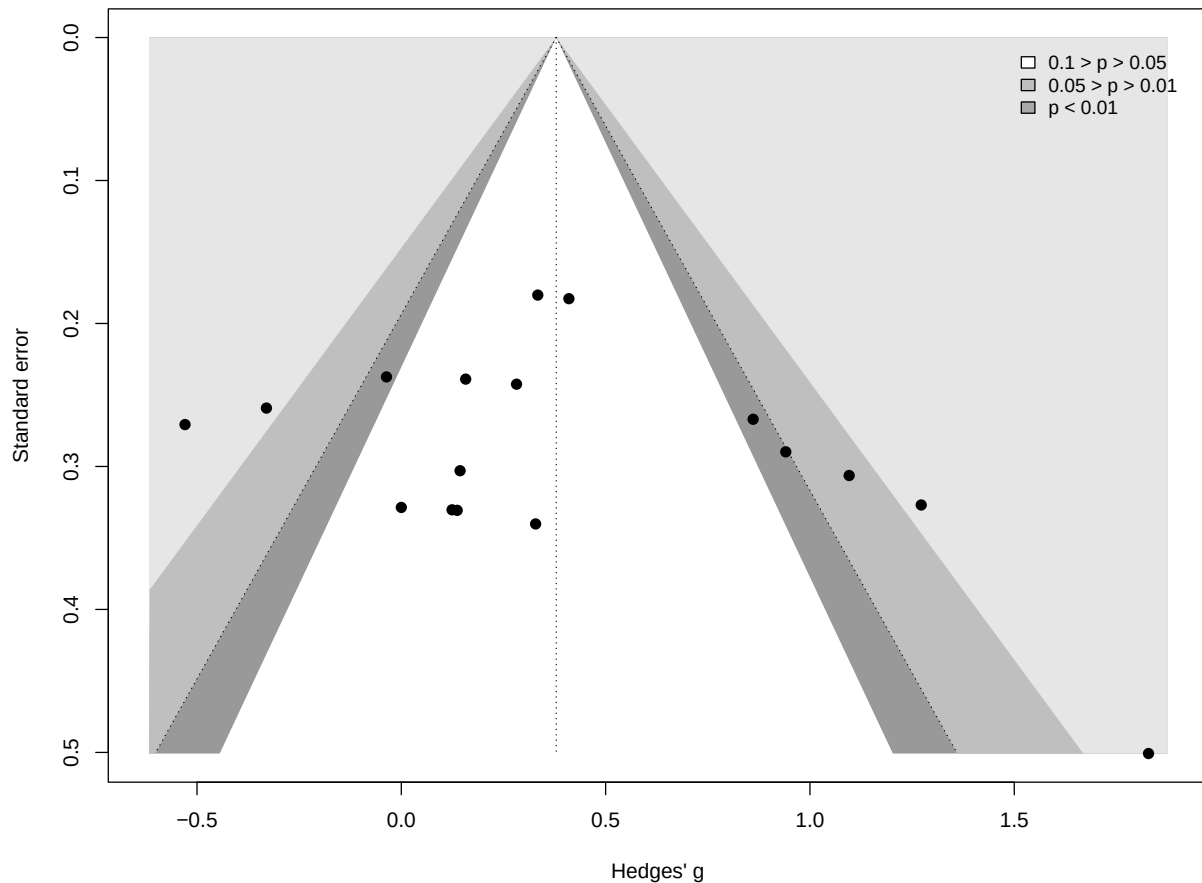


Table 7.2.: DFBETAS influence diagnostics

Short_reference	g	dfbetas
Ibfelt et al. (2002)	0.137	-0.1178026325
Kostrzewska-Nowak & Nowak (2018)a	-0.529	-0.8034118886
Kostrzewska-Nowak & Nowak (2018)b	-0.330	-0.6766632755
Kostrzewska-Nowak & Nowak (2020a)a	0.410	0.1818284116
Kostrzewska-Nowak & Nowak (2020a)b	0.334	0.0132850848
Kostrzewska-Nowak & Nowak (2020b)a	1.272	0.5951183620
Kostrzewska-Nowak & Nowak (2020b)b	0.941	0.4971720392
Kostrzewska-Nowak & Nowak (2020b)c	-0.036	-0.4523407882
Kostrzewska-Nowak & Nowak (2020b)d	0.158	-0.2088082167
Kostrzewska-Nowak & Nowak (2020b)e	0.282	-0.0547346426
Steensberg et al. (2001)	0.144	-0.1362231954
Tsubakihara et al. (2012)	0.861	0.5140815471
Lancaster et al. (2005)a	0.124	-0.1258018657
Lancaster et al. (2005)b	0.000	-0.2045859006

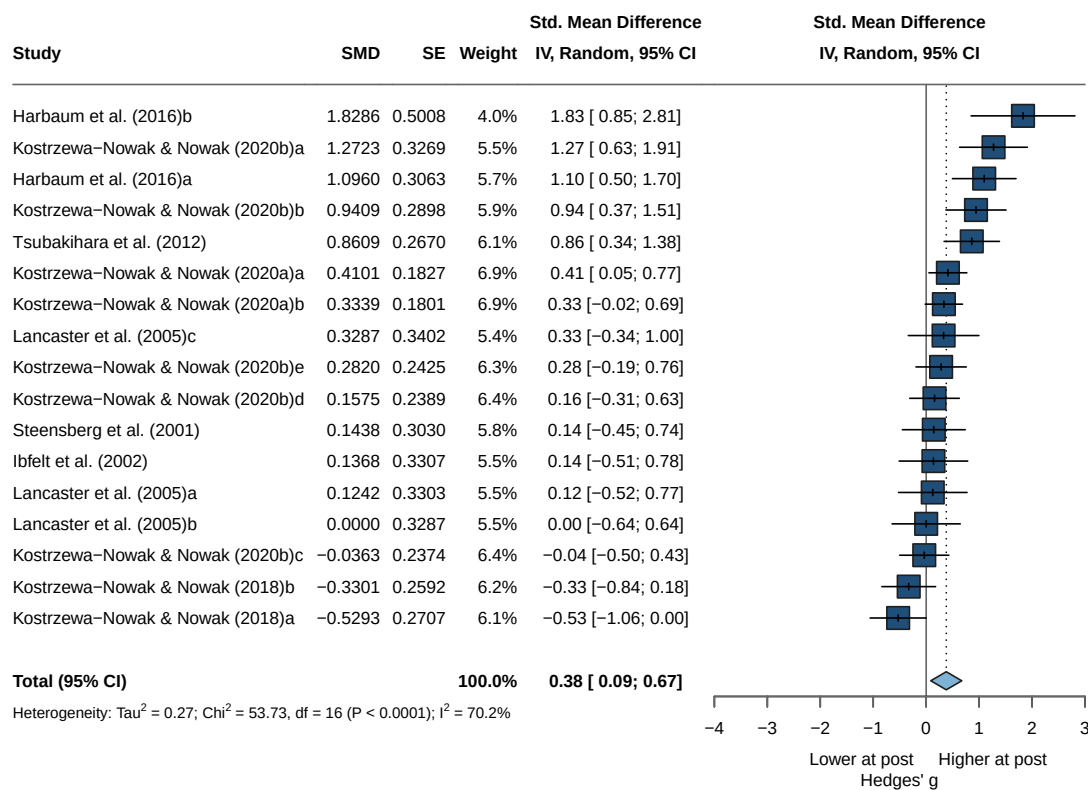
Table 7.2.: DFBETAS influence diagnostics

Short_reference	g	dfbetas
Lancaster et al. (2005)c	0.329	0.0003012762
Harbaum et al. (2016)a	1.096	0.5544242443
Harbaum et al. (2016)b	1.829	0.3938945044

 Tip

No influential case.

7.5. Forest Plot



7.6. Asymmetry Test

Egger's regression test (`method = "linreg"`) evaluates funnel asymmetry as a proxy for small-study/publication effects (one-tailed, $p < .10$). Where asymmetry is suggested, a trim-and-fill analysis estimates potentially

missing studies.

Linear regression test of funnel plot asymmetry

Test result: $t = 1.17$, $df = 15$, $p\text{-value} = 0.2595$

Bias estimate: 2.2167 (SE = 1.8916)

Details:

- multiplicative residual heterogeneity variance ($\tau^2 = 3.2814$)
- predictor: standard error
- weight: inverse variance
- reference: Egger et al. (1997), BMJ

i Note

No asymmetry -> no trim-and-fill

7.7. Moderator Analyses

7.7.1. Risk of Bias

Subgroup analysis by overall RoB rating tests whether study quality moderates the pooled effect; the between-subgroup statistic (Q_b) indicates whether subgroup means differ.

7.7.1.1. Model

Number of studies: $k = 17$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.2749$ [0.0825; 0.7087]; $\tau = 0.5243$ [0.2873; 0.8418]

$I^2 = 70.2\%$ [51.2%; 81.8%]; $H = 1.83$ [1.43; 2.35]

Test of heterogeneity:

Q	d.f.	p-value
53.73	16	< 0.0001

Results for subgroups (random effects model (HK)):

	k	SMD	95% CI	τ^2	τ
RoB = Serious	6	0.7080	[0.1099; 1.3062]	0.2926	0.5409
RoB = Moderate	3	-0.2584	[-1.0905; 0.5736]	0.0421	0.2053
RoB = Low	8	0.3689	[-0.0172; 0.7550]	0.1352	0.3677

	Q	I^2
RoB = Serious	14.63	65.8%
RoB = Moderate	2.84	29.5%
RoB = Low	17.08	59.0%

Test for subgroup differences (random effects model (HK)):

Q	d.f.	p-value
---	------	---------

Between groups 11.31 2 0.0035

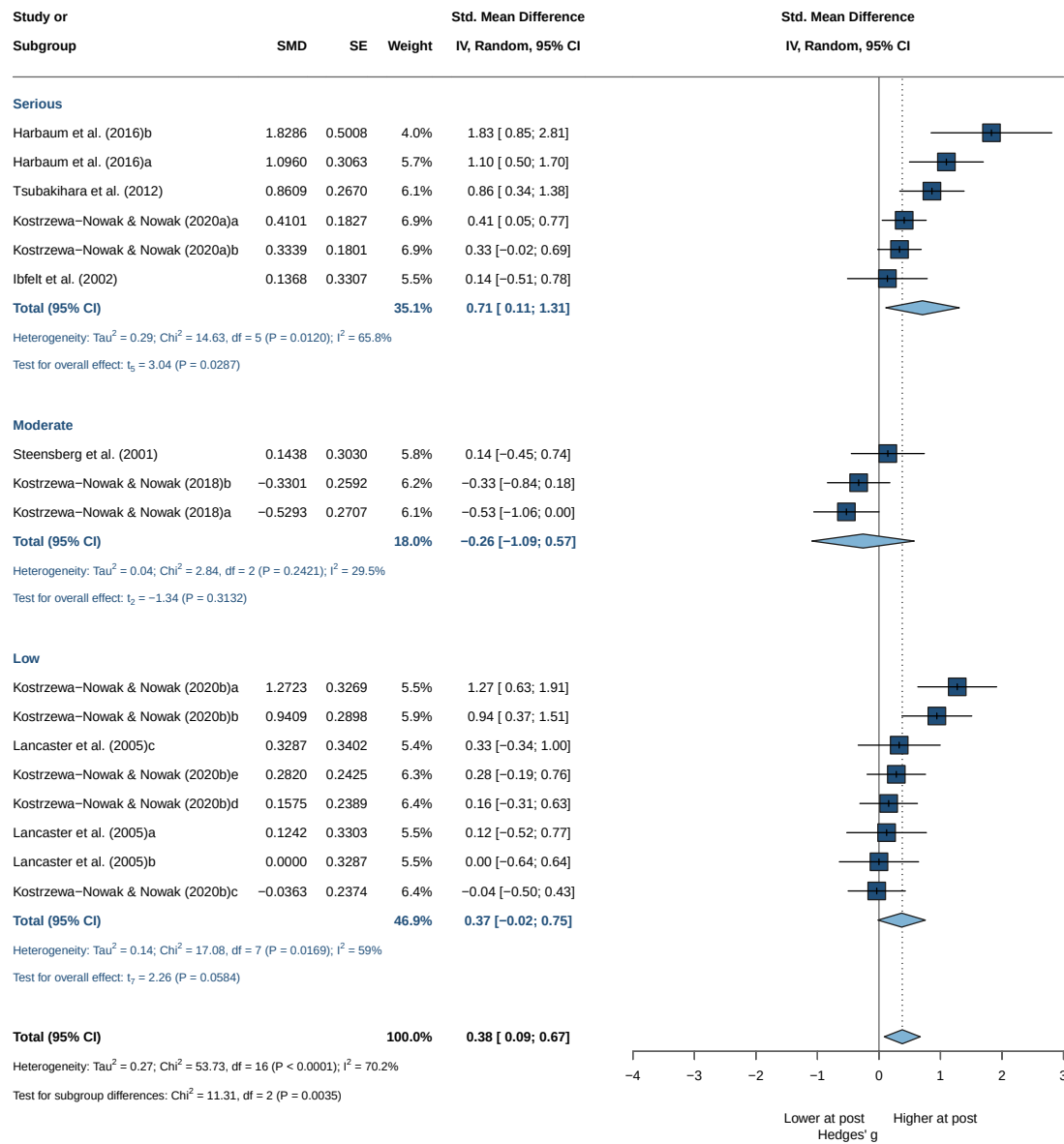
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

! Important

Significant results

7.7.1.2. Forrest Plot



7.7.1.3. Head-to-Head

Head-to-head RoB comparisons. Because the overall RoB subgroup test is significant, the individual rating levels are compared pairwise.

7.7.1.4. Moderate vs Serious

Number of studies: $k = 9$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.5204$ [0.2137; 1.9229]; $\tau = 0.7214$ [0.4622; 1.3867]

$I^2 = 98.0\%$ [97.3%; 98.6%]; $H = 7.11$ [6.08; 8.31]

Test of heterogeneity:

Q	d.f.	p-value
403.91	8	< 0.0001

Results for subgroups (random effects model (HK)):

	k	SMD	95% CI	τ^2	τ
RoB = Serious	6	0.7540	[0.1167; 1.3914]	0.3759	0.6131
RoB = Moderate	3	-0.2420	[-1.0969; 0.6129]	0.1135	0.3368

	Q	I^2
RoB = Serious	133.26	96.2%
RoB = Moderate	33.35	94.0%

Test for subgroup differences (random effects model (HK)):

	Q	d.f.	p-value
Between groups	9.83	1	0.0017

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

! Important

Significant

7.7.1.5. Moderate vs Low

Number of studies: $k = 11$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.2563$ [0.1206; 0.8046]; $\tau = 0.5063$ [0.3473; 0.8970]

$I^2 = 97.2\%$ [96.1%; 97.9%]; $H = 5.96$ [5.10; 6.97]

Test of heterogeneity:

Q	d.f.	p-value
355.37	10	< 0.0001

Results for subgroups (random effects model (HK)):

	k	SMD	95% CI	τ^2	τ
--	---	-----	--------	----------	--------

7. Th2 — Immediate Post-Exercise

```
RoB = Moderate    3 -0.2420 [-1.0969; 0.6129] 0.1135 0.3368
RoB = Low         8  0.3817 [-0.0113; 0.7748] 0.2141 0.4627
               Q    I^2
RoB = Moderate   33.35 94.0%
RoB = Low       193.09 96.4%
```

Test for subgroup differences (random effects model (HK)):

```
               Q d.f. p-value
Between groups 5.80    1  0.0161
```

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

! Important

Significant

7.7.1.6. Low vs Serious

Number of studies: $k = 14$

Quantifying heterogeneity (with 95% CIs):

```
tau^2 = 0.2999 [0.1367; 0.7776]; tau = 0.5476 [0.3697; 0.8818]
I^2 = 96.3% [95.0%; 97.2%]; H = 5.18 [4.46; 6.03]
```

Test of heterogeneity:

```
               Q d.f. p-value
349.48    13 < 0.0001
```

Results for subgroups (random effects model (HK)):

```
               k    SMD          95% CI tau^2    tau
RoB = Serious    6 0.7540 [ 0.1167; 1.3914] 0.3759 0.6131
RoB = Low        8 0.3817 [-0.0113; 0.7748] 0.2141 0.4627
               Q    I^2
RoB = Serious 133.26 96.2%
RoB = Low    193.09 96.4%
```

Test for subgroup differences (random effects model (HK)):

```
               Q d.f. p-value
Between groups 1.56    1  0.2123
```

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ

- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

i Note

Not significant

7.7.2. Sex

7.7.2.1. Model

Number of studies: $k = 17$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.2749$ [0.0825; 0.7087]; $\tau = 0.5243$ [0.2873; 0.8418]

$I^2 = 70.2\%$ [51.2%; 81.8%]; $H = 1.83$ [1.43; 2.35]

Test of heterogeneity:

Q	d.f.	p-value
53.73	16	< 0.0001

Results for subgroups (random effects model (HK)):

	k	SMD	95% CI	τ^2	τ
sex = male	14	0.2235	[-0.0321; 0.4792]	0.1284	0.3584
sex = female	1	0.8609	[0.3377; 1.3842]	--	--
sex = mixed	2	1.3552	[-3.0956; 5.8059]	0.0960	0.3099

	Q	I^2
sex = male	32.25	59.7%
sex = female	0.00	--
sex = mixed	1.56	35.8%

Test for subgroup differences (random effects model (HK)):

	Q	d.f.	p-value
Between groups	12.61	2	0.0018

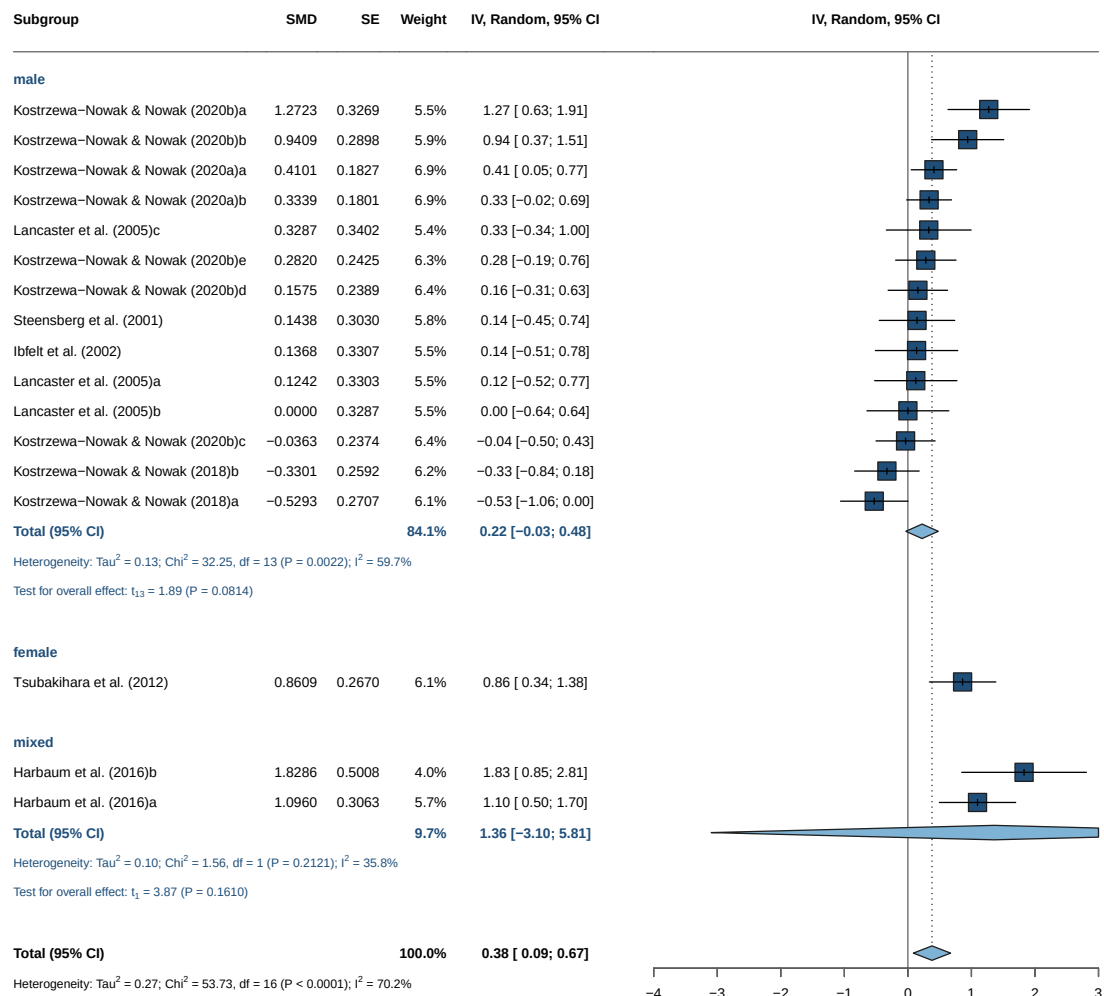
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

! Important

Significant results but 15 vs 1 vs 2

7.7.2.2. Forrest Plot



7.7.3. Intensity

7.7.3.1. Model

Number of studies: $k = 16$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.2839$ [0.0797; 0.7558]; $\tau = 0.5328$ [0.2822; 0.8694]

$I^2 = 69.7\%$ [49.4%; 81.8%]; $H = 1.82$ [1.41; 2.35]

Test of heterogeneity:

Q	d.f.	p-value
49.50	15	< 0.0001

Results for subgroups (random effects model (HK)):

	k	SMD	95% CI	τ^2
intensity = Vigorous	5	0.1443	[0.0026; 0.2860]	0
intensity = Maximal	11	0.4478	[-0.0115; 0.9070]	0.4316
		τ	Q	I^2
intensity = Vigorous	0	0.49	0.0%	

intensity = Maximal 0.6570 47.65 79.0%

Test for subgroup differences (random effects model (HK)):

Q d.f. p-value
Between groups 2.04 1 0.1529

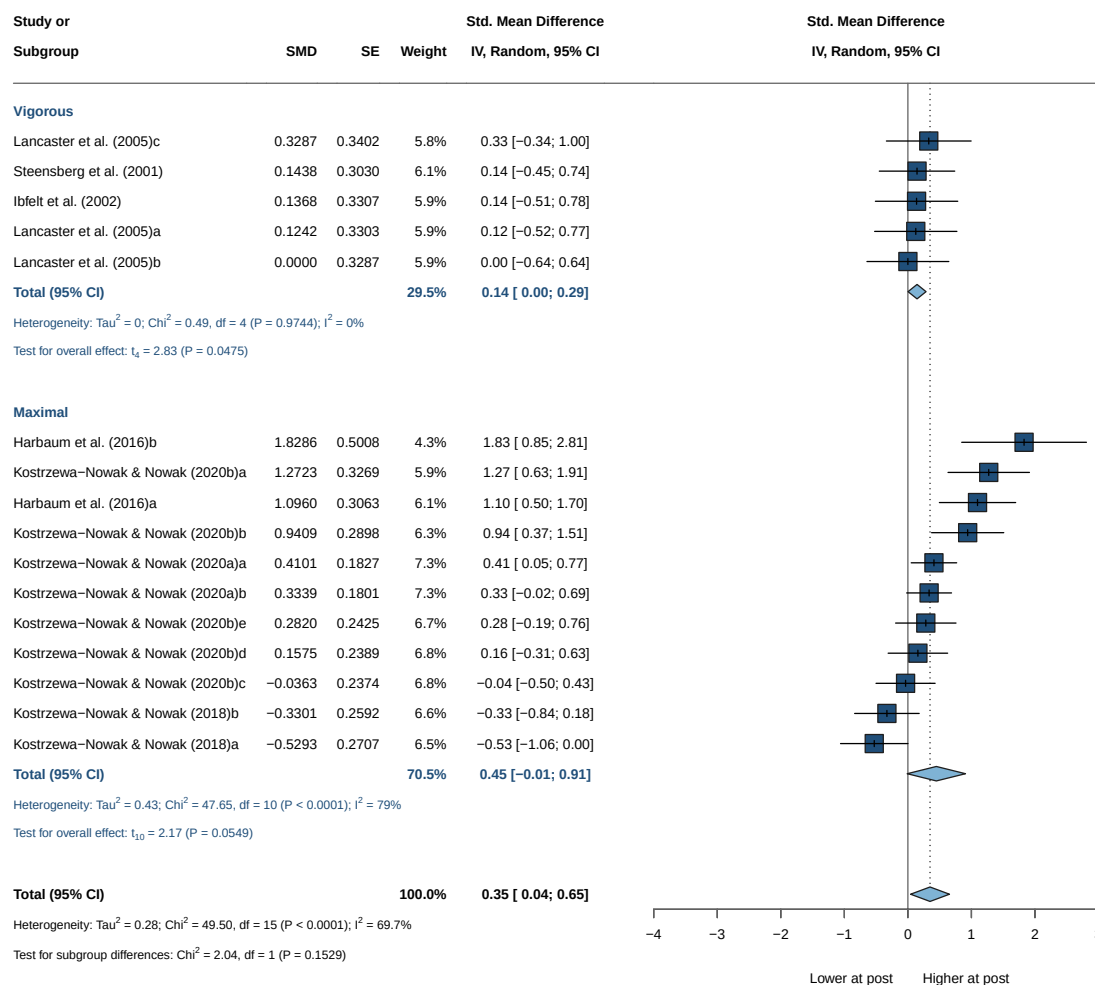
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

Note

Not significant

7.7.3.2. Forrest Plot



7.7.4. Staining

Number of studies: $k = 17$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.2749$ [0.0825; 0.7087]; $\tau = 0.5243$ [0.2873; 0.8418]

$I^2 = 70.2\%$ [51.2%; 81.8%]; $H = 1.83$ [1.43; 2.35]

Test of heterogeneity:

Q	d.f.	p-value
53.73	16	< 0.0001

Results for subgroups (random effects model (HK)):

	k	SMD	95% CI	τ^2	τ
staining = intra	14	0.2235	[-0.0321; 0.4792]	0.1284	0.3584
staining = extra	3	1.1539	[0.0415; 2.2663]	0.1161	0.3407

	Q	I^2
staining = intra	32.25	59.7%
staining = extra	2.91	31.3%

Test for subgroup differences (random effects model (HK)):

	Q	d.f.	p-value
Between groups	10.71	1	0.0011

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

! Important

Significant results

7.7.5. Continuous Moderators (Meta-Regression)

Continuous moderators are tested via weighted meta-regression (`metareg()`). A significant slope indicates the moderator explains part of the between-study heterogeneity (τ^2); bubble plots show the fitted line with 95% CI.

7.7.5.1. Age

Mixed-Effects Model ($k = 16$; τ^2 estimator: HE)

logLik	deviance	AIC	BIC	AICc
-10.1238	31.2182	26.2475	28.5653	28.2475

```

tau^2 (estimated amount of residual heterogeneity):    0.1586 (SE = 0.0936)
tau (square root of estimated tau^2 value):           0.3982
I^2 (residual heterogeneity / unaccounted variability): 68.97%
H^2 (unaccounted variability / sampling variability):  3.22
R^2 (amount of heterogeneity accounted for):           44.13%

```

Test for Residual Heterogeneity:

QE(df = 14) = 37.4912, p-val = 0.0006

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 14) = 6.4163, p-val = 0.0239

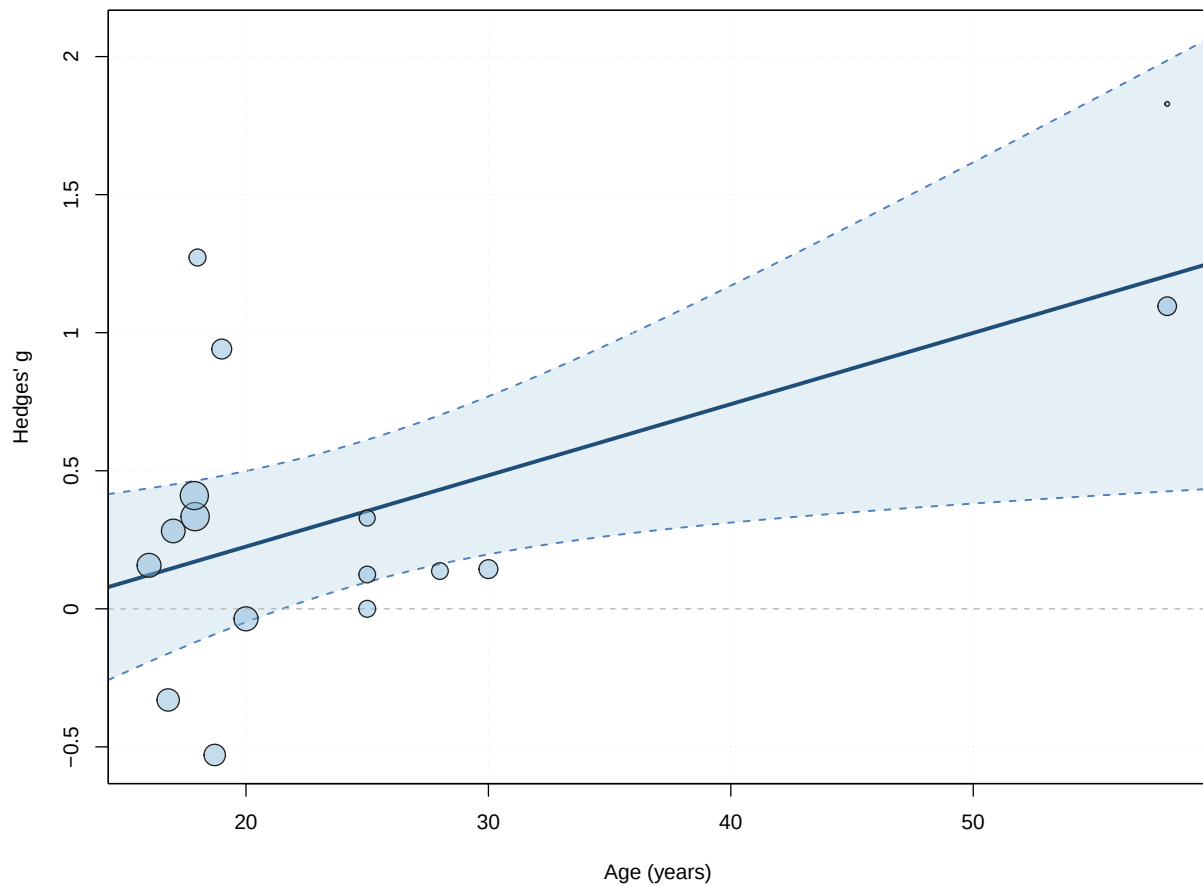
Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	-0.2905	0.2747	-1.0574	14	0.3082	-0.8798	
age	0.0258	0.0102	2.5330	14	0.0239	0.0040	
intrcpt	0.2988						
age	0.0476						*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

! Important

Significant



7.7.5.2. BMI

Mixed-Effects Model (k = 12; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-11.5110	32.7869	29.0221	30.4768	32.0221

tau² (estimated amount of residual heterogeneity): 0.4184 (SE = 0.2231)
 tau (square root of estimated tau² value): 0.6468
 I² (residual heterogeneity / unaccounted variability): 87.04%
 H² (unaccounted variability / sampling variability): 7.71
 R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 10) = 50.4537, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 10) = 0.4072, p-val = 0.5377

Model Results:

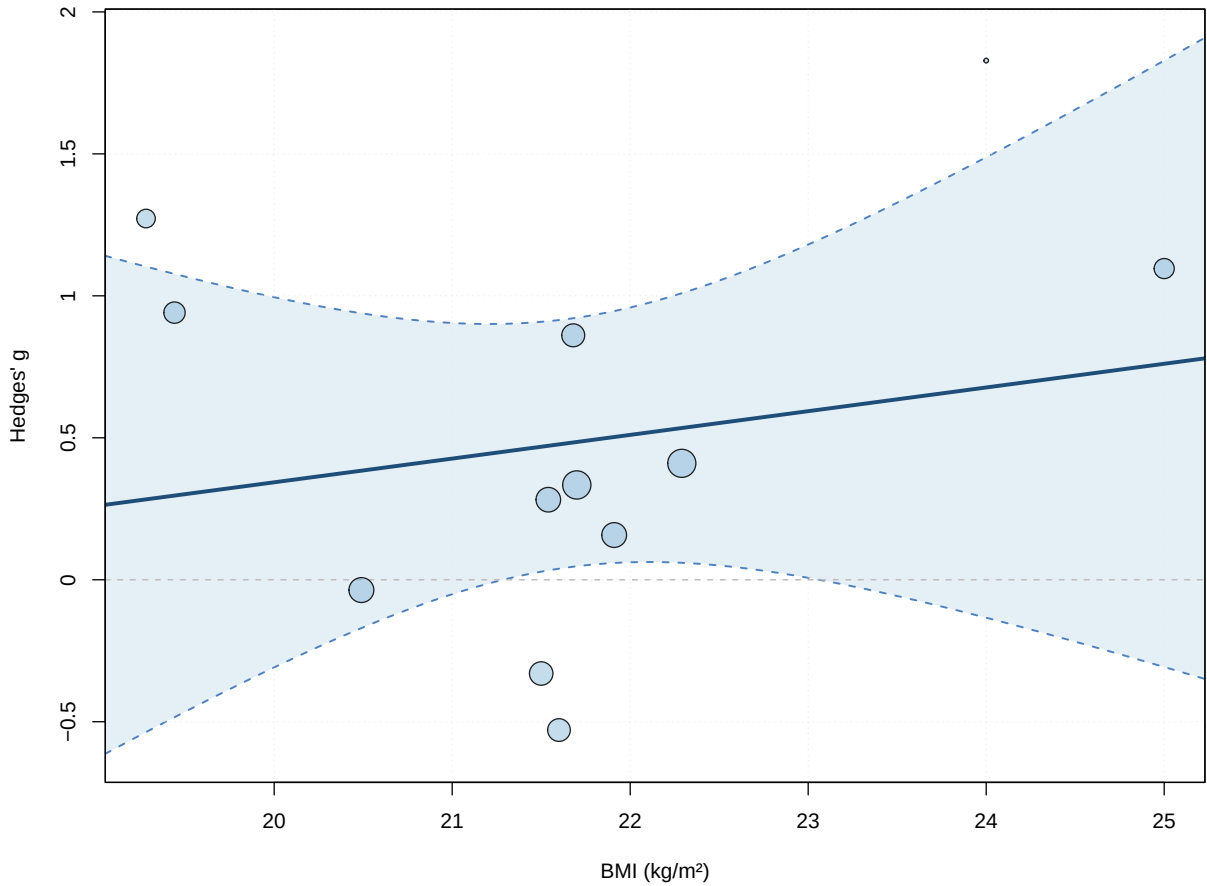
```
      estimate      se      tval  df      pval      ci.lb
intrcpt    -1.3265  2.8404   -0.4670  10   0.6505   -7.6554
bmi         0.0835  0.1308    0.6381  10   0.5377   -0.2080

      ci.ub
intrcpt    5.0023
bmi        0.3750

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

i Note

Not significant



7.7.5.3. Duration

Mixed-Effects Model (k = 15; tau^2 estimator: HE)

logLik	deviance	AIC	BIC	AICc
-13.4344	35.4902	32.8688	34.9929	35.0506

7. Th2 — Immediate Post-Exercise

```
tau^2 (estimated amount of residual heterogeneity):    0.3252 (SE = 0.1663)
tau (square root of estimated tau^2 value):           0.5702
I^2 (residual heterogeneity / unaccounted variability): 79.32%
H^2 (unaccounted variability / sampling variability):  4.84
R^2 (amount of heterogeneity accounted for):           0.00%
```

Test for Residual Heterogeneity:

QE(df = 13) = 53.3069, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 13) = 0.5003, p-val = 0.4919

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.5088	0.2399	2.1212	13	0.0537	-0.0094	
duration	-0.0020	0.0028	-0.7073	13	0.4919	-0.0079	
intrcpt	1.0270						
duration	0.0040						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

7.7.5.4. Dose

Mixed-Effects Model (k = 14; tau^2 estimator: HE)

logLik	deviance	AIC	BIC	AICc
-12.4175	32.6531	30.8349	32.7521	33.2349

```
tau^2 (estimated amount of residual heterogeneity):    0.3219 (SE = 0.1725)
tau (square root of estimated tau^2 value):           0.5673
I^2 (residual heterogeneity / unaccounted variability): 78.97%
H^2 (unaccounted variability / sampling variability):  4.76
R^2 (amount of heterogeneity accounted for):           3.33%
```

Test for Residual Heterogeneity:

QE(df = 12) = 47.3298, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 12) = 1.0162, p-val = 0.3333

Model Results:

```

      estimate      se      tval  df      pval      ci.lb
intrcpt    0.5419  0.2552   2.1233  12  0.0552  -0.0142
dose      -0.0005  0.0005  -1.0080  12  0.3333  -0.0015
      ci.ub
intrcpt    1.0980  .
dose       0.0005

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Note

Not significant

7.7.5.5. IFN

Mixed-Effects Model (k = 7; tau² estimator: HE)

```

logLik deviance      AIC      BIC      AICc
-3.8105  14.8706  13.6209  13.4587  21.6209

```

```

tau^2 (estimated amount of residual heterogeneity):    0.1830 (SE = 0.1575)
tau (square root of estimated tau^2 value):           0.4277
I^2 (residual heterogeneity / unaccounted variability): 75.67%
H^2 (unaccounted variability / sampling variability):  4.11
R^2 (amount of heterogeneity accounted for):           0.00%

```

Test for Residual Heterogeneity:

QE(df = 5) = 15.1720, p-val = 0.0097

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 0.1169, p-val = 0.7463

Model Results:

```

      estimate      se      tval  df      pval      ci.lb
intrcpt    0.7247  0.8225   0.8811   5  0.4186  -1.3896
IFN        -0.1511  0.4421  -0.3419   5  0.7463  -1.2876
      ci.ub
intrcpt    2.8389
IFN        0.9853

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

i Note

Not significant

7.7.5.6. IL_4Mixed-Effects Model (k = 9; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-7.3445	22.0122	20.6889	21.2806	25.4889

tau² (estimated amount of residual heterogeneity): 0.3176 (SE = 0.2182)
 tau (square root of estimated tau² value): 0.5636
 I² (residual heterogeneity / unaccounted variability): 82.89%
 H² (unaccounted variability / sampling variability): 5.84
 R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 7) = 27.0308, p-val = 0.0003

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 7) = 0.3834, p-val = 0.5554

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.5539	0.2387	2.3206	7	0.0534	-0.0105	
IL_4	0.1255	0.2026	0.6192	7	0.5554	-0.3536	
intrcpt	1.1183						
IL_4	0.6045						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

7.7.5.7. IL_6Mixed-Effects Model (k = 14; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-11.6641	33.3733	29.3282	31.2454	31.7282

tau² (estimated amount of residual heterogeneity): 0.3033 (SE = 0.1621)


```

tau (square root of estimated tau^2 value):      0.5507
I^2 (residual heterogeneity / unaccounted variability): 81.28%
H^2 (unaccounted variability / sampling variability):  5.34
R^2 (amount of heterogeneity accounted for):      8.75%

```

Test for Residual Heterogeneity:

```
QE(df = 12) = 41.9420, p-val < .0001
```

Test of Moderators (coefficient 2):

```
F(df1 = 1, df2 = 12) = 1.9487, p-val = 0.1880
```

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.5182	0.1867	2.7751	12	0.0168	0.1114	
IL_6	-0.1428	0.1023	-1.3960	12	0.1880	-0.3656	
intrcpt	0.9251	*					
IL_6	0.0801						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

7.7.5.8. IL_10

Mixed-Effects Model (k = 11; tau^2 estimator: HE)

logLik	deviance	AIC	BIC	AICc
-10.9326	30.8268	27.8652	29.0589	31.2937

```

tau^2 (estimated amount of residual heterogeneity):  0.4665 (SE = 0.2620)
tau (square root of estimated tau^2 value):          0.6830
I^2 (residual heterogeneity / unaccounted variability): 87.31%
H^2 (unaccounted variability / sampling variability):  7.88
R^2 (amount of heterogeneity accounted for):          0.00%

```

Test for Residual Heterogeneity:

```
QE(df = 9) = 46.2625, p-val < .0001
```

Test of Moderators (coefficient 2):

```
F(df1 = 1, df2 = 9) = 0.3720, p-val = 0.5570
```

Model Results:

7. Th2 — Immediate Post-Exercise

```

      estimate      se      tval  df      pval      ci.lb
intrcpt    0.4614  0.2143   2.1530   9  0.0597  -0.0234
IL_10     -0.3677  0.6029  -0.6099   9  0.5570  -1.7315
      ci.ub
intrcpt    0.9461  .
IL_10     0.9961

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note

Not significant

7.7.5.9. IL_2

Mixed-Effects Model (k = 7; tau² estimator: HE)

```

logLik  deviance      AIC      BIC      AICc
-3.9212  15.0921   13.8425   13.6802   21.8425

```

```

tau^2 (estimated amount of residual heterogeneity):    0.1931 (SE = 0.1621)
tau (square root of estimated tau^2 value):           0.4394
I^2 (residual heterogeneity / unaccounted variability): 78.08%
H^2 (unaccounted variability / sampling variability):   4.56
R^2 (amount of heterogeneity accounted for):           0.00%

```

Test for Residual Heterogeneity:

QE(df = 5) = 15.3694, p-val = 0.0089

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 0.0000, p-val = 0.9986

Model Results:

```

      estimate      se      tval  df      pval      ci.lb
intrcpt    0.4500  0.7121  0.6319   5  0.5552  -1.3805
IL_2       0.0007  0.3851  0.0019   5  0.9986  -0.9892
      ci.ub
intrcpt    2.2806
IL_2       0.9906

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note

Not significant

7.8. Multivariate Models

7.8.1. Intensity * Duration

Mixed-Effects Model (k = 14; tau² estimator: HE)

```
tau^2 (estimated amount of residual heterogeneity):    0.2579 (SE = 0.1595)
tau (square root of estimated tau^2 value):           0.5079
I^2 (residual heterogeneity / unaccounted variability): 75.59%
H^2 (unaccounted variability / sampling variability):  4.10
R^2 (amount of heterogeneity accounted for):           22.54%
```

Test for Residual Heterogeneity:

QE(df = 10) = 34.9688, p-val = 0.0001

Test of Moderators (coefficients 2:4):

F(df1 = 3, df2 = 10) = 1.6514, p-val = 0.2396

Model Results:

	estimate	se	tval	df
intrcpt	2.1195	0.8475	2.5008	10
intensityVigorous	-1.9999	1.7640	-1.1337	10
duration	-0.0926	0.0456	-2.0281	10
intensityVigorous:duration	0.0928	0.0470	1.9753	10
	pval	ci.lb	ci.ub	
intrcpt	0.0314	0.2311	4.0079	*
intensityVigorous	0.2834	-5.9302	1.9305	
duration	0.0700	-0.1943	0.0091	.
intensityVigorous:duration	0.0765	-0.0119	0.1974	.

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

7.8.2. IL_10 * IL_4

Mixed-Effects Model (k = 9; tau² estimator: HE)

```
tau^2 (estimated amount of residual heterogeneity):    0.3863 (SE = 0.3079)
tau (square root of estimated tau^2 value):           0.6215
I^2 (residual heterogeneity / unaccounted variability): 83.42%
H^2 (unaccounted variability / sampling variability):  6.03
R^2 (amount of heterogeneity accounted for):           0.00%
```

Test for Residual Heterogeneity:

QE(df = 5) = 23.7108, p-val = 0.0002

Test of Moderators (coefficients 2:4):

F(df1 = 3, df2 = 5) = 0.3848, p-val = 0.7691

Model Results:

	estimate	se	tval	df	pval	ci.lb
intrcpt	0.3992	0.3244	1.2307	5	0.2732	-0.4346
IL_10	-0.5850	0.7088	-0.8253	5	0.4468	-2.4069
IL_4	0.1813	0.2347	0.7725	5	0.4747	-0.4220
IL_10:IL_4	1.1806	1.5451	0.7641	5	0.4793	-2.7911
	ci.ub					
intrcpt	1.2330					
IL_10	1.2370					
IL_4	0.7845					
IL_10:IL_4	5.1524					

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

8. Th2 — ~2 h Post-Exercise

8.1. Data Import

The ~2 h window (effect_size/Th2/pre_post+2h/Th2_pre_post+2h_r_050.csv) is reported for the main analysis, outlier check, funnel, and forest only; the limited data do not support moderator modelling. No study is excluded a priori.

8.2. Primary Meta-Analysis

8.2.1. {meta}

Number of studies: k = 5

	SMD	95% CI	t
Random effects model (HK)	0.0744	[-0.0938; 0.2426]	1.23
Prediction interval		[-0.3288; 0.4775]	
	p-value		
Random effects model (HK)	0.2869		
Prediction interval			

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0$ [0.0000; 0.0495]; $\tau = 0$ [0.0000; 0.2226]
 $I^2 = 0.0\%$ [0.0%; 79.2%]; $H = 1.00$ [1.00; 2.19]

Test of heterogeneity:

Q	d.f.	p-value
0.70	4	0.9518

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 4)
- Prediction interval based on t-distribution (df = 4)

8.2.2. {metafor}

Random-Effects Model (k = 5; τ^2 estimator: HE)

```

tau^2 (estimated amount of total heterogeneity): 0 (SE = 0.0752)
tau (square root of estimated tau^2 value):      0
I^2 (total heterogeneity / total variability):    0.00%
H^2 (total variability / sampling variability):    1.00

```

Test for Heterogeneity:

Q(df = 4) = 0.6963, p-val = 0.9518

Model Results:

```

estimate      se      tval  df      pval      ci.lb      ci.ub
  0.0744  0.0606  1.2277   4  0.2869  -0.0938  0.2426

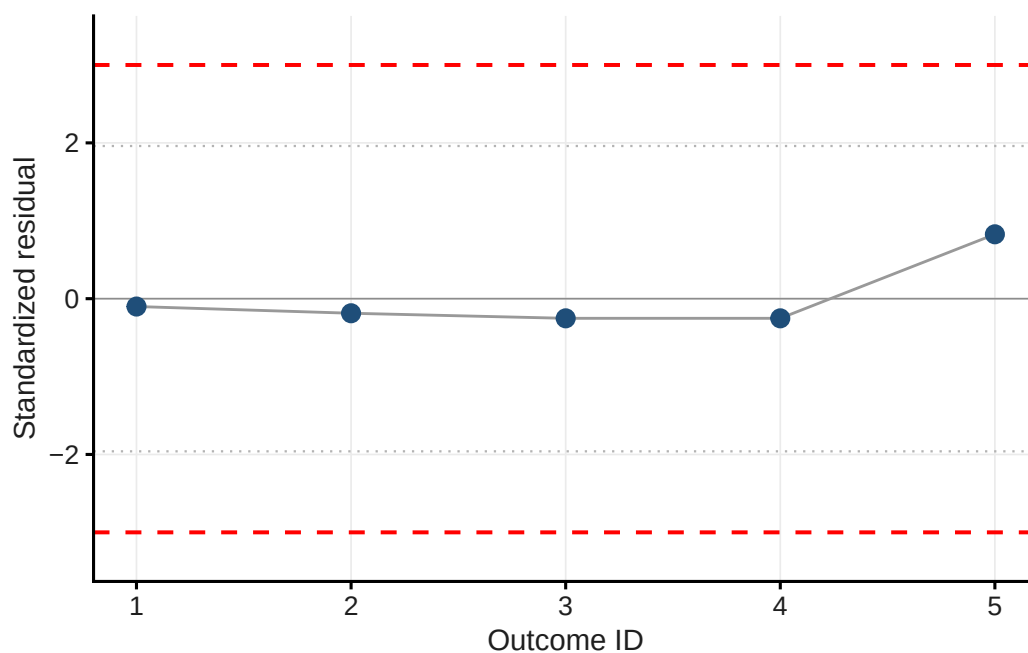
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

8.3. Outlier Detection

Table 8.1.: Standardised residuals

Short_reference	g	std_res
Ibfelt et al. (2002)	0.045	-0.101
Steensberg et al. (2001)	0.025	-0.187
Lancaster et al. (2005)a	0.000	-0.252
Lancaster et al. (2005)b	0.000	-0.252
Lancaster et al. (2005)c	0.329	0.827



i Note

One outcome has standardized residuals around -2 (Kostrzewa-Nowak et al. (2019)).

8.4. Funnel Plot & Influential Cases

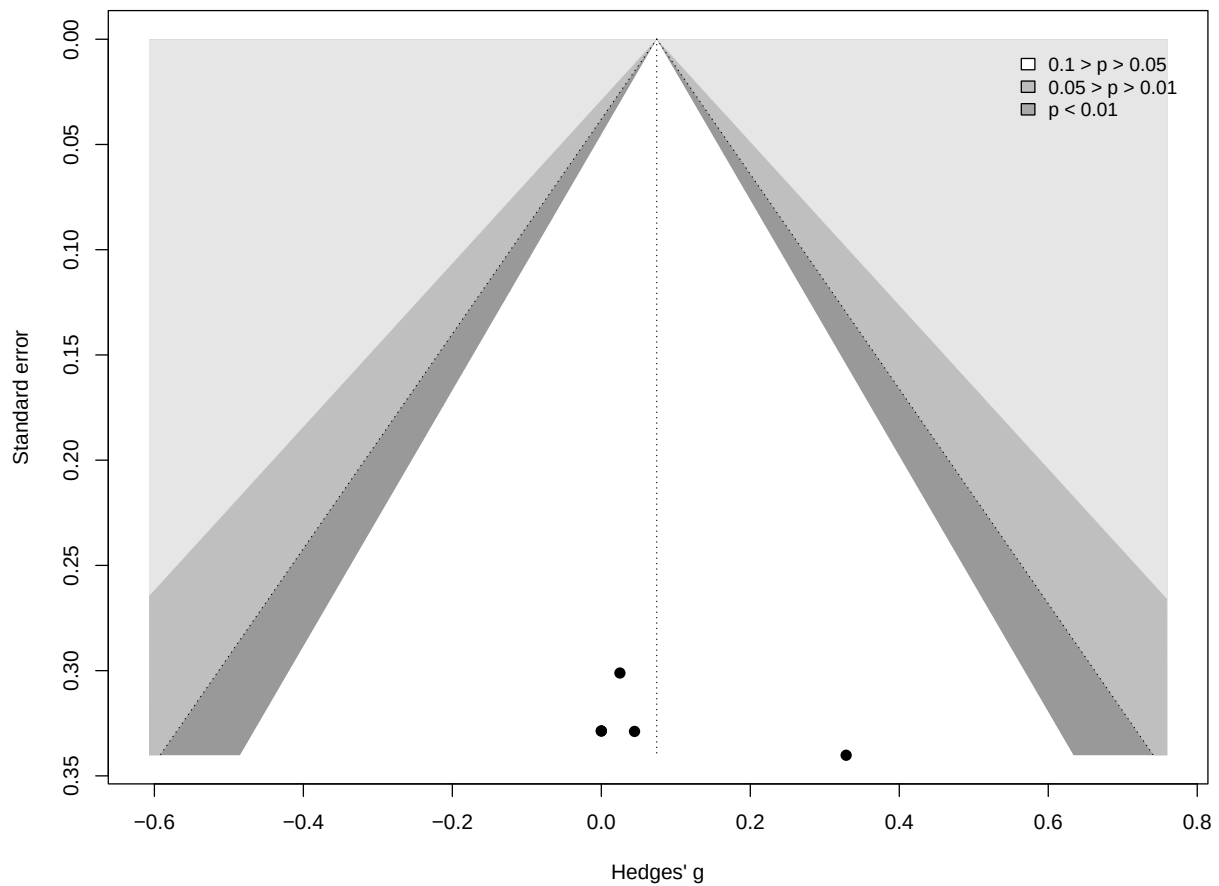


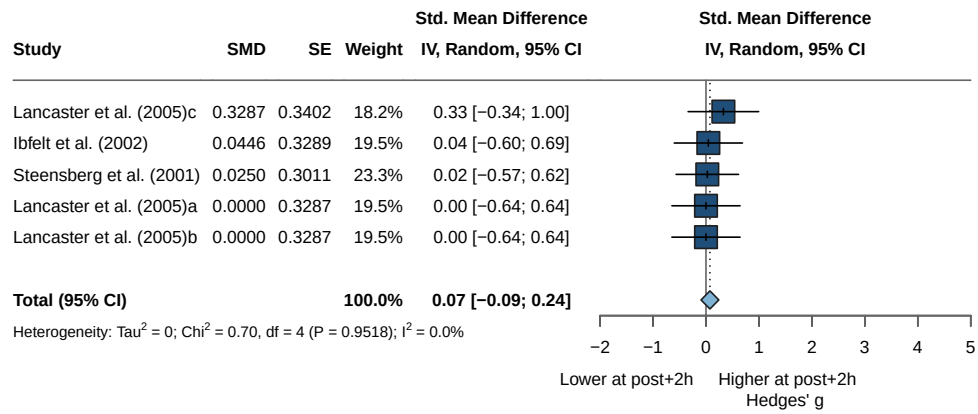
Table 8.2.: DFBETAS influence diagnostics

Short_reference	g	dfbetas
Ibfelt et al. (2002)	0.045	-0.04966493
Steensberg et al. (2001)	0.025	-0.10307622
Lancaster et al. (2005)a	0.000	-0.12421976
Lancaster et al. (2005)b	0.000	-0.12421976
Lancaster et al. (2005)c	0.329	0.39010883

i Note

No influential case.

8.5. Forest Plot



8.6. Asymmetry Test

Note: the number of studies in this window is too small to reliably test for small-study effects (minimum $k = 10$). No Egger's test or trim-and-fill is performed.

9. Th2 — 17 h Post-Exercise

9.1. Data Import

9.2. Primary Meta-Analysis

Kostrzewa-Nowak et al. (2019) is removed before pooling and the model re-fitted.

9.2.1. {meta}

Number of studies: $k = 11$

	SMD	95% CI	t
Random effects model (HK)	0.2468	[-0.4359; 0.9295]	0.81
Prediction interval		[-2.8310; 3.3247]	
	p-value		
Random effects model (HK)	0.4392		
Prediction interval			

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 1.7347$ [0.0804; 4.1148]; $\tau = 1.3171$ [0.2835; 2.0285]
 $I^2 = 72.2\%$ [48.9%; 84.9%]; $H = 1.90$ [1.40; 2.57]

Test of heterogeneity:

Q	d.f.	p-value
35.94	10	< 0.0001

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 10)
- Prediction interval based on t-distribution (df = 10)

9.2.2. {metafor}

Random-Effects Model ($k = 11$; τ^2 estimator: HE)

τ^2 (estimated amount of total heterogeneity): 1.7347 (SE = 0.9828)
 τ (square root of estimated τ^2 value): 1.3171

I^2 (total heterogeneity / total variability): 96.28%
 H^2 (total variability / sampling variability): 26.85

Test for Heterogeneity:
 $Q(df = 10) = 35.9359$, $p\text{-val} < .0001$

Model Results:

estimate	se	tval	df	pval	ci.lb	ci.ub
0.2468	0.3064	0.8056	10	0.4392	-0.4359	0.9295

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

9.2.3. Adapted Model {meta}

Number of studies: $k = 10$

	SMD	95% CI	t
Random effects model (HK)	0.4009	[0.0582; 0.7436]	2.65
Prediction interval		[-0.6056; 1.4073]	
	p-value		
Random effects model (HK)	0.0266		
Prediction interval			

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.1740$ [0.0376; 0.7263]; $\tau = 0.4171$ [0.1939; 0.8522]
 $I^2 = 69.1\%$ [40.3%; 83.9%]; $H = 1.80$ [1.29; 2.50]

Test of heterogeneity:

Q	d.f.	p-value
29.08	9	0.0006

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model ($df = 9$)
- Prediction interval based on t-distribution ($df = 9$)

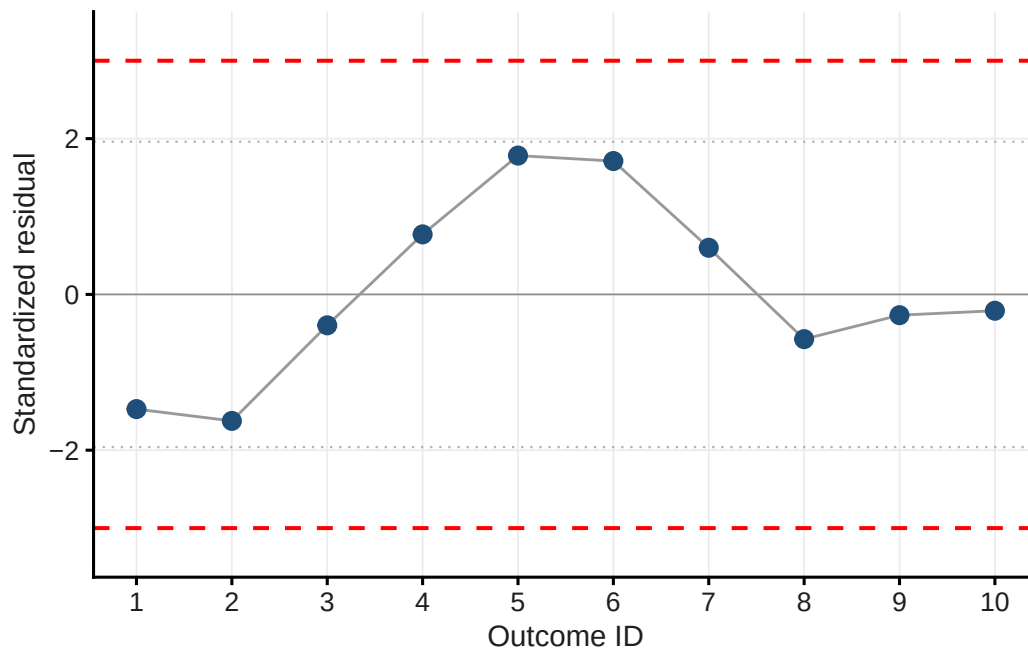
9.3. Outlier Detection

Table 9.1.: Standardised residuals

Short_reference	g	std_res
Kostrzewa-Nowak & Nowak (2018)a	-0.199	-1.472
Kostrzewa-Nowak & Nowak (2018)b	-0.250	-1.624

Table 9.1.: Standardised residuals

Short_reference	g	std_res
Kostrzewa-Nowak & Nowak (2020a)a	0.233	-0.397
Kostrzewa-Nowak & Nowak (2020a)b	0.707	0.771
Kostrzewa-Nowak & Nowak (2020b)a	1.162	1.782
Kostrzewa-Nowak & Nowak (2020b)b	1.133	1.712
Kostrzewa-Nowak & Nowak (2020b)c	0.664	0.599
Kostrzewa-Nowak & Nowak (2020b)d	0.150	-0.574
Kostrzewa-Nowak & Nowak (2020b)e	0.282	-0.267
Steensberg et al. (2001)	0.300	-0.210



i Note

No outlier.

9.4. Funnel Plot & Influential Cases

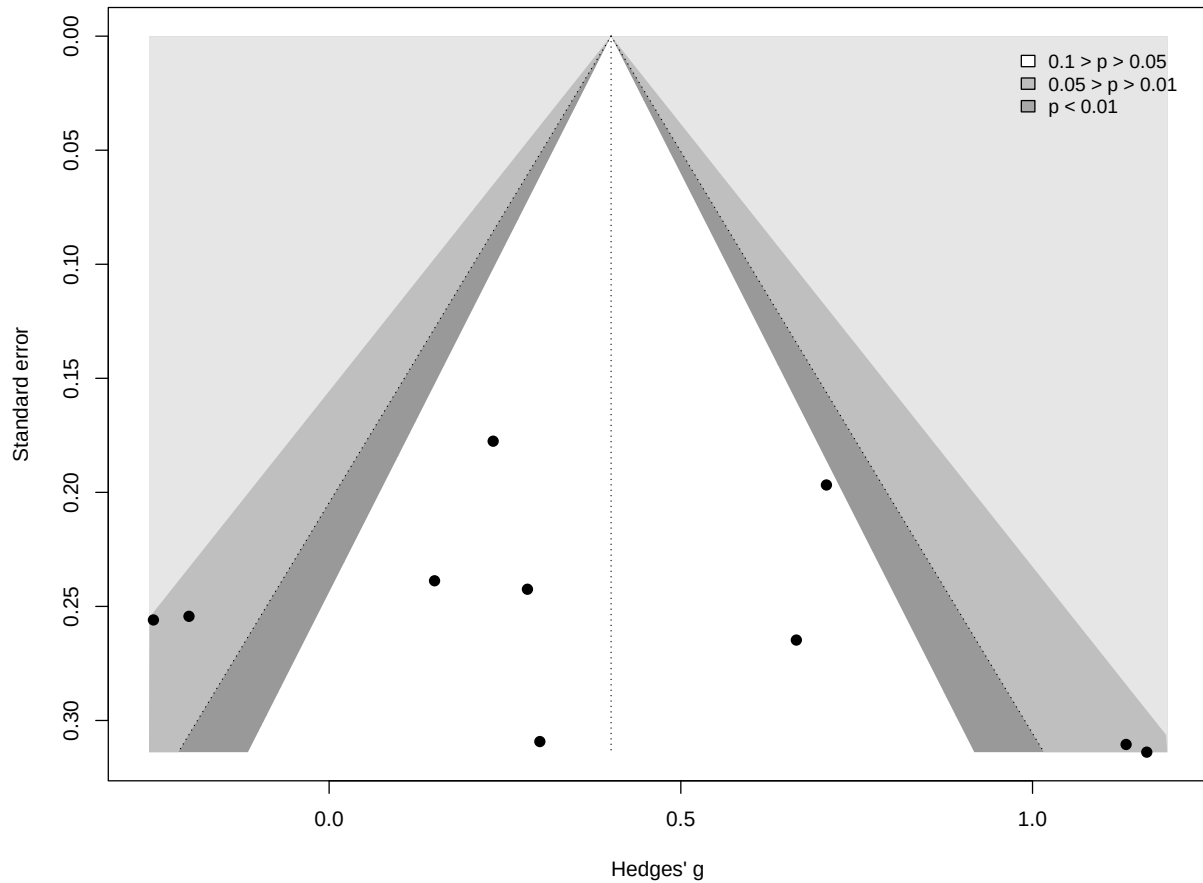


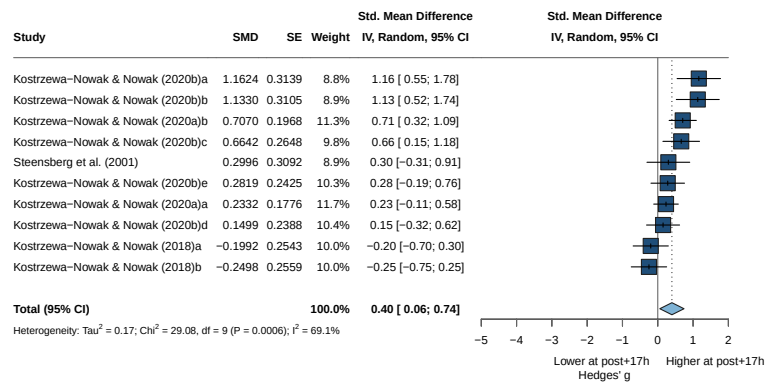
Table 9.2.: DFBETAS influence diagnostics

Short_reference	g	dfbetas
Kostrzewa-Nowak & Nowak (2018)a	-0.199	-0.7487610
Kostrzewa-Nowak & Nowak (2018)b	-0.250	-0.8040583
Kostrzewa-Nowak & Nowak (2020a)a	0.233	-0.4154425
Kostrzewa-Nowak & Nowak (2020a)b	0.707	0.7911110
Kostrzewa-Nowak & Nowak (2020b)a	1.162	0.6594062
Kostrzewa-Nowak & Nowak (2020b)b	1.133	0.6497221
Kostrzewa-Nowak & Nowak (2020b)c	0.664	0.3526656
Kostrzewa-Nowak & Nowak (2020b)d	0.150	-0.3340006
Kostrzewa-Nowak & Nowak (2020b)e	0.282	-0.1299465
Steensberg et al. (2001)	0.300	-0.0613988

Note

No influential case.

9.5. Forest Plot



9.6. Asymmetry Test

Linear regression test of funnel plot asymmetry

Test result: $t = 0.76$, $df = 8$, $p\text{-value} = 0.4679$

Bias estimate: 2.3883 (SE = 3.1343)

Details:

- multiplicative residual heterogeneity variance ($\tau^2 = 3.3889$)
- predictor: standard error
- weight: inverse variance
- reference: Egger et al. (1997), BMJ

i Note

No asymmetry -> no trim-and-fill

9.7. Moderator Analyses

9.7.1. Risk of Bias

9.7.1.1. Model

Number of studies: $k = 10$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.1740$ [0.0376; 0.7263]; $\tau = 0.4171$ [0.1939; 0.8522]

$I^2 = 69.1\%$ [40.3%; 83.9%]; $H = 1.80$ [1.29; 2.50]

Test of heterogeneity:

Q	d.f.	p-value
29.08	9	0.0006

Results for subgroups (random effects model (HK)):

	k	SMD	95% CI	τ^2	τ
RoB = Moderate	3	-0.0845	[-0.7928; 0.6239]	0.0169	0.1301
RoB = Serious	2	0.4625	[-2.5460; 3.4710]	0.0771	0.2777
RoB = Low	5	0.6449	[0.0667; 1.2230]	0.1432	0.3785

	Q	I^2
RoB = Moderate	2.16	7.5%
RoB = Serious	3.20	68.7%
RoB = Low	11.47	65.1%

Test for subgroup differences (random effects model (HK)):

	Q	d.f.	p-value
Between groups	8.52	2	0.0141

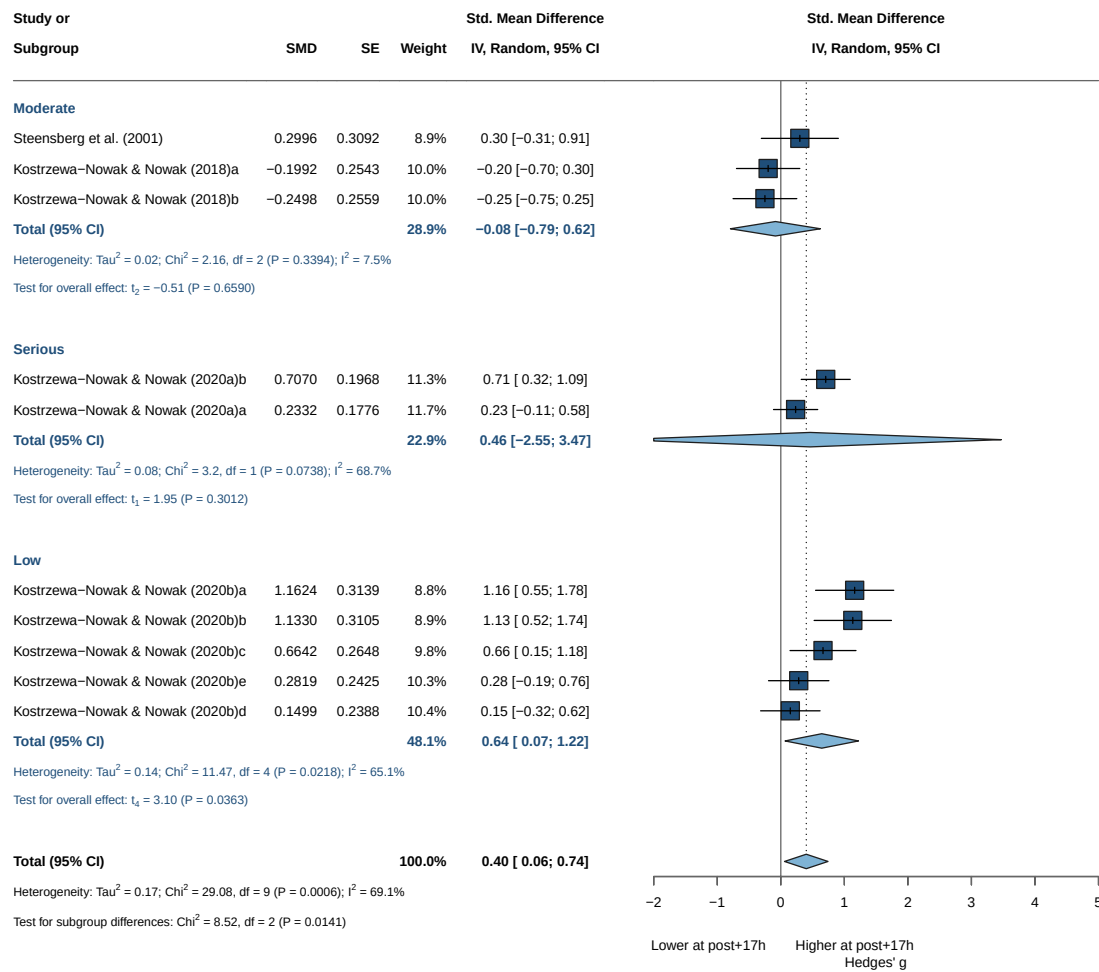
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

i Note

Not significant results

9.7.1.2. Forrest Plot



Sex: all retained arms are male, so no sex subgroup analysis is performed.

9.7.2. Intensity

Number of studies: $k = 10$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.1740$ [0.0376; 0.7263]; $\tau = 0.4171$ [0.1939; 0.8522]

$I^2 = 69.1\%$ [40.3%; 83.9%]; $H = 1.80$ [1.29; 2.50]

Test of heterogeneity:

Q	d.f.	p-value
29.08	9	0.0006

Results for subgroups (random effects model (HK)):

	k	SMD	95% CI	τ^2
intensity = Maximal	9	0.4129	[0.0242; 0.8016]	0.2053
intensity = Vigorous	1	0.2996	[-0.3065; 0.9057]	--
		τ	Q	I^2
intensity = Maximal		0.4531	29.02	72.4%
intensity = Vigorous		--	0.00	--

Test for subgroup differences (random effects model (HK)):

	Q d.f.	p-value
Between groups	0.10 1	0.7477

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

i Note

Not significant but 10 vs 1

9.7.3. Continuous Moderators (Meta-Regression)

9.7.3.1. Age

Mixed-Effects Model (k = 10; τ^2 estimator: HE)

logLik	deviance	AIC	BIC	AICc
-6.5029	22.1591	19.0057	19.9135	23.0057

τ^2 (estimated amount of residual heterogeneity):	0.2063 (SE = 0.1360)
τ (square root of estimated τ^2 value):	0.4542
I^2 (residual heterogeneity / unaccounted variability):	77.93%
H^2 (unaccounted variability / sampling variability):	4.53
R^2 (amount of heterogeneity accounted for):	0.00%

Test for Residual Heterogeneity:

QE(df = 8) = 28.7965, p-val = 0.0003

Test of Moderators (coefficient 2):

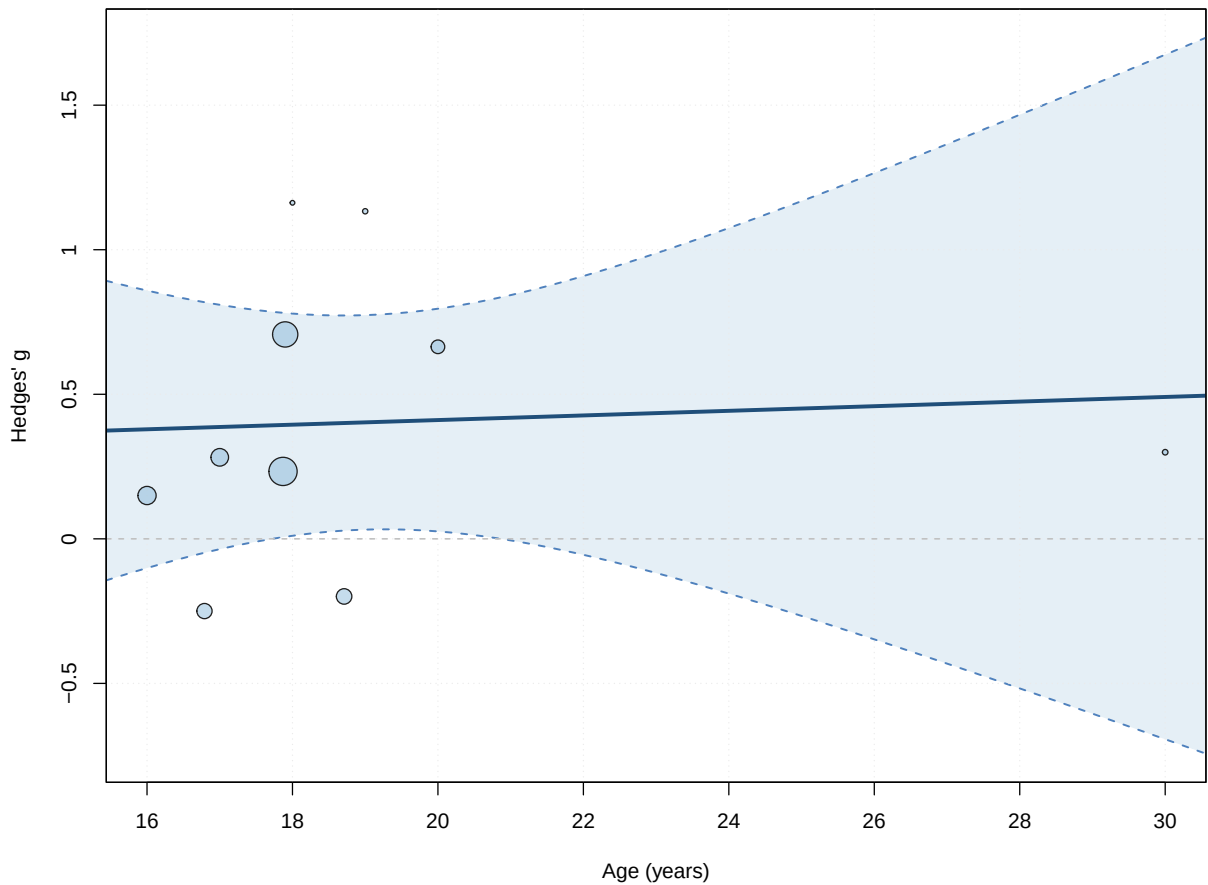
F(df1 = 1, df2 = 8) = 0.0326, p-val = 0.8613

Model Results:

	estimate	se	tval	df	pval	ci.lb
intrcpt	0.2511	0.8554	0.2936	8	0.7766	-1.7214
age	0.0080	0.0443	0.1805	8	0.8613	-0.0941
	ci.ub					
intrcpt	2.2236					
age	0.1100					

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note
Not significant



9.7.3.2. BMI

Mixed-Effects Model (k = 9; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-2.3155	13.2751	10.6311	11.2227	15.4311

tau ² (estimated amount of residual heterogeneity):	0.0517 (SE = 0.0618)
tau (square root of estimated tau ² value):	0.2273
I ² (residual heterogeneity / unaccounted variability):	47.38%
H ² (unaccounted variability / sampling variability):	1.90
R ² (amount of heterogeneity accounted for):	74.84%

Test for Residual Heterogeneity:
QE(df = 7) = 15.1650, p-val = 0.0339

Test of Moderators (coefficient 2):

9. Th2 — 17 h Post-Exercise

$F(df1 = 1, df2 = 7) = 8.4875, p\text{-val} = 0.0226$

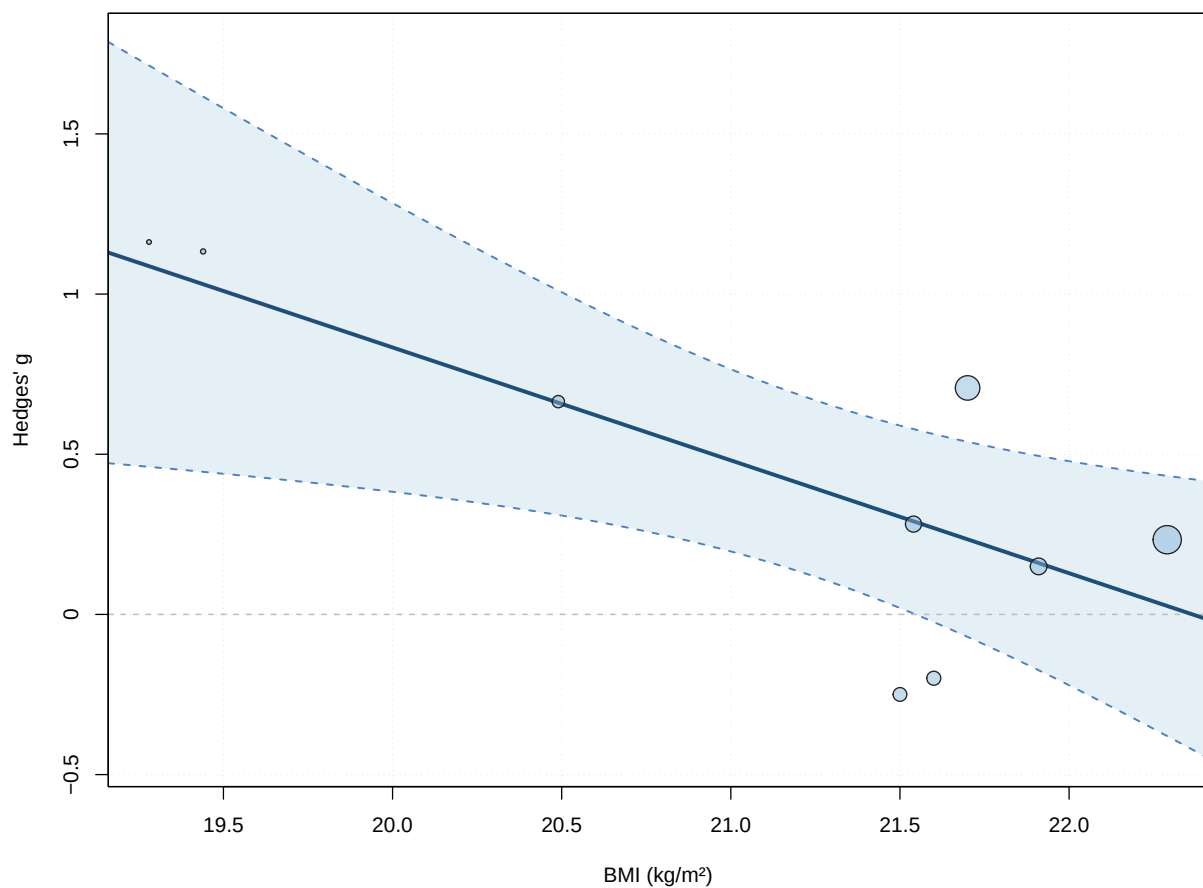
Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	7.8813	2.5727	3.0635	7	0.0182	1.7980	
bmi	-0.3524	0.1210	-2.9133	7	0.0226	-0.6384	
intrcpt	13.9647	*					
bmi	-0.0664	*					

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

! Important

Significant



9.7.3.3. Duration

Mixed-Effects Model ($k = 8$; τ^2 estimator: HE)

logLik	deviance	AIC	BIC	AICc
-5.9984	18.1177	17.9969	18.2352	23.9969

tau² (estimated amount of residual heterogeneity): 0.2675 (SE = 0.1968)
 tau (square root of estimated tau² value): 0.5172
 I² (residual heterogeneity / unaccounted variability): 79.17%
 H² (unaccounted variability / sampling variability): 4.80
 R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 6) = 25.3804, p-val = 0.0003

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 6) = 0.0249, p-val = 0.8798

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.4150	0.2673	1.5525	6	0.1715	-0.2391	
duration	-0.0008	0.0049	-0.1578	6	0.8798	-0.0127	
intrcpt	1.0691						
duration	0.0112						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

9.7.3.4. Dose

Mixed-Effects Model (k = 8; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-5.9984	18.1177	17.9969	18.2352	23.9969

tau² (estimated amount of residual heterogeneity): 0.2675 (SE = 0.1968)
 tau (square root of estimated tau² value): 0.5172
 I² (residual heterogeneity / unaccounted variability): 79.17%
 H² (unaccounted variability / sampling variability): 4.80
 R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 6) = 25.3804, p-val = 0.0003

Test of Moderators (coefficient 2):

$F(df1 = 1, df2 = 6) = 0.0249, p\text{-val} = 0.8798$

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.4198	0.2876	1.4593	6	0.1948	-0.2841	
dose	-0.0001	0.0007	-0.1578	6	0.8798	-0.0019	
intrcpt	1.1236						
dose	0.0017						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

9.7.3.5. IFN

Mixed-Effects Model (k = 7; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-2.4686	11.7933	10.9373	10.7750	18.9373

tau ² (estimated amount of residual heterogeneity):	0.1048 (SE = 0.1101)
tau (square root of estimated tau ² value):	0.3238
I ² (residual heterogeneity / unaccounted variability):	62.72%
H ² (unaccounted variability / sampling variability):	2.68
R ² (amount of heterogeneity accounted for):	5.44%

Test for Residual Heterogeneity:

$QE(df = 5) = 10.3245, p\text{-val} = 0.0665$

Test of Moderators (coefficient 2):

$F(df1 = 1, df2 = 5) = 1.4551, p\text{-val} = 0.2817$

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.0498	0.4627	0.1077	5	0.9184	-1.1396	
IFN	0.1801	0.1493	1.2063	5	0.2817	-0.2037	
intrcpt	1.2392						
IFN	0.5640						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

9.7.3.6. IL_4Mixed-Effects Model (k = 7; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-2.0026	10.8611	10.0052	9.8429	18.0052

tau² (estimated amount of residual heterogeneity): 0.0779 (SE = 0.0935)
 tau (square root of estimated tau² value): 0.2790
 I² (residual heterogeneity / unaccounted variability): 55.08%
 H² (unaccounted variability / sampling variability): 2.23
 R² (amount of heterogeneity accounted for): 29.77%

Test for Residual Heterogeneity:

QE(df = 5) = 10.3727, p-val = 0.0653

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 2.1405, p-val = 0.2033

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.7575	0.1881	4.0262	5	0.0101	0.2739	
IL_4	-0.3349	0.2289	-1.4630	5	0.2033	-0.9234	
intrcpt	1.2411	*					
IL_4	0.2535						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

9.7.3.7. IL_6Mixed-Effects Model (k = 9; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-4.7669	18.1779	15.5338	16.1255	20.3338

tau² (estimated amount of residual heterogeneity): 0.1548 (SE = 0.1178)

9. Th2 — 17 h Post-Exercise

```
tau (square root of estimated tau^2 value):      0.3934
I^2 (residual heterogeneity / unaccounted variability): 72.75%
H^2 (unaccounted variability / sampling variability): 3.67
R^2 (amount of heterogeneity accounted for):      24.62%
```

Test for Residual Heterogeneity:

```
QE(df = 7) = 21.5167, p-val = 0.0031
```

Test of Moderators (coefficient 2):

```
F(df1 = 1, df2 = 7) = 2.7235, p-val = 0.1429
```

Model Results:

	estimate	se	tval	df	pval	ci.lb
intrcpt	0.6354	0.2046	3.1055	7	0.0172	0.1516
IL_6	-0.1053	0.0638	-1.6503	7	0.1429	-0.2562
	ci.ub					
intrcpt	1.1193	*				
IL_6	0.0456					

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

! Important

Significant

9.7.3.8. IL_10

Mixed-Effects Model (k = 9; tau^2 estimator: HE)

logLik	deviance	AIC	BIC	AICc
-5.8893	20.4226	17.7785	18.3702	22.5785

```
tau^2 (estimated amount of residual heterogeneity): 0.2170 (SE = 0.1510)
tau (square root of estimated tau^2 value):         0.4658
I^2 (residual heterogeneity / unaccounted variability): 79.03%
H^2 (unaccounted variability / sampling variability): 4.77
R^2 (amount of heterogeneity accounted for):         0.00%
```

Test for Residual Heterogeneity:

```
QE(df = 7) = 26.1288, p-val = 0.0005
```

Test of Moderators (coefficient 2):

```
F(df1 = 1, df2 = 7) = 0.6854, p-val = 0.4351
```

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.5383	0.2288	2.3529	7	0.0509	-0.0027	
IL_10	-0.1244	0.1503	-0.8279	7	0.4351	-0.4798	
intrcpt	1.0793						
IL_10	0.2309						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

9.7.3.9. IL_2Mixed-Effects Model (k = 7; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-3.2017	13.2593	12.4033	12.2410	20.4033

tau² (estimated amount of residual heterogeneity): 0.1367 (SE = 0.1309)tau (square root of estimated tau² value): 0.3697I² (residual heterogeneity / unaccounted variability): 68.35%H² (unaccounted variability / sampling variability): 3.16R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 5) = 13.5802, p-val = 0.0185

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 0.1962, p-val = 0.6763

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.4754	0.2972	1.5994	5	0.1706	-0.2887	
IL_2	0.0608	0.1372	0.4430	5	0.6763	-0.2919	
intrcpt	1.2395						
IL_2	0.4134						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

9.8. Multivariate Models

Mixed-Effects Model (k = 8; tau² estimator: HE)

```
tau^2 (estimated amount of residual heterogeneity):    0.2675 (SE = 0.1968)
tau (square root of estimated tau^2 value):          0.5172
I^2 (residual heterogeneity / unaccounted variability): 79.17%
H^2 (unaccounted variability / sampling variability):  4.80
R^2 (amount of heterogeneity accounted for):          0.00%
```

Test for Residual Heterogeneity:

QE(df = 6) = 25.3804, p-val = 0.0003

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 6) = 0.0249, p-val = 0.8798

Model Results:

	estimate	se	tval	df	pval
intrcpt	0.3996	0.2176	1.8360	6	0.1160
intensityVigorous	-0.1000	0.6338	-0.1578	6	0.8798
	ci.lb	ci.ub			
intrcpt	-0.1330	0.9322			
intensityVigorous	-1.6510	1.4509			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

Part III.

Th1/Th2 Ratio

10. Th1/Th2 Ratio — Immediate Post-Exercise

10.1. Data Import

The base analysis uses the pre-computed Hedges' g file for the Th1/Th2 ratio at $r = 0.5$ (`effect_size/Ratio/pre_post/Ratio_0.5`). Sensitivity analyses can substitute the $r = 0.3 / 0.4 / 0.6 / 0.7$ files in the same folder. The pipeline column names are mapped to the analysis conventions (`VE_g`, `Sex`, `Age`, `BMI`) by `load_es()`.

10.2. Primary Meta-Analysis

`metagen()` fits a random-effects model on the pre-computed effect sizes (g , SE_g) using the **Hedges–Olkin (HE)** ² estimator and the **Knapp–Hartung** small-sample correction (`hakn = TRUE`). **Kostrzewa-Nowak et al. (2019)** is removed before pooling and the model re-fitted; `metafor::rma()` cross-validates.

10.2.1. Full Model {meta}

Number of studies: $k = 18$

	SMD	95% CI	t
Random effects model (HK)	0.3544	[-0.0597; 0.7685]	1.81
Prediction interval		[-1.7206; 2.4294]	
	p-value		
Random effects model (HK)	0.0887		
Prediction interval			

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.9096$ [0.2029; 1.5996]; $\tau = 0.9537$ [0.4504; 1.2647]
 $I^2 = 80.5\%$ [70.0%; 87.3%]; $H = 2.26$ [1.83; 2.81]

Test of heterogeneity:

Q	d.f.	p-value
87.11	17	< 0.0001

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model ($df = 17$)
- Prediction interval based on t-distribution ($df = 17$)

10.2.2. Full Model {metafor}

Random-Effects Model (k = 18; tau² estimator: HE)

tau² (estimated amount of total heterogeneity): 0.9096 (SE = 0.4257)
 tau (square root of estimated tau² value): 0.9537
 I² (total heterogeneity / total variability): 91.77%
 H² (total variability / sampling variability): 12.15

Test for Heterogeneity:

Q(df = 17) = 87.1136, p-val < .0001

Model Results:

estimate	se	tval	df	pval	ci.lb	ci.ub
0.3544	0.1963	1.8055	17	0.0887	-0.0597	0.7685

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

10.2.3. Adapted Model {meta}

Number of studies: k = 17

	SMD	95% CI	t
Random effects model (HK)	0.3005	[-0.0558; 0.6568]	1.79
Prediction interval		[-1.1139; 1.7149]	

	p-value
Random effects model (HK)	0.0928
Prediction interval	

Quantifying heterogeneity (with 95% CIs):

tau² = 0.4156 [0.1703; 1.0546]; tau = 0.6447 [0.4126; 1.0270]
 I² = 80.6% [69.8%; 87.5%]; H = 2.27 [1.82; 2.83]

Test of heterogeneity:

Q	d.f.	p-value
82.37	16	< 0.0001

Details of meta-analysis methods:

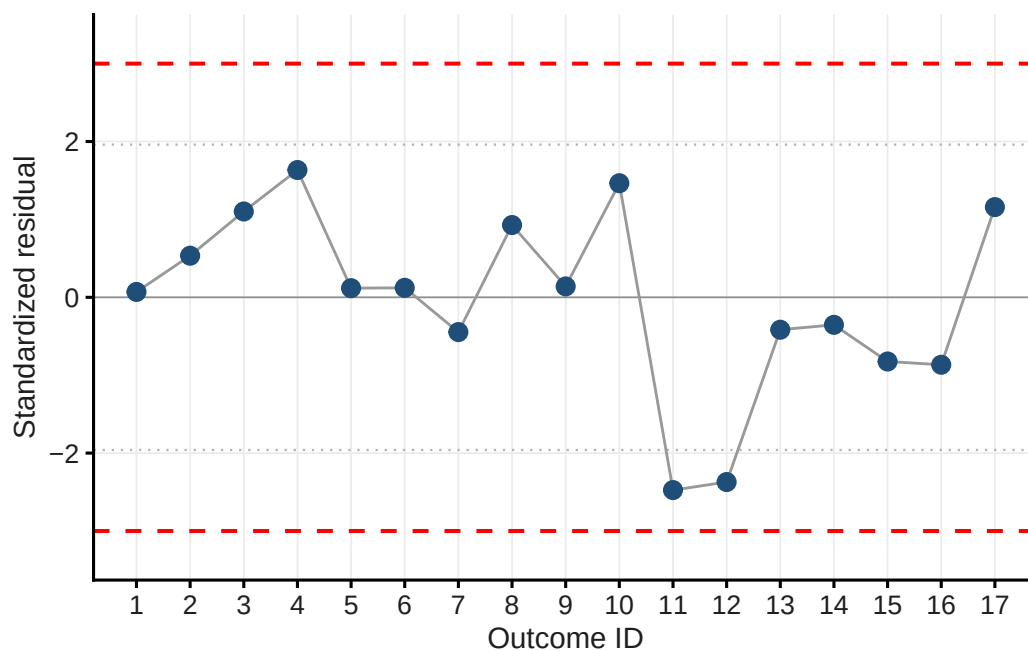
- Inverse variance method
- Hedges estimator for tau²
- Q-Profile method for confidence interval of tau² and tau
- Calculation of I² based on Q
- Hartung-Knapp adjustment for random effects model (df = 16)
- Prediction interval based on t-distribution (df = 16)

10.3. Outlier Detection

Leave-one-out standardised residuals are computed via `metaoutliers()`, which is passed the **variance** (`m5$seTE^2`). Studies with $|z| > 3$ are flagged as outliers; $|z| > 1.96$ warrants inspection.

Table 10.1.: Standardised residuals

Short_reference	g	std_res
Ibfelt et al. (2002)	0.347	0.069
Kostrzewska-Nowak & Nowak (2018)a	0.637	0.534
Kostrzewska-Nowak & Nowak (2018)b	0.993	1.102
Kostrzewska-Nowak & Nowak (2020a)a	1.209	1.634
Kostrzewska-Nowak & Nowak (2020a)b	0.374	0.117
Kostrzewska-Nowak & Nowak (2020b)a	0.378	0.122
Kostrzewska-Nowak & Nowak (2020b)b	0.028	-0.447
Kostrzewska-Nowak & Nowak (2020b)c	0.875	0.928
Kostrzewska-Nowak & Nowak (2020b)d	0.389	0.140
Kostrzewska-Nowak & Nowak (2020b)e	1.206	1.465
Steensberg et al. (2001)	-1.309	-2.475
Tsubakihara et al. (2012)	-0.959	-2.372
Lancaster et al. (2005)a	0.033	-0.416
Lancaster et al. (2005)b	0.073	-0.353
Lancaster et al. (2005)c	-0.226	-0.824
Harbaum et al. (2016)a	-0.215	-0.866
Harbaum et al. (2016)b	1.065	1.158



i Note

No outlier.

10.4. Funnel Plot & Influential Cases

The contour-enhanced funnel plot shows study estimates against their standard errors; shaded regions mark significance zones ($\alpha = 0.10, 0.05, 0.01$). The `dfbetas` diagnostic from `metafor::rma.mv()` identifies **influential cases** — distinct from statistical outliers and not necessarily excluded.

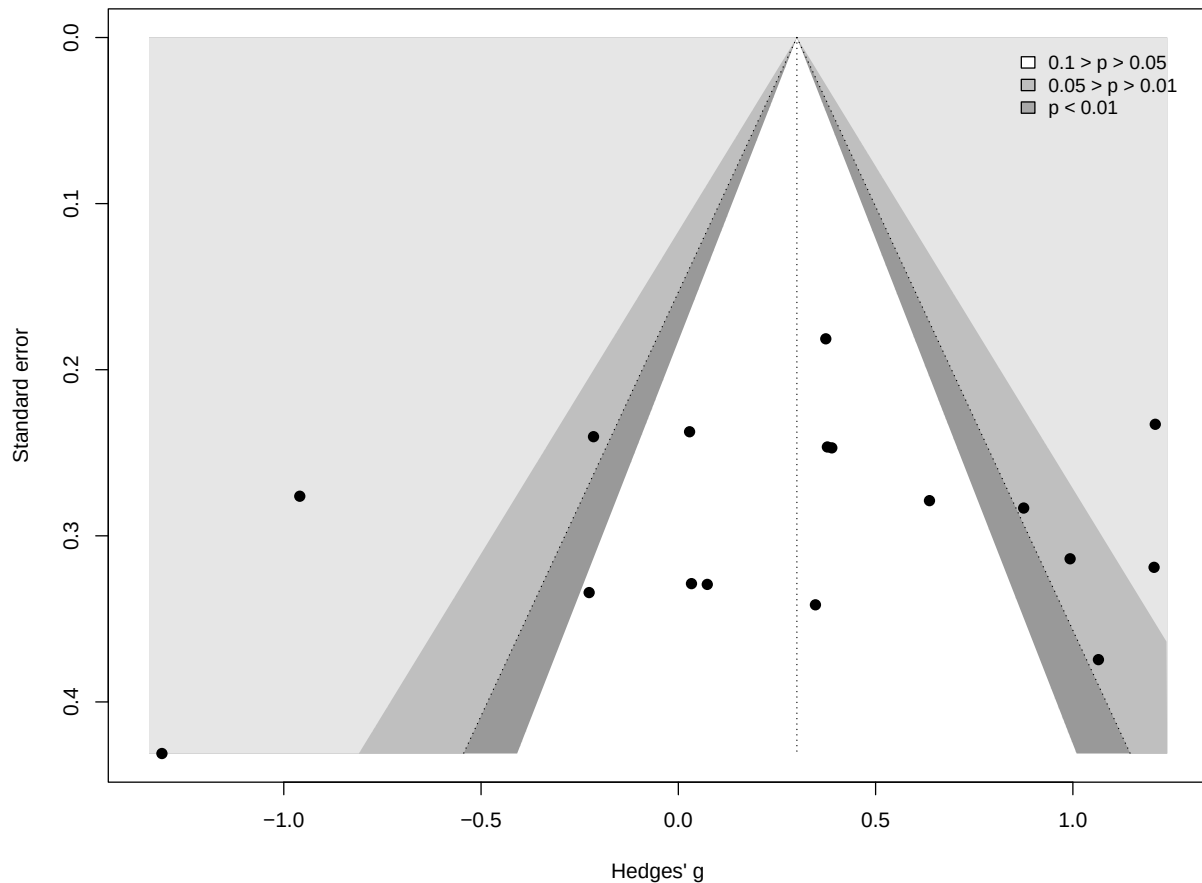


Table 10.2.: DFBETAS influence diagnostics

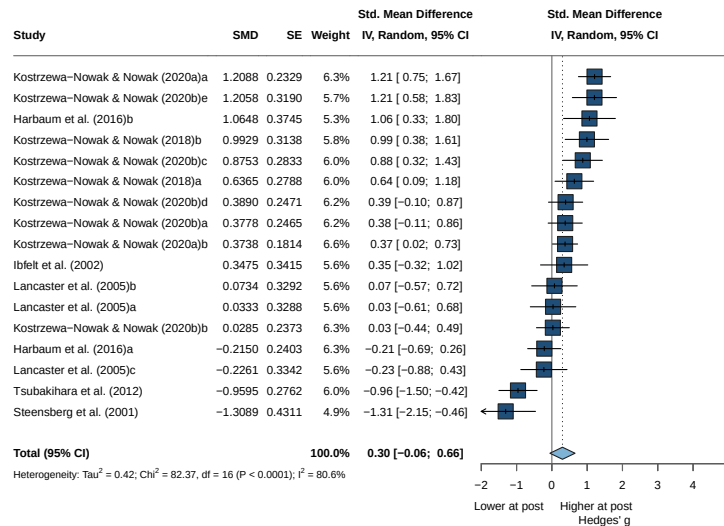
Short_reference	g	dfbetas
Ibfelt et al. (2002)	0.347	0.01004021
Kostrzewska-Nowak & Nowak (2018)a	0.637	0.27911537
Kostrzewska-Nowak & Nowak (2018)b	0.993	0.47102525
Kostrzewska-Nowak & Nowak (2020a)a	1.209	1.18025273
Kostrzewska-Nowak & Nowak (2020a)b	0.374	0.10134265
Kostrzewska-Nowak & Nowak (2020b)a	0.378	0.05596990
Kostrzewska-Nowak & Nowak (2020b)b	0.028	-0.38958693
Kostrzewska-Nowak & Nowak (2020b)c	0.875	0.48056733
Kostrzewska-Nowak & Nowak (2020b)d	0.389	0.06893600
Kostrzewska-Nowak & Nowak (2020b)e	1.206	0.60156479
Steensberg et al. (2001)	-1.309	-0.60449272
Tsubakihara et al. (2012)	-0.959	-1.20154886
Lancaster et al. (2005)a	0.033	-0.19186890
Lancaster et al. (2005)b	0.073	-0.16552634

Table 10.2.: DFBETAS influence diagnostics

Short_reference	g	dfbetas
Lancaster et al. (2005)c	-0.226	-0.34724912
Harbaum et al. (2016)a	-0.215	-0.68475128
Harbaum et al. (2016)b	1.065	0.36143534

One value was above 1 and one below -1 , so both were identified as influential cases. Neither is an outlier; all outcomes are retained and the sources of heterogeneity are examined in the moderator analysis.

10.5. Forest Plot



10.6. Asymmetry Test

Egger's regression test (`method = "linreg"`) evaluates funnel asymmetry as a proxy for small-study/publication effects (one-tailed, $p < .10$). Where asymmetry is suggested, a trim-and-fill analysis estimates potentially missing studies.

Linear regression test of funnel plot asymmetry

Test result: $t = -0.53$, $df = 15$, $p\text{-value} = 0.6026$

Bias estimate: -1.4500 (SE = 2.7260)

Details:

- multiplicative residual heterogeneity variance ($\tau^2 = 5.3895$)
- predictor: standard error
- weight: inverse variance
- reference: Egger et al. (1997), BMJ

i Note

No asymmetry -> no trim-and-fill

10.7. Moderator Analyses

10.7.1. Risk of Bias

Subgroup analysis by overall RoB rating tests whether study quality moderates the pooled effect; the between-subgroup statistic (Q_b) indicates whether subgroup means differ.

10.7.1.1. Model

Number of studies: $k = 17$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.4156$ [0.1703; 1.0546]; $\tau = 0.6447$ [0.4126; 1.0270]
 $I^2 = 80.6\%$ [69.8%; 87.5%]; $H = 2.27$ [1.82; 2.83]

Test of heterogeneity:

Q	d.f.	p-value
82.37	16	< 0.0001

Results for subgroups (random effects model (HK)):

	k	SMD	95% CI	τ^2	τ
RoB = Serious	6	0.2952	[-0.5524; 1.1427]	0.5743	0.7578
RoB = Moderate	3	0.1355	[-2.9084; 3.1793]	1.4143	1.1892
RoB = Low	8	0.3483	[-0.0432; 0.7398]	0.1457	0.3818

	Q	I^2
RoB = Serious	44.98	88.9%
RoB = Moderate	20.00	90.0%
RoB = Low	17.11	59.1%

Test for subgroup differences (random effects model (HK)):

	Q	d.f.	p-value
Between groups	0.10	2	0.9517

Details of meta-analysis methods:

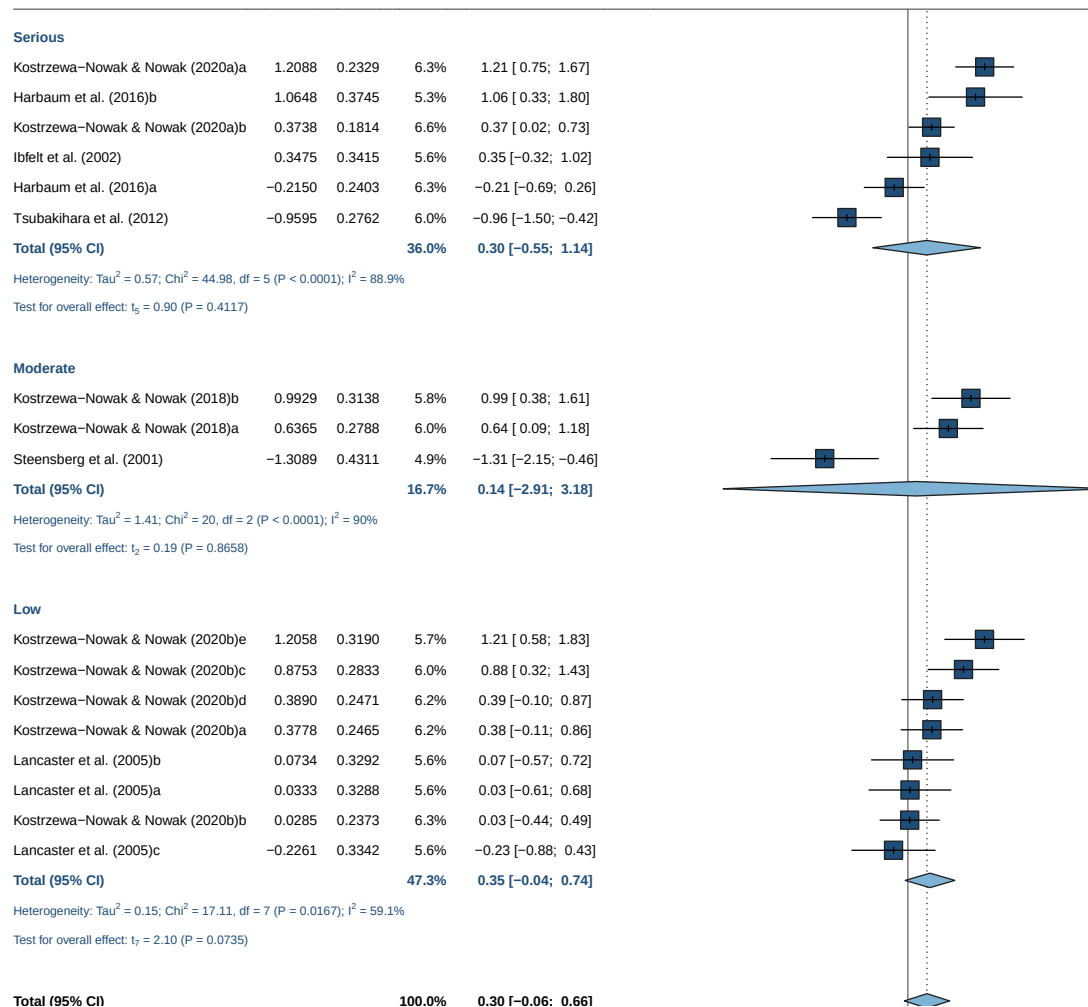
- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model

- Prediction interval based on t-distribution

i Note

Not significant results

10.7.1.2. Forrest Plot



10.7.2. Sex

10.7.2.1. Model

Number of studies: $k = 17$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.4156$ [0.1703; 1.0546]; $\tau = 0.6447$ [0.4126; 1.0270]

$I^2 = 80.6\%$ [69.8%; 87.5%]; $H = 2.27$ [1.82; 2.83]

Test of heterogeneity:

Q	d.f.	p-value
82.37	16	< 0.0001

10. Th1/Th2 Ratio — Immediate Post-Exercise

Results for subgroups (random effects model (HK)):

	k	SMD	95% CI	tau ²	tau
sex = male	14	0.3858	[0.0247; 0.7469]	0.3417	0.5846
sex = female	1	-0.9595	[-1.5008; -0.4182]	--	--
sex = mixed	2	0.3927	[-7.7274; 8.5127]	0.7199	0.8484

	Q	I ²
sex = male	49.15	73.5%
sex = female	0.00	--
sex = mixed	8.27	87.9%

Test for subgroup differences (random effects model (HK)):

	Q	d.f.	p-value
Between groups	17.68	2	0.0001

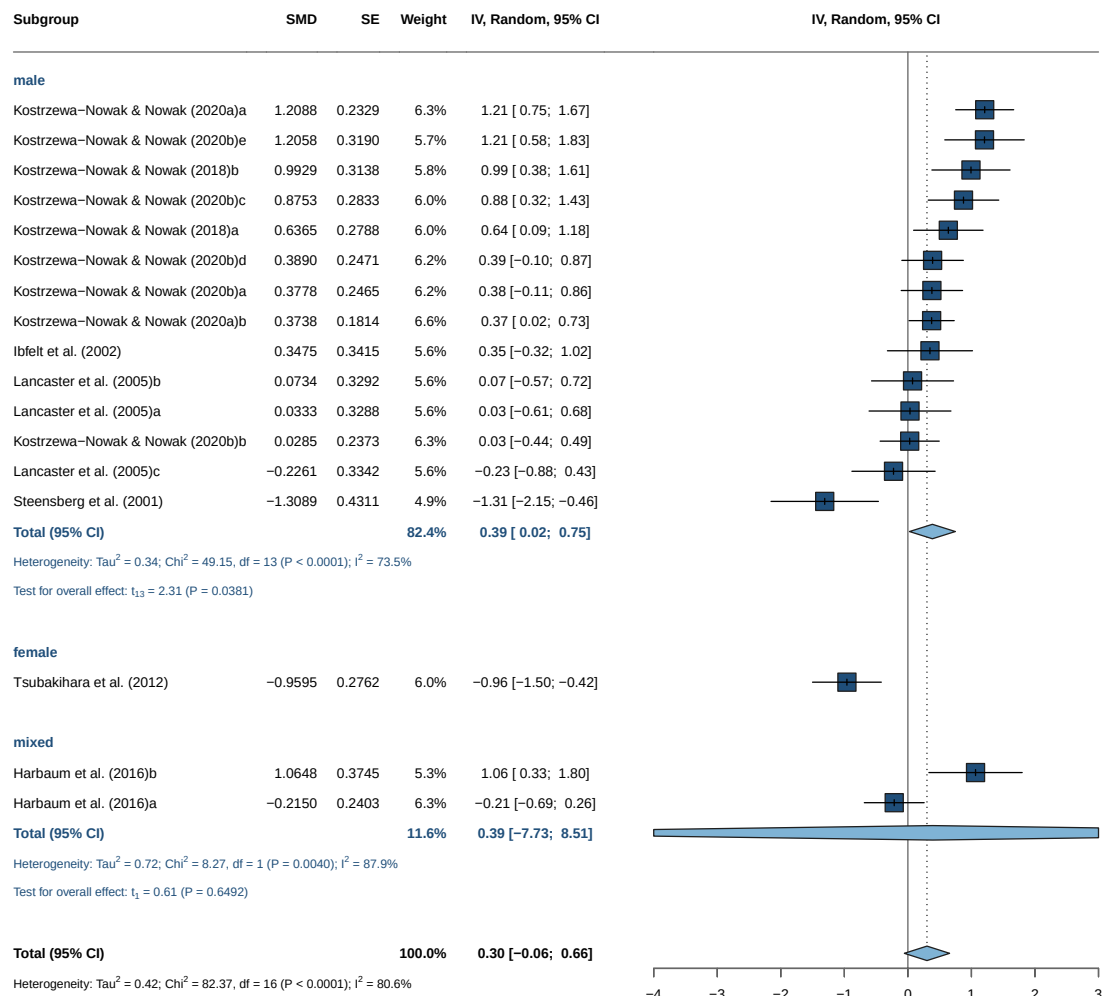
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for tau²
- Q-Profile method for confidence interval of tau² and tau
- Calculation of I² based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

! Important

Significant results but 15 vs 1 vs 2

10.7.2.2. Forrest Plot



10.7.3. Intensity

10.7.3.1. Model

Number of studies: $k = 16$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.3382$ [0.1128; 0.8975]; $\tau = 0.5815$ [0.3358; 0.9474]

$I^2 = 74.7\%$ [58.6%; 84.5%]; $H = 1.99$ [1.56; 2.54]

Test of heterogeneity:

Q	d.f.	p-value
59.18	15	< 0.0001

Results for subgroups (random effects model (HK)):

	k	SMD	95% CI	τ^2
intensity = Vigorous	5	-0.1814	[-0.9439; 0.5811]	0.2883
intensity = Maximal	11	0.5997	[0.2761; 0.9232]	0.1564
		τ	Q	I^2
intensity = Vigorous		0.5370	10.14	60.6%

10. Th1/Th2 Ratio — Immediate Post-Exercise

intensity = Maximal 0.3955 34.33 70.9%

Test for subgroup differences (random effects model (HK)):

Q d.f. p-value
Between groups 6.32 1 0.0119

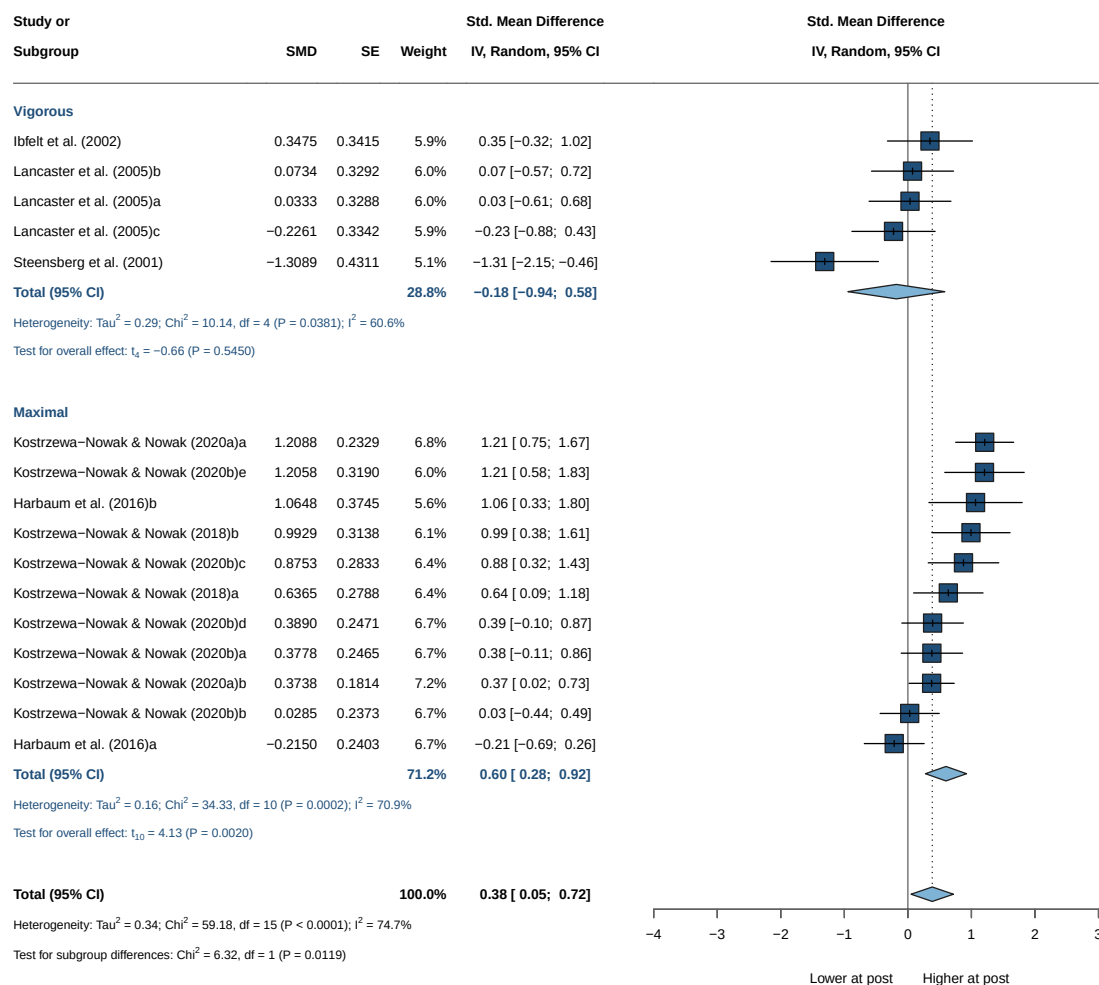
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

! Important

Significant

10.7.3.2. Forrest Plot



10.7.4. Staining

Number of studies: $k = 17$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.4156$ [0.1703; 1.0546]; $\tau = 0.6447$ [0.4126; 1.0270]

$I^2 = 80.6\%$ [69.8%; 87.5%]; $H = 2.27$ [1.82; 2.83]

Test of heterogeneity:

Q d.f. p-value

82.37 16 < 0.0001

Results for subgroups (random effects model (HK)):

	k	SMD	95% CI	τ^2
staining = intra	14	0.3858	[0.0247; 0.7469]	0.3417
staining = extra	3	-0.0594	[-2.5758; 2.4570]	0.9569

	τ	Q	I^2
staining = intra	0.5846	49.15	73.5%
staining = extra	0.9782	18.93	89.4%

Test for subgroup differences (random effects model (HK)):

	Q	d.f.	p-value
Between groups	0.54	1	0.4643

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

Note

Not significant results

10.7.5. Continuous Moderators (Meta-Regression)

Continuous moderators are tested via weighted meta-regression (`metareg()`). A significant slope indicates the moderator explains part of the between-study heterogeneity (τ^2); bubble plots show the fitted line with 95% CI.

10.7.5.1. Age

Mixed-Effects Model ($k = 16$; τ^2 estimator: HE)

logLik	deviance	AIC	BIC	AICc
-14.5497	39.4622	35.0995	37.4173	37.0995

10. Th1/Th2 Ratio — Immediate Post-Exercise

```
tau^2 (estimated amount of residual heterogeneity):    0.3523 (SE = 0.1679)
tau (square root of estimated tau^2 value):           0.5936
I^2 (residual heterogeneity / unaccounted variability): 81.90%
H^2 (unaccounted variability / sampling variability):  5.52
R^2 (amount of heterogeneity accounted for):           0.00%
```

Test for Residual Heterogeneity:

QE(df = 14) = 53.0007, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 14) = 0.7114, p-val = 0.4131

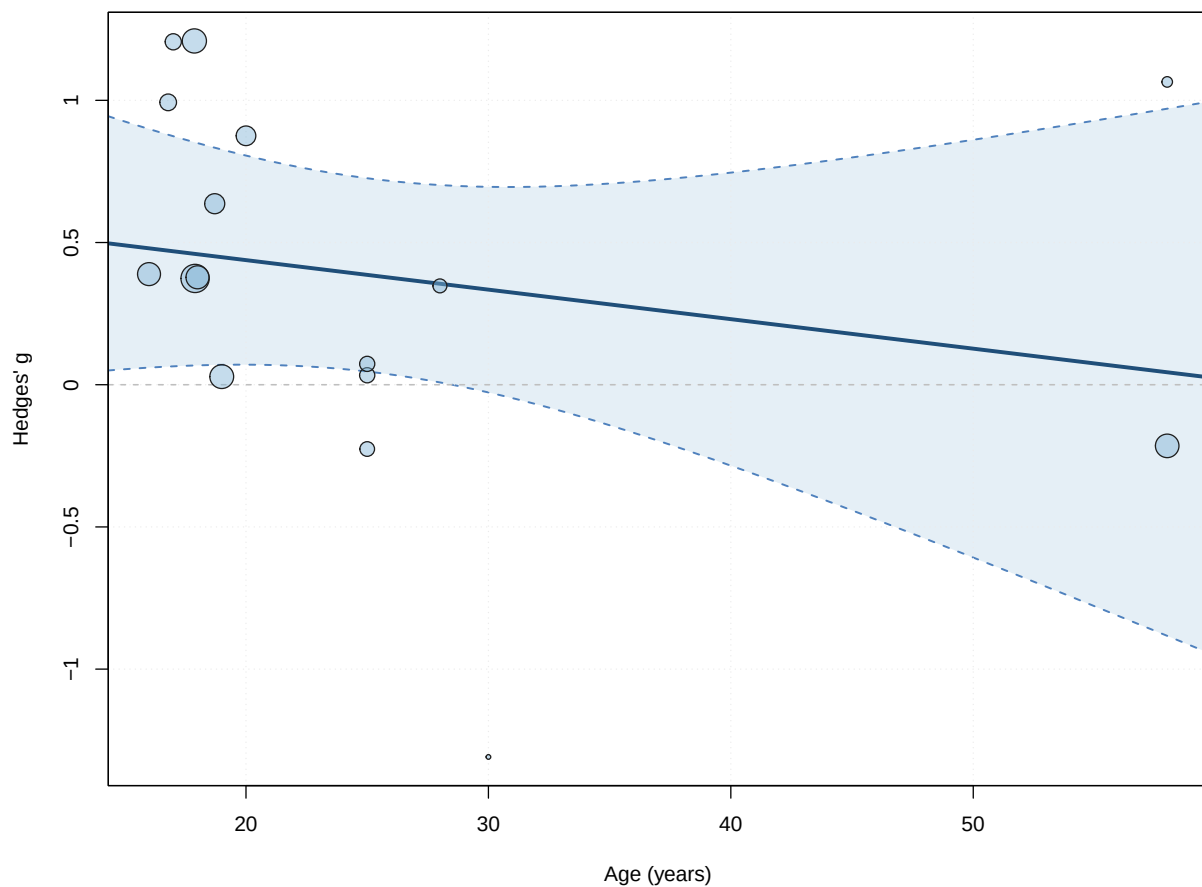
Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.6457	0.3495	1.8473	14	0.0859	-0.1040	
age	-0.0104	0.0123	-0.8435	14	0.4131	-0.0368	
intrcpt	1.3953						
age	0.0160						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant



10.7.5.2. BMI

Mixed-Effects Model ($k = 12$; τ^2 estimator: HE)

logLik	deviance	AIC	BIC	AICc
-11.3566	32.5305	28.7132	30.1679	31.7132

τ^2 (estimated amount of residual heterogeneity): 0.3880 (SE = 0.2073)
 τ (square root of estimated τ^2 value): 0.6229
 I^2 (residual heterogeneity / unaccounted variability): 85.24%
 H^2 (unaccounted variability / sampling variability): 6.77
 R^2 (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 10) = 61.5799, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 10) = 0.0011, p-val = 0.9744

Model Results:

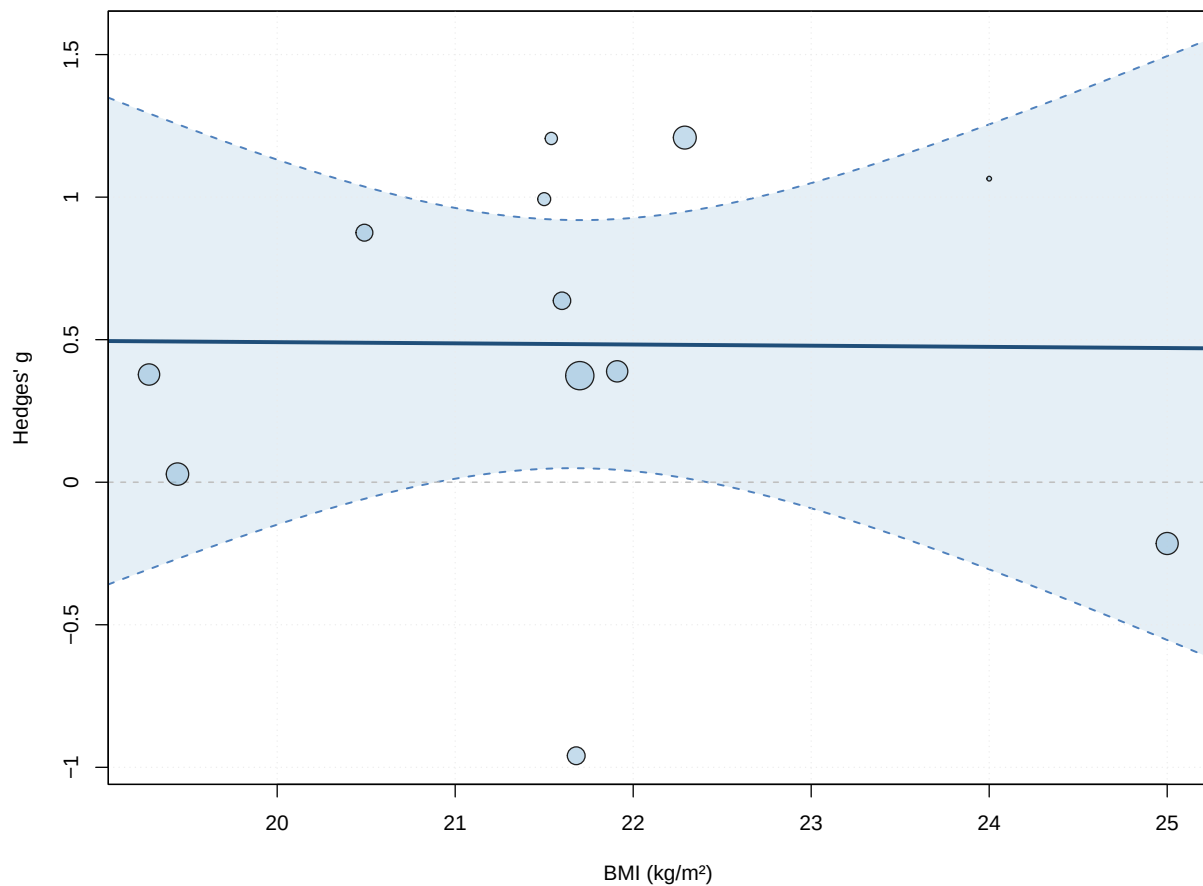
10. Th1/Th2 Ratio — Immediate Post-Exercise

	estimate	se	tval	df	pval	ci.lb
intrcpt	0.5737	2.7249	0.2106	10	0.8375	-5.4977
bmi	-0.0041	0.1254	-0.0329	10	0.9744	-0.2834
	ci.ub					
intrcpt	6.6452					
bmi	0.2752					

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note

Not significant



10.7.5.3. Duration

Mixed-Effects Model (k = 15; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-11.2968	31.0389	28.5936	30.7177	30.7754


```

tau^2 (estimated amount of residual heterogeneity):    0.2107 (SE = 0.1206)
tau (square root of estimated tau^2 value):           0.4590
I^2 (residual heterogeneity / unaccounted variability): 71.01%
H^2 (unaccounted variability / sampling variability):  3.45
R^2 (amount of heterogeneity accounted for):           49.32%

```

Test for Residual Heterogeneity:

QE(df = 13) = 43.6946, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 13) = 9.0460, p-val = 0.0101

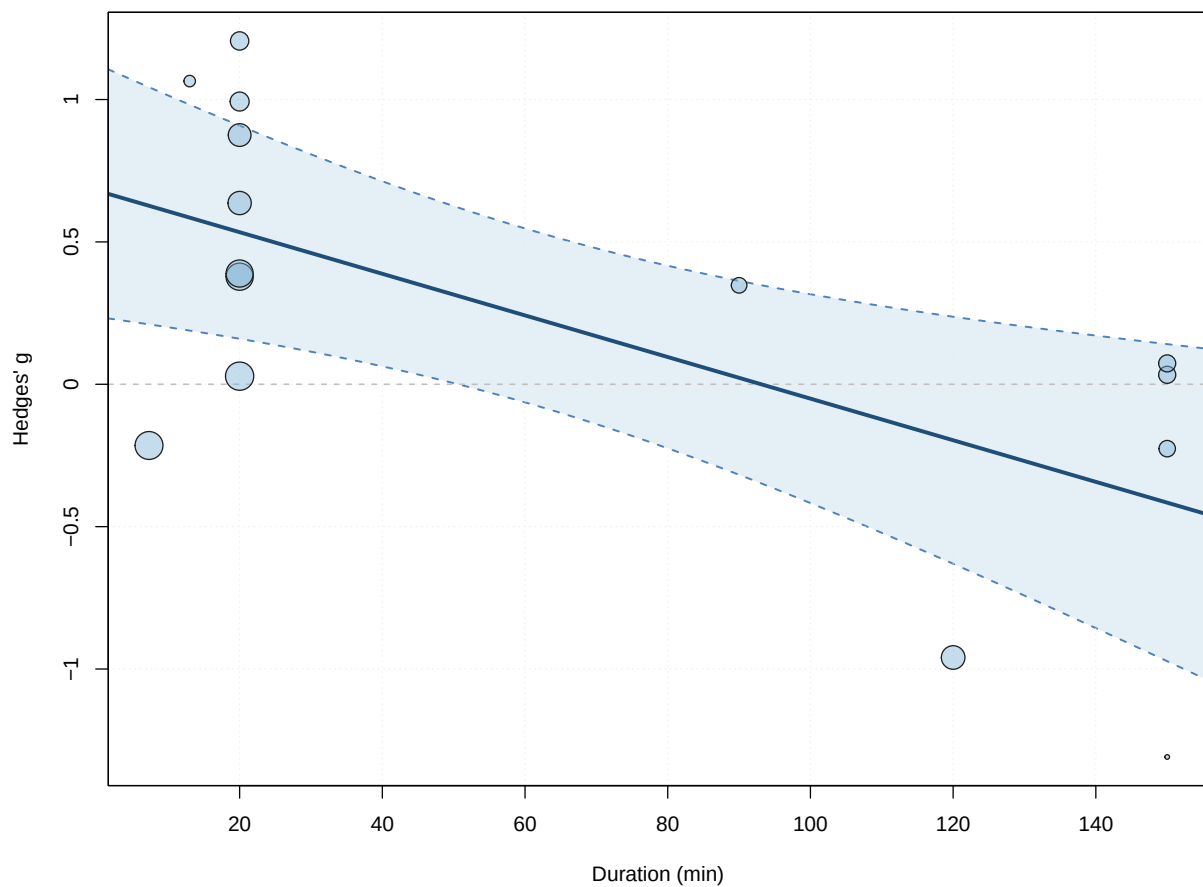
Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.6802	0.2053	3.3142	13	0.0056	0.2368	
duration	-0.0073	0.0024	-3.0077	13	0.0101	-0.0126	
intrcpt	1.1237	**					
duration	-0.0021	*					

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

! Important

Significant



10.7.5.4. Dose

Mixed-Effects Model (k = 14; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-9.3620	26.4336	24.7240	26.6412	27.1240

tau² (estimated amount of residual heterogeneity): 0.1656 (SE = 0.1075)
 tau (square root of estimated tau² value): 0.4070
 I² (residual heterogeneity / unaccounted variability): 65.73%
 H² (unaccounted variability / sampling variability): 2.92
 R² (amount of heterogeneity accounted for): 51.12%

Test for Residual Heterogeneity:
 QE(df = 12) = 34.1013, p-val = 0.0007

Test of Moderators (coefficient 2):
 F(df1 = 1, df2 = 12) = 7.2942, p-val = 0.0193

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.7315	0.2022	3.6180	12	0.0035	0.2910	
dose	-0.0010	0.0004	-2.7008	12	0.0193	-0.0019	
intrcpt	1.1721	**					
dose	-0.0002	*					

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

10.7.5.5. IFNMixed-Effects Model (k = 7; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-4.0987	14.9426	14.1973	14.0350	22.1973

tau² (estimated amount of residual heterogeneity): 0.1868 (SE = 0.1615)tau (square root of estimated tau² value): 0.4322I² (residual heterogeneity / unaccounted variability): 74.03%H² (unaccounted variability / sampling variability): 3.85R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 5) = 19.9174, p-val = 0.0013

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 0.0002, p-val = 0.9881

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.6319	0.8621	0.7330	5	0.4965	-1.5842	
IFN	-0.0073	0.4649	-0.0156	5	0.9881	-1.2024	
intrcpt	2.8480						
IFN	1.1878						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

10.7.5.6. IL_4Mixed-Effects Model (k = 9; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-6.2496	20.3845	18.4992	19.0909	23.2992

tau² (estimated amount of residual heterogeneity): 0.2223 (SE = 0.1585)
 tau (square root of estimated tau² value): 0.4715
 I² (residual heterogeneity / unaccounted variability): 77.77%
 H² (unaccounted variability / sampling variability): 4.50
 R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 7) = 30.1193, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 7) = 0.3341, p-val = 0.5814

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.4967	0.2145	2.3160	7	0.0537	-0.0104	
IL_4	0.1043	0.1804	0.5780	7	0.5814	-0.3224	
intrcpt	1.0038						
IL_4	0.5310						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

10.7.5.7. IL_6Mixed-Effects Model (k = 14; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-8.9389	28.0846	23.8778	25.7949	26.2778

tau² (estimated amount of residual heterogeneity): 0.1675 (SE = 0.1022)
 tau (square root of estimated tau² value): 0.4092
 I² (residual heterogeneity / unaccounted variability): 70.18%
 H² (unaccounted variability / sampling variability): 3.35
 R² (amount of heterogeneity accounted for): 7.58%

Test for Residual Heterogeneity:

QE(df = 12) = 36.6447, p-val = 0.0003

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 12) = 1.8665, p-val = 0.1969

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.3688	0.1537	2.4001	12	0.0335	0.0340	
IL_6	0.1207	0.0884	1.3662	12	0.1969	-0.0718	
intrcpt	0.7036						*
IL_6	0.3132						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

10.7.5.8. IL_10

Mixed-Effects Model (k = 11; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-7.0987	23.2792	20.1975	21.3911	23.6260

tau² (estimated amount of residual heterogeneity): 0.1760 (SE = 0.1201)

tau (square root of estimated tau² value): 0.4195

I² (residual heterogeneity / unaccounted variability): 71.18%

H² (unaccounted variability / sampling variability): 3.47

R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 9) = 34.2151, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 9) = 0.0965, p-val = 0.7631

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.5975	0.1530	3.9061	9	0.0036	0.2515	
IL_10	0.1301	0.4189	0.3107	9	0.7631	-0.8174	

10. Th1/Th2 Ratio — Immediate Post-Exercise

```
intrcpt  0.9435  **
IL_10    1.0777
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note

Not significant

10.7.5.9. IL_2

Mixed-Effects Model (k = 7; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-3.8027	14.3505	13.6053	13.4430	21.6053

```
tau^2 (estimated amount of residual heterogeneity):    0.1669 (SE = 0.1434)
tau (square root of estimated tau^2 value):           0.4086
I^2 (residual heterogeneity / unaccounted variability): 74.80%
H^2 (unaccounted variability / sampling variability):  3.97
R^2 (amount of heterogeneity accounted for):           0.00%
```

Test for Residual Heterogeneity:

QE(df = 5) = 19.7007, p-val = 0.0014

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 0.4392, p-val = 0.5368

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.1414	0.7407	0.1909	5	0.8561	-1.7626	
IL_2	0.2684	0.4049	0.6627	5	0.5368	-0.7725	
intrcpt	2.0453						
IL_2	1.3093						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note

Not significant

10.8. Multivariate Models

Multivariate meta-regression tests interactions between candidate moderators. None of the models below yields independent predictors.

10.8.1. Intensity * Duration

Mixed-Effects Model (k = 14; tau² estimator: HE)

```
tau^2 (estimated amount of residual heterogeneity):    0.1901 (SE = 0.1290)
tau (square root of estimated tau^2 value):           0.4360
I^2 (residual heterogeneity / unaccounted variability): 68.47%
H^2 (unaccounted variability / sampling variability):  3.17
R^2 (amount of heterogeneity accounted for):           43.88%
```

Test for Residual Heterogeneity:

QE(df = 10) = 26.5082, p-val = 0.0031

Test of Moderators (coefficients 2:4):

F(df1 = 3, df2 = 10) = 2.9355, p-val = 0.0857

Model Results:

	estimate	se	tval	df
intrcpt	-0.2459	0.6993	-0.3516	10
intensityVigorous	1.5730	1.5618	1.0071	10
duration	0.0452	0.0381	1.1868	10
intensityVigorous:duration	-0.0561	0.0394	-1.4245	10

	pval	ci.lb	ci.ub
intrcpt	0.7324	-1.8040	1.3122
intensityVigorous	0.3376	-1.9070	5.0529
duration	0.2627	-0.0397	0.1300
intensityVigorous:duration	0.1848	-0.1438	0.0316

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

10.8.2. IL_10 * IL_4

Mixed-Effects Model (k = 9; tau² estimator: HE)

```
tau^2 (estimated amount of residual heterogeneity):    0.3297 (SE = 0.2594)
tau (square root of estimated tau^2 value):           0.5742
```

10. Th1/Th2 Ratio — Immediate Post-Exercise

I² (residual heterogeneity / unaccounted variability): 81.69%
H² (unaccounted variability / sampling variability): 5.46
R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 5) = 30.1034, p-val < .0001

Test of Moderators (coefficients 2:4):

F(df1 = 3, df2 = 5) = 0.0923, p-val = 0.9611

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.5401	0.3177	1.6997	5	0.1499	-0.2767	
IL_10	0.0951	0.6813	0.1395	5	0.8945	-1.6563	
IL_4	0.0953	0.2254	0.4229	5	0.6899	-0.4842	
IL_10:IL_4	-0.2817	1.4854	-0.1897	5	0.8570	-4.1000	
intrcpt	1.3568						
IL_10	1.8464						
IL_4	0.6748						
IL_10:IL_4	3.5365						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note

Not significant

10.8.3. IL_2 * IFN

Mixed-Effects Model (k = 7; tau² estimator: HE)

tau² (estimated amount of residual heterogeneity): 0.2707 (SE = 0.2724)
tau (square root of estimated tau² value): 0.5203
I² (residual heterogeneity / unaccounted variability): 81.40%
H² (unaccounted variability / sampling variability): 5.38
R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 3) = 17.0604, p-val = 0.0007

Test of Moderators (coefficients 2:4):

F(df1 = 3, df2 = 3) = 0.2470, p-val = 0.8596

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	5.7989	9.5568	0.6068	3	0.5868	-24.6150	
IL_2	-2.8869	5.5986	-0.5156	3	0.6417	-20.7043	
IFN	-3.1991	5.2880	-0.6050	3	0.5879	-20.0280	
IL_2:IFN	1.7440	3.0132	0.5788	3	0.6033	-7.8454	
intrcpt	36.2129						
IL_2	14.9304						
IFN	13.6298						
IL_2:IFN	11.3335						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

11. Th1/Th2 Ratio — ~2 h Post-Exercise

11.1. Data Import

The ~2 h window (effect_size/Ratio/pre_post+2h/Ratio_pre_post+2h_r_050.csv) is reported for the main analysis, outlier check, funnel, and forest only; the limited data do not support moderator modelling. No study is excluded a priori.

11.2. Primary Meta-Analysis

11.2.1. {meta}

Number of studies: k = 5

	SMD	95% CI	t
Random effects model (HK)	-0.5071	[-1.1495; 0.1353]	-2.19
Prediction interval		[-1.8150; 0.8007]	
	p-value		
Random effects model (HK)	0.0935		
Prediction interval			

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.1626$ [0.0000; 2.3056]; $\tau = 0.4033$ [0.0000; 1.5184]
 $I^2 = 44.6\%$ [0.0%; 79.7%]; $H = 1.34$ [1.00; 2.22]

Test of heterogeneity:

Q	d.f.	p-value
7.22	4	0.1245

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 4)
- Prediction interval based on t-distribution (df = 4)

11.2.2. {metafor}

Random-Effects Model (k = 5; τ^2 estimator: HE)

11. Th1/Th2 Ratio — ~2 h Post-Exercise

```
tau^2 (estimated amount of total heterogeneity): 0.1626 (SE = 0.2133)
tau (square root of estimated tau^2 value):      0.4033
I^2 (total heterogeneity / total variability):    55.30%
H^2 (total variability / sampling variability):    2.24
```

Test for Heterogeneity:

Q(df = 4) = 7.2248, p-val = 0.1245

Model Results:

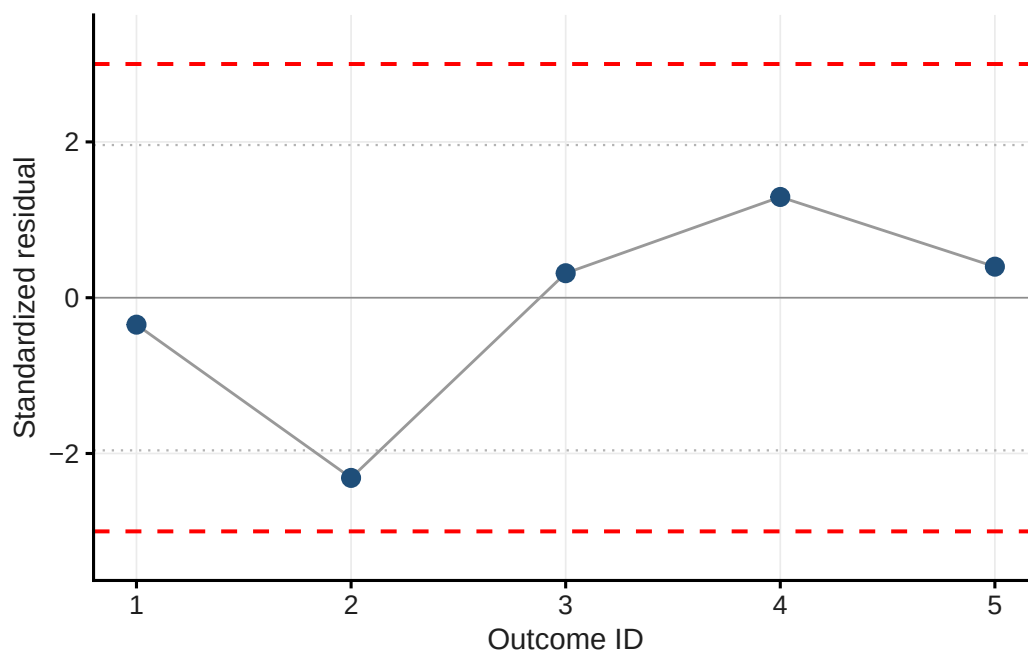
```
estimate      se      tval  df    pval    ci.lb    ci.ub
-0.5071  0.2314  -2.1916   4   0.0935  -1.1495   0.1353  .
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

11.3. Outlier Detection

Table 11.1.: Standardised residuals

Short_reference	g	std_res
Ibfelt et al. (2002)	-0.679	-0.346
Steensberg et al. (2001)	-1.439	-2.313
Lancaster et al. (2005)a	-0.357	0.314
Lancaster et al. (2005)b	0.000	1.293
Lancaster et al. (2005)c	-0.317	0.398



i Note

No outlier.

11.4. Funnel Plot & Influential Cases

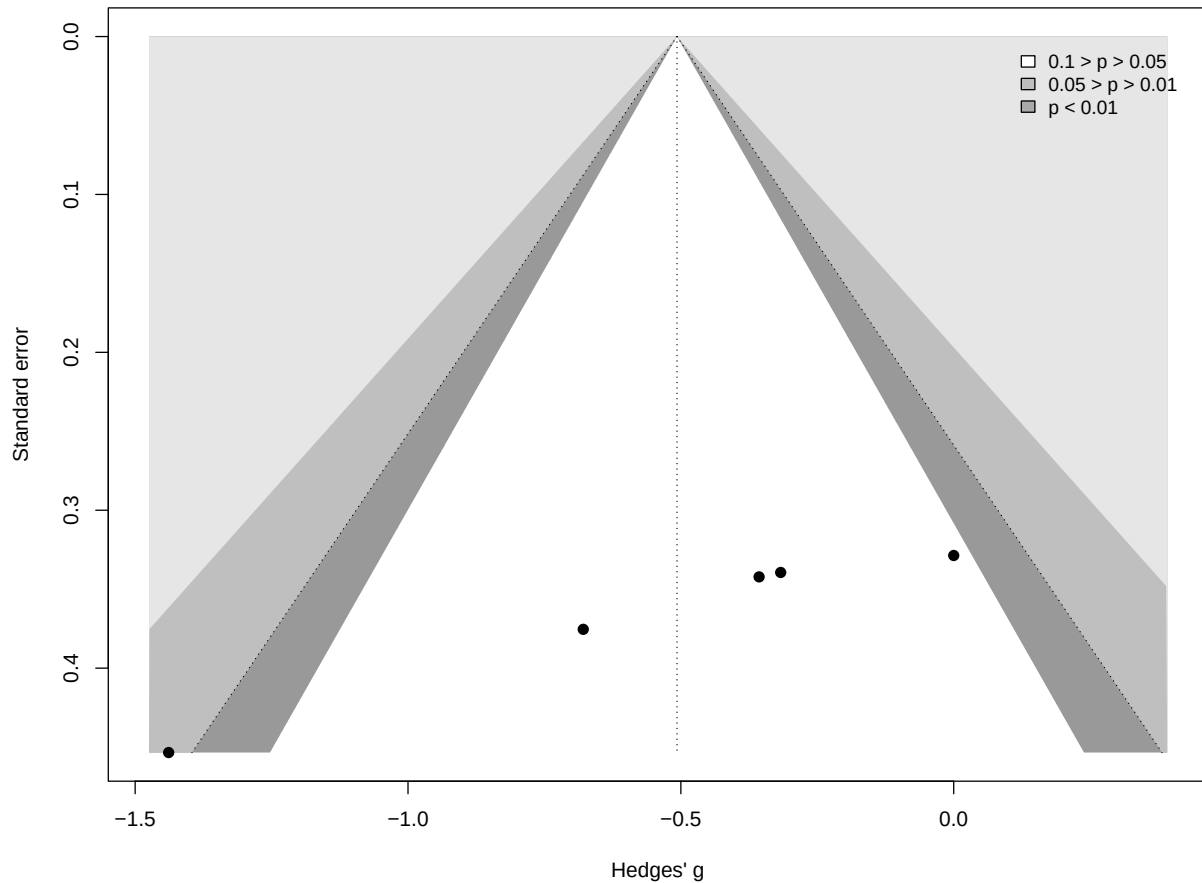
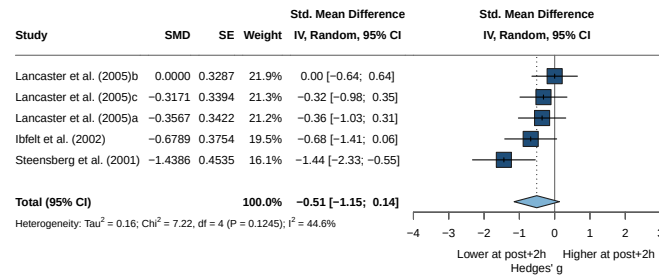


Table 11.2.: DFBETAS influence diagnostics

Short_reference	g	dfbetas
Ibfelt et al. (2002)	-0.679	-0.3097587
Steensberg et al. (2001)	-1.439	-0.8804841
Lancaster et al. (2005)a	-0.357	0.1798076
Lancaster et al. (2005)b	0.000	0.9012816
Lancaster et al. (2005)c	-0.317	0.2552408

One value above 1 and one below -1 were identified as influential cases; neither is an outlier, so all outcomes are retained.

11.5. Forest Plot



11.6. Asymmetry Test

Note: the number of studies in this window is too small to reliably test for small-study effects (minimum $k = 10$). No Egger's test or trim-and-fill is performed.

12. Th1/Th2 Ratio — 17 h Post-Exercise

12.1. Data Import

12.2. Primary Meta-Analysis

Kostrzewa-Nowak et al. (2019) is removed before pooling and the model re-fitted.

12.2.1. {meta}

Number of studies: $k = 11$

	SMD	95% CI	t
Random effects model (HK)	0.6433	[-0.0265; 1.3131]	2.14
Prediction interval		[-2.4114; 3.6980]	
	p-value		
Random effects model (HK)	0.0580		
Prediction interval			

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 1.7065$ [0.1105; 3.8540]; $\tau = 1.3063$ [0.3324; 1.9632]
 $I^2 = 73.5\%$ [51.6%; 85.5%]; $H = 1.94$ [1.44; 2.62]

Test of heterogeneity:

Q	d.f.	p-value
37.71	10	< 0.0001

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model ($df = 10$)
- Prediction interval based on t-distribution ($df = 10$)

12.2.2. {metafor}

Random-Effects Model ($k = 11$; τ^2 estimator: HE)

τ^2 (estimated amount of total heterogeneity): 1.7065 (SE = 1.0895)
 τ (square root of estimated τ^2 value): 1.3063

12. Th1/Th2 Ratio — 17 h Post-Exercise

I² (total heterogeneity / total variability): 95.92%
H² (total variability / sampling variability): 24.49

Test for Heterogeneity:
Q(df = 10) = 37.7068, p-val < .0001

Model Results:

estimate	se	tval	df	pval	ci.lb	ci.ub
0.6433	0.3006	2.1401	10	0.0580	-0.0265	1.3131

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

12.2.3. Adapted Model {meta}

Number of studies: k = 10

	SMD	95% CI	t
Random effects model (HK)	0.5129	[0.0830; 0.9427]	2.70
Prediction interval		[-0.8512; 1.8770]	
	p-value		
Random effects model (HK)	0.0244		
Prediction interval			

Quantifying heterogeneity (with 95% CIs):
tau² = 0.3238 [0.0746; 1.2164]; tau = 0.5690 [0.2731; 1.1029]
I² = 72.7% [48.5%; 85.6%]; H = 1.92 [1.39; 2.63]

Test of heterogeneity:
Q d.f. p-value
33.01 9 0.0001

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for tau²
- Q-Profile method for confidence interval of tau² and tau
- Calculation of I² based on Q
- Hartung-Knapp adjustment for random effects model (df = 9)
- Prediction interval based on t-distribution (df = 9)

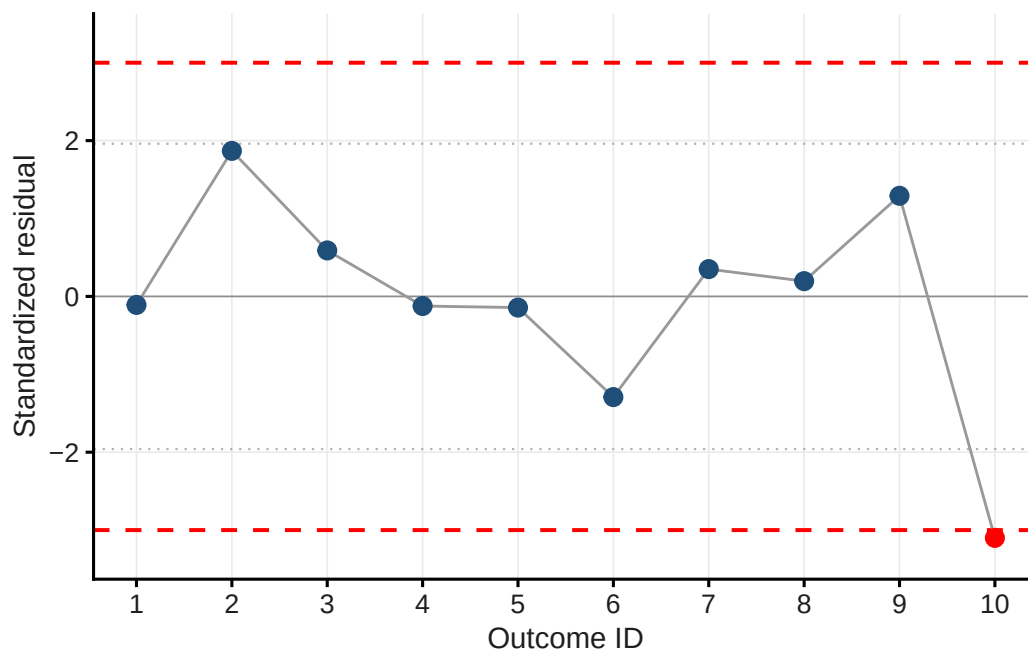
12.3. Outlier Detection

Table 12.1.: Standardised residuals

Short_reference	g	std_res
Kostrzewa-Nowak & Nowak (2018)a	0.458	-0.109
Kostrzewa-Nowak & Nowak (2018)b	1.463	1.868

Table 12.1.: Standardised residuals

Short_reference	g	std_res
Kostrzewa-Nowak & Nowak (2020a)a	0.785	0.590
Kostrzewa-Nowak & Nowak (2020a)b	0.454	-0.123
Kostrzewa-Nowak & Nowak (2020b)a	0.442	-0.144
Kostrzewa-Nowak & Nowak (2020b)b	-0.051	-1.292
Kostrzewa-Nowak & Nowak (2020b)c	0.685	0.350
Kostrzewa-Nowak & Nowak (2020b)d	0.609	0.196
Kostrzewa-Nowak & Nowak (2020b)e	1.141	1.291
Steensberg et al. (2001)	-0.854	-3.100



i Note

No outlier.

12.4. Funnel Plot & Influential Cases

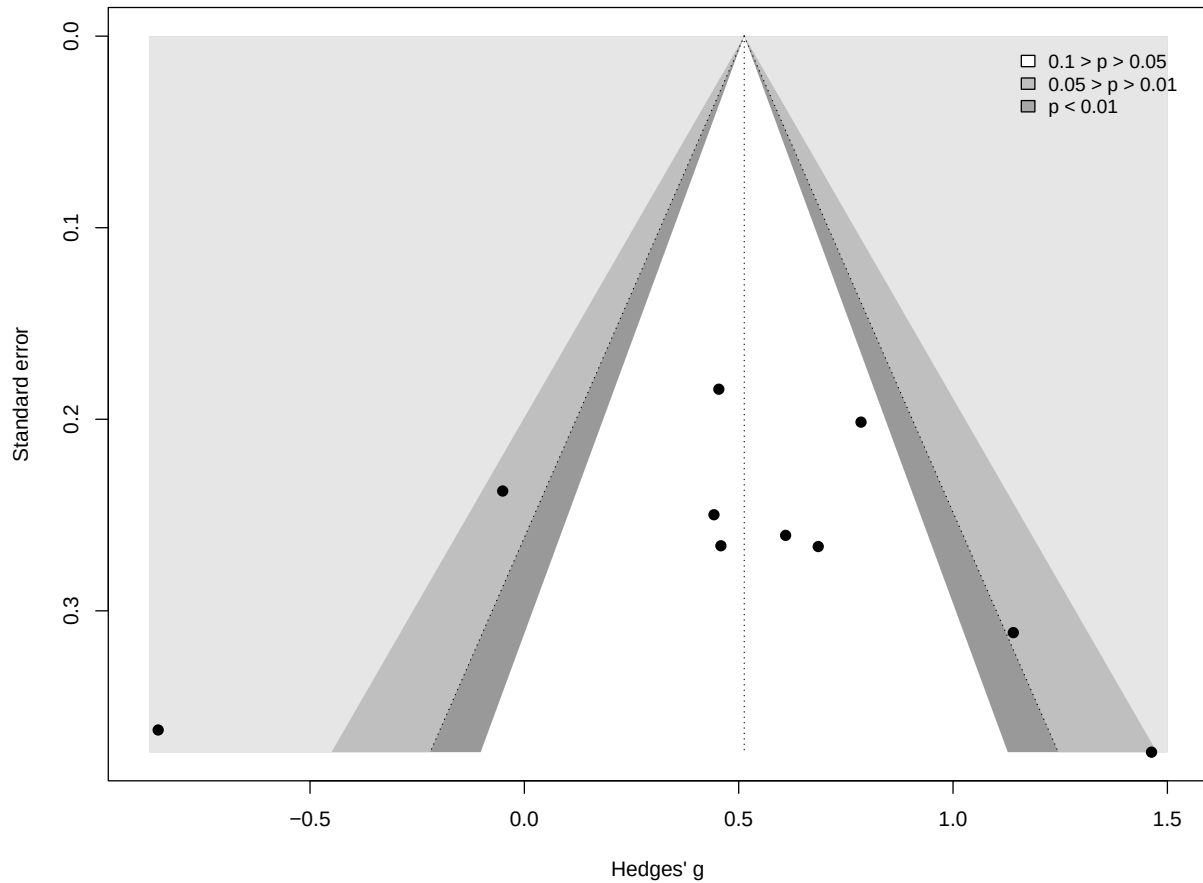


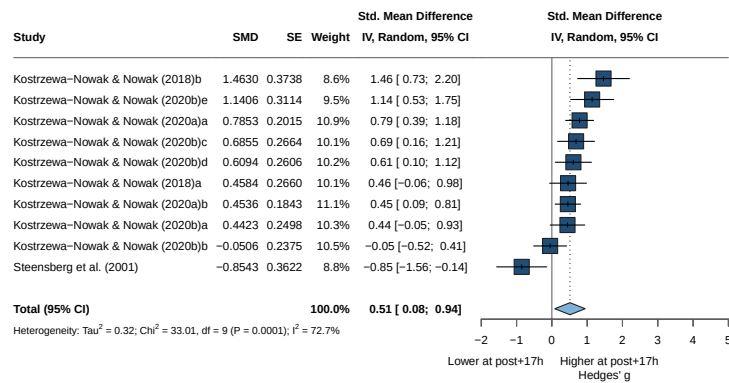
Table 12.2.: DFBETAS influence diagnostics

Short_reference	g	dfbetas
Kostrzewska-Nowak & Nowak (2018)a	0.458	-0.06601183
Kostrzewska-Nowak & Nowak (2018)b	1.463	0.57178772
Kostrzewska-Nowak & Nowak (2020a)a	0.785	0.64171703
Kostrzewska-Nowak & Nowak (2020a)b	0.454	-0.16812828
Kostrzewska-Nowak & Nowak (2020b)a	0.442	-0.09893333
Kostrzewska-Nowak & Nowak (2020b)b	-0.051	-0.90092632
Kostrzewska-Nowak & Nowak (2020b)c	0.685	0.21588205
Kostrzewska-Nowak & Nowak (2020b)d	0.609	0.12756806
Kostrzewska-Nowak & Nowak (2020b)e	1.141	0.55656301
Steensberg et al. (2001)	-0.854	-0.87675665

i Note

No influential case.

12.5. Forest Plot



12.6. Asymmetry Test

Linear regression test of funnel plot asymmetry

Test result: $t = 0.02$, $df = 8$, $p\text{-value} = 0.9847$

Bias estimate: 0.0606 (SE = 3.0606)

Details:

- multiplicative residual heterogeneity variance ($\tau^2 = 4.1263$)
- predictor: standard error
- weight: inverse variance
- reference: Egger et al. (1997), BMJ

i Note

No asymmetry -> no trim-and-fill

12.7. Moderator Analyses

12.7.1. Risk of Bias

12.7.1.1. Model

Number of studies: $k = 10$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.3238$ [0.0746; 1.2164]; $\tau = 0.5690$ [0.2731; 1.1029]

$I^2 = 72.7\%$ [48.5%; 85.6%]; $H = 1.92$ [1.39; 2.63]

Test of heterogeneity:

Q	d.f.	p-value
33.01	9	0.0001

Results for subgroups (random effects model (HK)):

	k	SMD	95% CI	τ^2	τ
RoB = Moderate	3	0.3550	[-2.5088; 3.2188]	1.2364	1.1119
RoB = Serious	2	0.6095	[-1.4940; 2.7129]	0.0177	0.1331
RoB = Low	5	0.5399	[0.0105; 1.0692]	0.1145	0.3384

	Q	I^2
RoB = Moderate	20.08	90.0%
RoB = Serious	1.48	32.2%
RoB = Low	10.32	61.2%

Test for subgroup differences (random effects model (HK)):

	Q	d.f.	p-value
Between groups	0.19	2	0.9112

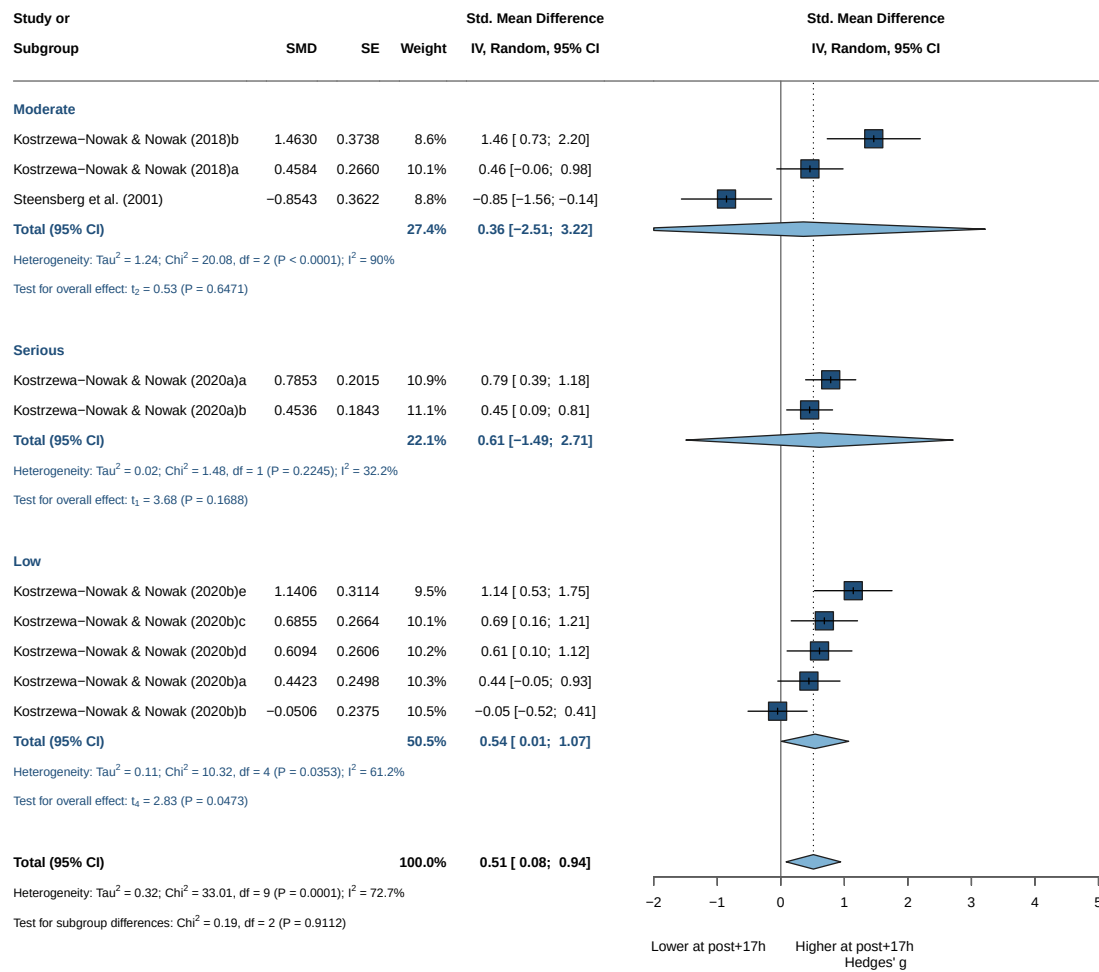
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

i Note

Not significant results

12.7.1.2. Forrest Plot



12.7.2. Intensity

12.7.2.1. Model

Number of studies: $k = 10$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.3238$ [0.0746; 1.2164]; $\tau = 0.5690$ [0.2731; 1.1029]

$I^2 = 72.7\%$ [48.5%; 85.6%]; $H = 1.92$ [1.39; 2.63]

Test of heterogeneity:

Q	d.f.	p-value
33.01	9	0.0001

Results for subgroups (random effects model (HK)):

	k	SMD	95% CI	τ^2
intensity = Maximal	9	0.6254	[0.3097; 0.9411]	0.1202
intensity = Vigorous	1	-0.8543	[-1.5641; -0.1444]	--
	tau	Q	I^2	
intensity = Maximal	0.3466	18.06	55.7%	

12. Th1/Th2 Ratio — 17 h Post-Exercise

intensity = Vigorous -- 0.00 --

Test for subgroup differences (random effects model (HK)):

	Q	d.f.	p-value
Between groups	14.60	1	0.0001

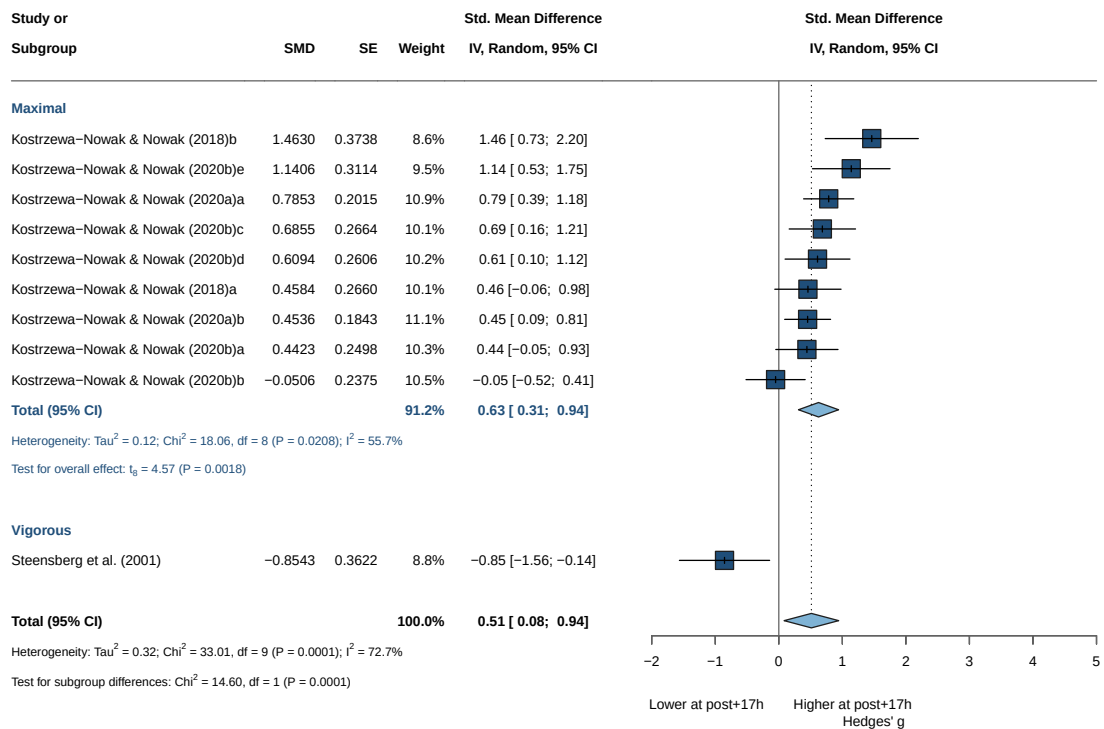
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for τ^2
- Q-Profile method for confidence interval of τ^2 and τ
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

! Important

Significant results but 10 vs 1

12.7.2.2. Forrest Plot



12.7.3. Continuous Moderators (Meta-Regression)

12.7.3.1. Age

Mixed-Effects Model (k = 10; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-3.3225	14.8176	12.6449	13.5527	16.6449

tau ² (estimated amount of residual heterogeneity):	0.0757 (SE = 0.0748)
tau (square root of estimated tau ² value):	0.2751
I ² (residual heterogeneity / unaccounted variability):	54.86%
H ² (unaccounted variability / sampling variability):	2.22
R ² (amount of heterogeneity accounted for):	76.63%

Test for Residual Heterogeneity:

QE(df = 8) = 14.6115, p-val = 0.0672

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 8) = 12.7719, p-val = 0.0073

Model Results:

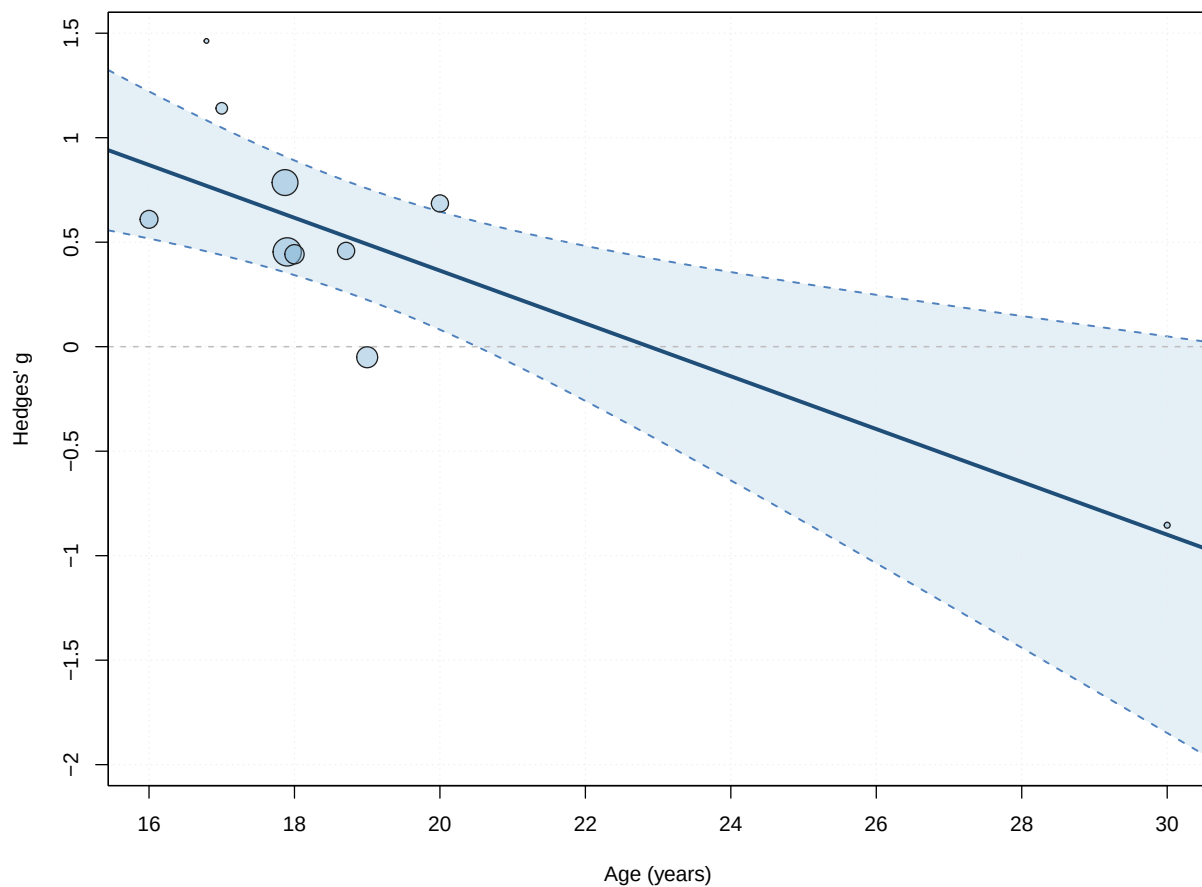
	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	2.8912	0.6757	4.2791	8	0.0027	1.3331	
age	-0.1264	0.0354	-3.5738	8	0.0073	-0.2079	
intrcpt	4.4493						**
age	-0.0448						**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

12. Th1/Th2 Ratio — 17 h Post-Exercise



12.7.3.2. BMI

Mixed-Effects Model (k = 9; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-2.9847	13.9489	11.9695	12.5612	16.7695

tau² (estimated amount of residual heterogeneity): 0.0868 (SE = 0.0866)
tau (square root of estimated tau² value): 0.2947
I² (residual heterogeneity / unaccounted variability): 57.84%
H² (unaccounted variability / sampling variability): 2.37
R² (amount of heterogeneity accounted for): 27.72%

Test for Residual Heterogeneity:

QE(df = 7) = 12.7253, p-val = 0.0791

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 7) = 2.7310, p-val = 0.1424

Model Results:

	estimate	se	tval	df	pval	ci.lb
intrcpt	-3.4303	2.4532	-1.3983	7	0.2047	-9.2312
bmi	0.1920	0.1162	1.6526	7	0.1424	-0.0827
	ci.ub					
intrcpt	2.3706					
bmi	0.4668					

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

12.7.3.3. DurationMixed-Effects Model (k = 8; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-4.6365	14.5351	15.2729	15.5113	21.2729

tau² (estimated amount of residual heterogeneity): 0.1642 (SE = 0.1421)tau (square root of estimated tau² value): 0.4052I² (residual heterogeneity / unaccounted variability): 68.62%H² (unaccounted variability / sampling variability): 3.19R² (amount of heterogeneity accounted for): 60.58%

Test for Residual Heterogeneity:

QE(df = 6) = 16.5428, p-val = 0.0111

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 6) = 7.1231, p-val = 0.0371

Model Results:

	estimate	se	tval	df	pval	ci.lb
intrcpt	0.8663	0.2241	3.8659	6	0.0083	0.3180
duration	-0.0115	0.0043	-2.6689	6	0.0371	-0.0220
	ci.ub					
intrcpt	1.4146	**				
duration	-0.0010	*				

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

12.7.3.4. Dose

Mixed-Effects Model (k = 8; tau^2 estimator: HE)

logLik	deviance	AIC	BIC	AICc
-4.6365	14.5351	15.2729	15.5113	21.2729

tau^2 (estimated amount of residual heterogeneity): 0.1642 (SE = 0.1421)
tau (square root of estimated tau^2 value): 0.4052
I^2 (residual heterogeneity / unaccounted variability): 68.62%
H^2 (unaccounted variability / sampling variability): 3.19
R^2 (amount of heterogeneity accounted for): 60.58%

Test for Residual Heterogeneity:
QE(df = 6) = 16.5428, p-val = 0.0111

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 6) = 7.1231, p-val = 0.0371

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.9372	0.2421	3.8715	6	0.0083	0.3448	
dose	-0.0017	0.0006	-2.6689	6	0.0371	-0.0033	
intrcpt	1.5295	**					
dose	-0.0001	*					

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

 Note

Not significant

12.7.3.5. MET

 Note

Only modelled when the MET column is present; the current $r = 0.5$ export does not include it, so this meta-regression is skipped.

12.7.3.6. IFN

Mixed-Effects Model (k = 7; tau^2 estimator: HE)

logLik	deviance	AIC	BIC	AICc
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-2.0723 11.1835 10.1447 9.9824 18.1447

tau² (estimated amount of residual heterogeneity): 0.0848 (SE = 0.0943)
 tau (square root of estimated tau² value): 0.2913
 I² (residual heterogeneity / unaccounted variability): 58.34%
 H² (unaccounted variability / sampling variability): 2.40
 R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 5) = 10.9775, p-val = 0.0518

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 0.4201, p-val = 0.5455

Model Results:

	estimate	se	tval	df	pval	ci.lb
intrcpt	0.8400	0.4512	1.8617	5	0.1217	-0.3198
IFN	-0.0933	0.1440	-0.6481	5	0.5455	-0.4634
	ci.ub					
intrcpt	1.9998					
IFN	0.2768					

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note

Not significant

12.7.3.7. IL_4

Mixed-Effects Model (k = 7; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-1.4940	10.0268	8.9880	8.8258	16.9880

tau² (estimated amount of residual heterogeneity): 0.0657 (SE = 0.0832)
 tau (square root of estimated tau² value): 0.2563
 I² (residual heterogeneity / unaccounted variability): 52.41%
 H² (unaccounted variability / sampling variability): 2.10
 R² (amount of heterogeneity accounted for): 8.99%

Test for Residual Heterogeneity:

QE(df = 5) = 8.9170, p-val = 0.1124

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 1.4107, p-val = 0.2883

12. Th1/Th2 Ratio — 17 h Post-Exercise

Model Results:

	estimate	se	tval	df	pval	ci.lb
intrcpt	0.4280	0.1697	2.5220	5	0.0530	-0.0083
IL_4	0.2553	0.2149	1.1877	5	0.2883	-0.2972
	ci.ub					
intrcpt	0.8643	.				
IL_4	0.8077					

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

12.7.3.8. IL_6

Mixed-Effects Model (k = 9; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-0.3049	8.5892	6.6098	7.2014	11.4098

tau ² (estimated amount of residual heterogeneity):	0.0027 (SE = 0.0374)
tau (square root of estimated tau ² value):	0.0522
I ² (residual heterogeneity / unaccounted variability):	4.44%
H ² (unaccounted variability / sampling variability):	1.05
R ² (amount of heterogeneity accounted for):	97.74%

Test for Residual Heterogeneity:

QE(df = 7) = 8.6202, p-val = 0.2811

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 7) = 7.8742, p-val = 0.0263

Model Results:

	estimate	se	tval	df	pval	ci.lb
intrcpt	0.3614	0.1208	2.9922	7	0.0202	0.0758
IL_6	0.1441	0.0514	2.8061	7	0.0263	0.0227
	ci.ub					
intrcpt	0.6469	*				
IL_6	0.2655	*				

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

! Important

Significant

12.7.3.9. IL_10

Mixed-Effects Model (k = 9; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-4.3135	16.6064	14.6270	15.2186	19.4270

tau² (estimated amount of residual heterogeneity): 0.1340 (SE = 0.1107)
 tau (square root of estimated tau² value): 0.3661
 I² (residual heterogeneity / unaccounted variability): 68.68%
 H² (unaccounted variability / sampling variability): 3.19
 R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 7) = 17.5453, p-val = 0.0142

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 7) = 0.3559, p-val = 0.5696

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.7042	0.1924	3.6597	7	0.0081	0.2492	
IL_10	-0.0754	0.1263	-0.5966	7	0.5696	-0.3741	
intrcpt	1.1592						**
IL_10	0.2233						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

12.7.3.10. IL_2

Mixed-Effects Model (k = 7; tau² estimator: HE)

logLik	deviance	AIC	BIC	AICc
-2.1646	11.3680	10.3292	10.1669	18.3292

tau² (estimated amount of residual heterogeneity): 0.0869 (SE = 0.0971)

12. Th1/Th2 Ratio — 17 h Post-Exercise

tau (square root of estimated tau² value): 0.2947
I² (residual heterogeneity / unaccounted variability): 58.04%
H² (unaccounted variability / sampling variability): 2.38
R² (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 5) = 11.4142, p-val = 0.0438

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 0.2537, p-val = 0.6359

Model Results:

	estimate	se	tval	df	pval	ci.lb	ci.ub
intrcpt	0.6698	0.2567	2.6093	5	0.0477	0.0099	
IL_2	-0.0585	0.1162	-0.5037	5	0.6359	-0.3573	
intrcpt	1.3296						*
IL_2	0.2402						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i Note

Not significant

12.8. Multivariate Models

Mixed-Effects Model (k = 8; tau² estimator: HE)

tau² (estimated amount of residual heterogeneity): 0.1642 (SE = 0.1421)
tau (square root of estimated tau² value): 0.4052
I² (residual heterogeneity / unaccounted variability): 68.62%
H² (unaccounted variability / sampling variability): 3.19
R² (amount of heterogeneity accounted for): 60.58%

Test for Residual Heterogeneity:

QE(df = 6) = 16.5428, p-val = 0.0111

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 6) = 7.1231, p-val = 0.0371

Model Results:

	estimate	se	tval	df	pval
intrcpt	0.6369	0.1810	3.5193	6	0.0125
intensityVigorous	-1.4912	0.5587	-2.6689	6	0.0371

	ci.lb	ci.ub	
intrcpt	0.1941	1.0797	*
intensityVigorous	-2.8583	-0.1240	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

! Important

Model significant but no independent predictors

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